Development and Validation of a Questionnaire for Assessment of Happiness: The Happiness Rating Scale.

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2 authors:

Ram B Singh
Halberg Hospital And Research Institute,Moradabad, India

Harpal Buttar
University of Ottawa

Statins Administration for Primary Prevention of Stroke and Coronary Artery Disease: A Scientific Statement of the International College of Cardiology

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Ram Kushal Singh
Deen Dayal Upadhyay Gorakhpur University

Jan Fedacko
Pavol Jozef Šafárik University in Košice

Pasquale Palmiero
Azienda Sanitaria Locale Brindisi

Germaine Cornélissen
University of Minnesota Twin Cities

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School of Forensic and Applied Sciences
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Preston, Lancashire, England
jsingh3@uclan.ac.uk

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Department of Epidemiology
Harvard School of Public Health
Boston, USA
kjoshipu@hsph.harvard.edu

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Division of Cardiology
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Taipei Veterans General Hospital
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Statins Administration for Primary Prevention of Stroke and Coronary Artery Disease: A Scientific Statement of the International College of Cardiology

Ram B Singh[1], Jan Fedacko[2], Galaleldin Nagib Elkilany[3,*], Pasquale Palmiero[4], Krasimira Hristova[5], and Germaine Cornelissen[6]

[1]Halberg Hospital and Research Institute, Moradabad, India
[2]PJ Safaric University, Kosice, Slovakia
[3]Tanta University, Egypt, Kings College Hospital London & Almousa DCH, Dubai, UAE
[4]ASL BRINDISI, Cardiology Equipe, District of Brindisi, Brindisi, Italy
[5]National Heart Hospital University, Sofia, Bulgaria
[6]Halberg Chronobiology Center, University of Minnesota Medical School, Minneapolis, USA

Introduction

Statins have been criticized as a double-edged sword [1]. However, continuous research has proven their efficacy. If they are used carefully, adverse effects may be avoided. In a recent review, five major guidelines on administration of statins for primary prevention of atherosclerotic cardiovascular disease (CVD) have been discussed [2]. The guidelines are proposed by various health agencies: the UK National Institute for Health and Care Excellence (NICE; 2014) [3], US Preventive Services Task Force statement (USPSTF; 2016) [4], Canadian Cardiovascular Society (CCS; 2016) [5], European Society of Cardiology/European Atherosclerosis Society (ESC/EAS; 2016) [6], and American College of Cardiology/American Heart Association (ACC/AHA; 2018) [7]. These guidelines are based on the published evidence from multiple randomized clinical trials (RCTs) of statin therapy for primary prevention of CVDs. However, interpretation of these trials is different by each guideline and hence their recommendations for who should be treated with statins differ markedly [8-11]. This viewpoint aims to discuss causes underlying these differences and the possible influence of behavioral factors in the guidelines on primary prevention of CVDs.

Adverse Effects of Statins and Cardiovascular Mortality

Some experts believe that anti-inflammatory drugs, such as statins, may be required right after birth if inflammation is predominant; the objective is to

* Correspondence: Galaleldin Nagib Elkilany, MD, FESC, FISCU Professor and Consultant Cardiologist – SCU at Tanta University , Egypt , Kings College Hospital London & Almousa DCH, Dubai, UAE.
E-mail: galal.elkilany@gmail.com

prevent fatty streaking in the vascular system [1]. Voices from the public and the medical community argue strongly against the widespread use of statins for the primary prevention of atherosclerotic CVDs [1]. The main concerns are about adverse effects, and in some trials about lack of a total benefit in terms of mortality, cost, and a philosophical aversion to drug therapy. However, now there is sufficient evidence to refute each of these concerns [2]. Recent trials provide extensive evidence that statins reduce cardiovascular events and total mortality in individuals at lower risk of cardiovascular events than was previously recognized, and do so with an excellent margin of safety [2-6]. These studies indicate that of several primary prevention statin trials on several thousand participants, statins significantly reduce all-cause mortality, fatal and nonfatal cardiovascular disease, CAD, stroke and coronary revascularization. These benefits in risk reduction occur in the absence of an increased risk of complications, such as cancer, myalgia, rhabdomyolysis, liver enzyme elevation, renal dysfunction, or arthritism but the last endpoint, revascularization, continues to be open to bias. This analysis also included the JUPITER trial in which case fatality was greater in the intervention group compared to the control group [1].

Guidelines for Prevention of Cardiovascular Diseases

The 2013 AHA/ACC cholesterol guideline was published after a rigorous systematic review of higher-quality randomized trials, and systematic reviews and meta-analyses of randomized trials of drug therapy to reduce atherosclerotic CVD events [1]. The evidence provided by the 2013 Cochrane meta-analysis was reviewed as part for the recently released 2019 American College of Cardiology and American Heart Association guideline on the treatment of blood cholesterol to prevent atherosclerotic CVDs in adults [7]. Mortenson et al. [2] recently reviewed five major guidelines on statin therapy for primary prevention of atherosclerotic CVDs: the National Institute for Health and Care Excellence (NICE; 2014), US Preventive Services Task Force (USPSTF; 2016), Canadian Cardiovascular Society (CCS; 2016), European Society of Cardiology/European Atherosclerosis Society (ESC/EAS; 2016), and American College of Cardiology/American Heart Association (ACC/AHA; 2018). The aim was to compare the sensitivity, specificity, and estimated number needed to treat (NNT_{10}) to prevent 1 event in 10 years according to statin criteria from the five guidelines. The analyses were performed in the Copenhagen General Population Study, including 45,750 individuals aged 40 to 75 years with a mean follow-up of 10.9 years. The median age at start was 56 years, and 43% of participants were men (n = 19,870 of 45,750). There were 4,156 CVD events during the follow-up of 10.9 years. Overall, 44% of individuals in the Copenhagen General Population Study were statin-eligible with CCS (n = 19,953 of 45,750), 42% with ACC/AHA (n = 19,400 of 45,750), 40% with NICE (n = 19,400 of 45,750), 31% with USPSTF (n = 13,966 of 45,750), and 15% with ESC/EAS (n = 6,870 of 45,750). Sensitivity and specificity for CVD events respectively were 68% (n = 2,815 of 4,156) and 59% (n = 24,456 of 41,594) for CCS, 70% (n = 2,889 of 4,156) and 60% (n = 25,083 of 41,594) for ACC/AHA, 68% (n = 2,815 of 4,156) and 63% (n = 26,213 of 41,594) for NICE, 57% (n = 2,377 of 4,156) and 72% (n = 30,005 of 41,594) for USPSTF, and 24% (n = 1,001 of 4,156) and 86% (n = 35,725 of 41,594) for ESC/EAS. The NNT_{10} to prevent 1 CVD using moderate-intensity and high-intensity statin therapy, respectively, was 32 and 21 for CCS criteria, 30 and 20 for ACC/AHA criteria, 30 and 20 for NICE criteria, 27 and 18 for USPSTF criteria, and 29 and 20 for ESC/EAS criteria. The relevance is that with similar NNT_{10} to prevent 1 event, the CCS, ACC/AHA, and NICE guidelines correctly assign statin therapy to many more of the subjects, who on follow-up develop CVDs, compared with the USPSTF and ESC/EAS guidelines. Therefore, the results indicate that the CCS, ACC/AHA, or NICE guidelines may be preferred for primary prevention. However, this choice is not final because due consideration is not given to diet and lifestyle factors, which can independently influence LDL-C concentration.

Role of Confounders on Results of Statins Trials

Some experts emphasize the role of numerous confounders among subjects included for assessment of cholesterol targets due to the presence of behavioral risk factors and protective factors [12-15]. These confounders, sedentary behavior and mental stress, as well as depression, may add to adverse effects of statins on long-term use and can predispose to new-onset incident diabetes, which is clear from a recent meta-analysis suggesting that statin therapy was associated with a risk of incident diabetes [11]. The study showed that relative reduction in low-density lipoprotein cholesterol (LDL-C) was a good indicator of the risk of new-onset diabetes. The overall risk of incident diabetes increased by 11% and the group with intensive LDL-C reduction by statin had an 18% increase in the likelihood of developing diabetes. The risks of incident diabetes were 13% and 29% in the subgroups with 30–40% and 40–50% reductions in LDL-C, respectively. These results showed that LDL-C decrease may provide a dynamic risk assessment parameter for new-onset diabetes. LDL-C lowering may be positively related to the risk of new-onset diabetes. Blood glucose monitoring is important if there is 30% decline in LDL cholesterol during lipid lowering therapy in order to detect incident diabetes in these populations [11]. However, not all statins have the same effect; rosuvastatin or atorvastatin are more, while pitavastatin is least commonly associated with diabetes. More studies are needed to determine whether intensive therapy with statins can cause new-onset diabetes, which may possibly be due to decreased insulin sensitivity due to epithelial cell damage of beta cells of the pancreas. The guidelines recommend using different prediction models for CVD risk assessment as well as different risk thresholds and LDL-C criteria for prescription of statin therapy. Previous comparisons of guidelines regarding their potential for overall CVD prevention in the general population showed that recommending statins to more individuals for primary prevention prevents more events than recommending statin use to fewer individuals [8, 9]. In most of these guidelines, the ability to correctly assign statin therapy has not been compared, perhaps because the higher preventive potential with the more statin-liberal guidelines comes with the downside of treating many people at low risk of CVDs, leading to higher numbers required to treat (NNT) to prevent 1 CVD event, as compared to the more statin-conservative guidelines.

Effects on CVDs by Alternative Therapies

It should be noted that atherosclerotic CVDs can be reduced by other therapies without decreasing LDL-C, because many of these patients may have behavioral risk factors, such as mental disorders independent of blood cholesterol (Table 1) [12-19]. These important risk factors and protective factors can act as confounders, which may complicate results of cholesterol-lowering trials. For the first time, Mortensen et al. [2] compared the ability of the five major statin guidelines to correctly assign statin therapy for primary prevention of CVDs. The findings indicate that statin eligibility and sensitivity for cardiovascular events differ markedly, but the estimated NNT_{10} to prevent 1 event in 10 years is nearly identical for the five guidelines. These findings are crucial for clinical practice, as compared to the more statin-conservative guidelines. The purpose of statin guidelines is to reduce the burden of CVDs in the population; therefore, recommendations to treat target as many individuals as possible who are destined to develop CVDs. In view of the adequately documented long-term safety, and available cheap generics, most guidelines have lowered the treatment threshold since 2013, based on evidence from clinical trials.

Thus, the expanded indication for statins may result in treatment of many lower-risk individuals, leading to high NNTs to prevent 1 CVD event, compared to more statin-conservative guidelines, which may not be correct. In various studies, anti-depressant drugs, prayer, and optimism have been shown to bring about significant decline in CVDs [12, 20-22]. It has also been observed that a new agent, inclisiran, reduces mean LDL-C exposure with an infrequent dosing regimen, thereby offering sustained LDL-C lowering with infrequent use of this agent

[20]. Through potent and durable lowering of LDL-C with a twice-a-year dosing regimen, inclisiran could offer a sustained lipid-lowering therapeutic option in the future. The future would tell about the adverse effects of this agent and whether it can replace statins. Statins can also decrease sympathetic activity, which is an additional beneficial effect compared to other agents. Statins have also been found to provide benefits among elderly patients [23]. A recent study, IMPROVE-IT, of 18,144 patients enrolled (13,728 men [75.7%]; mean [SD] age, 64.1 [9.8] years), included 5,173 (28.5%) patients 65 to 74 years of age, and 2,798 (15.4%) patients 75 years of age or older. Administration of simvastatin-ezetimibe was associated with a significant decrease in the primary endpoint, compared to simvastatin-placebo: a 0.9% decrease for patients younger than 65 years (HR=0.97; 95%CI: 0.90-1.05) and a 0.8% decrease for patients 65 to 74 years of age (HR = 0.96; 95%CI: 0.87-1.06), with the greatest absolute risk reduction of 8.7% for patients 75 years or older (HR,= 0.80; 95%CI: 0.70-0.90) (*P* = 0.02 for interaction). Adverse events did not increase with simvastatin-ezetimibe vs. simvastatin-placebo among younger or older patients. Patients hospitalized for ACS derived benefit from higher-intensity therapy to lower lipid levels with simvastatin-ezetimibe compared with simvastatin monotherapy, with the greatest absolute risk reduction among patients 75 years or older. Addition of ezetimibe to simvastatin was not associated with any significant increase in safety issues among older patients [23]. These results may have implications for guideline recommendations regarding lowering of lipid concentrations in elderly populations.

Table 1. Frequency of depression and its characteristics in patients with acute coronary syndrome and controls (modified from [15])

Condition	Acute Coronary Syndrome (n = 4,35)	Control Subjects (n = 495)
Depression (severe and moderate)	60 (13.8)**	15 (3.0)
Possible depression	26 (5.9)*	16(3.2)
Total of depression subjects	86 (19.8)**	31 (6.3)
Anxiety; memory dysfunction, fear, weeping	41 (9.4)**	17 (3.9)
Excess of thinking and difficulty in sleeping.	96 (22.0)**	52 (10.5)
Depressed mood without desire to get up in the morning.	58 (13.4)**	24 (4.8)
Depressed mood without desire to do routine work.	52 (11.9)*	15 (3.0)
Tingling and heaviness in head	62 (12.2)**	24 (4.8)
Agitation	41 (9.4)**	10 (2.0)
Known diabetes mellitus (by records)	105 (24.1)**	52 (10.5)

Chronotherapy with Statins Administration

In view of the present need for cholesterol lowering by statin therapy and its adverse effects, it has been proposed that statins may be administered at a personalized optimal circadian stage, because the benefits could be increased and adverse effects can be markedly minimized [24, 25]. Statin–induced mitochondrial dysfunction and coenzyme Q_{10} deficiency may require coenzyme Q_{10} therapy for prevention of its adverse effects [25]. The new-onset diabetes observed to occur frequently in the statin group, consistent with another meta-analysis of statin trials, can also be reduced if statins are administered when beta cells of the pancreas are highly active. Halberg, “the Lord of Time” [26], proposed that any adverse effects of any therapeutic agent can be eliminated or reduced in extent and bioactivity can be increased by rescheduling the treatment along the 24-hour scale [27]. The treatment may be modifications of routine activities, diet and/or daily exercise and/or statin or antihypertensive medication. For any non-drug or drug treatment, the rescheduling in kind and/or timing of administration can be determined in relation to the time of awakening or another marker rhythm, such as C-ABPM, wrist activity or a human metabolite timetable [27]. Statins are administered in the evening because synthesis of cholesterol occurs at night, which could be reduced by high concentration of statin achieved by evening administration.

In brief, there is evidence that statins are potent and effective agents with several pleiotropic effects for treatment of hypercholesterolemia and inflammation, commonly observed among patients with atherosclerotic CVDs. Statins have also been found to be protective among subjects above 75 years. Statins may have adverse effects if given in higher doses and in combinations, which may be reduced by chronotherapy. If the approach based on timing is used, the dosage of statins may be lowered to achieve greater therapeutic benefit, with fewer adverse effects. Consideration of behavioral risk factors and protective factors may help in decreasing the dosage of statins and improving its long-term benefits.

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Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Editorial - World Health Organization Cardiovascular Disease Risk Charts: Revised Models to Estimate Risk in 21 Global Regions

Pasquale Palmiero*
Cardiology Equipe, ASL Brindisi,
Brindisi, Italy

Atherosclerotic cardiovascular disease (CVD) is a leading cause of morbidity and mortality worldwide. The World Health Organization (WHO) supports the goal to reduce CVD mortality, at least one third by the year 2030 [1]. CVDs, such as coronary artery disease (CAD) and stroke, caused an estimated 17.8 million deaths in 2017, of which more than 75% lived in low-income and middle-income countries [2].

The first step for the reduction of the global burden of CVDs is the commitment, by WHO member states, to identify early a population of subjects aged 40 years or older, at high risk of CVD events, using risk prediction charts. After identification, every state has to provide counselling and drug treatments for at least 50% of identified people by the year 2025 [3–5].

The risk prediction models assess that the higher the cardiovascular risk, the greater the benefit in terms of events, so risk prediction charts are fundamental for CVD prevention, helping to identify people at high risk of CVD who should benefit the most from preventive interventions [6, 7]. Many different risk prediction models have been developed [8–13], and all of them can estimate individual risk over a 10-year period with measured levels of conventional risk factors for CVD [14]. However, all these models have important limitations for use in low-income and middle-income countries, because they are derived and validated based on a narrow set of studies. They might be directly applicable only to specific populations, living in high-income countries and may not predict the correct risk in the target population being screened [15–17].

Considering the need to update the 2007 WHO CVD risk prediction approaches, WHO organized a WHO CVD Risk Chart Working Group. Their work

* Correspondence: Dr Pasquale Palmiero, MD, PhD. Heart Station and Echocardiography Laboratory, ASL Brindisi, Italy. Email: pasqualepalmiero@iciscu.org

started by searching PubMed, Scientific Citation Index Expanded, and Embase, looking for existing risk prediction models for CVD in the context of primary prevention published in any language up to May 15, 2019, by the terms: “cardiovascular disease”, “risk score”, “risk equation”, “risk algorithm” and “risk prediction” [18]. Naturally, they found many studies and reviews describing risk prediction models to estimate CVD risk in a primary prevention context; but none of them was suitable to develop reliable risk models useful for low-income and middle-income countries. The use of powerful and diverse global data, simple and generalizable methods is required to account for differences in populations and acquisition of information that is readily available in many low-income and middle-income countries [19].

The newly developed risk models involve several features that seem more adequate compared with existing tools, starting from its foundation on powerful, extensive, and complementary datasets of global relevance, going to comprehensive contemporary estimation of CVD incidence and risk factor values to adapt the risk models to many different populations using a simpler and more generalizable approach. These models provide estimation for the combined outcome of fatal and non-fatal events; this is relevant to avoid underestimation of total CVD risk, particularly for individuals in populations where the death rate is low, as typically happens among younger individuals [17]. Finally, they include pragmatic models that do not assume availability of laboratory measurements that could be used as part of gradual approaches and used only as extra information.

These new WHO models for CVD risk prediction, adapted for the needs of low-income and middle-income countries, are useful to support CVD prevention and control, improving the accuracy, practicability, and sustainability of efforts to reduce the burden of CVD worldwide. For this purpose, WHO models for CVD risk prediction have been statistically adapted [20]. Nineteen different models of many different global regions used routinely available information; the aim of adaptation was to obtain that risk prediction models estimate risk for individuals in each region more accurately, using information about epidemiological trends in CVD within different global regions.

The WHO CVD Risk Chart Working Group is a cross-sectional collaboration of academics, politicians and final users of risk scores, which implemented these revised models for prediction of CVD risk adapted to low-income and middle-income countries. In this initiative of model revision, 10-year risk prediction models were derived for fatal and non-fatal cardiovascular disease, such as myocardial infarction and stroke, using individual participant data from the Emerging Risk Factors Collaboration. Models included information on age, smoking status, systolic blood pressure, history of diabetes, and total cholesterol. For derivation, participants aged 40–80 years without a known baseline history of cardiovascular disease were included, who were followed-up until the first myocardial infarction, fatal coronary heart disease, or stroke event. Models were recalibrated using age-specific and sex-specific incidences and risk factor values available from 21 global regions. For external validation, individual participant data were analyzed from studies distinct from those used in model derivation. Models were illustrated by analyzing data on a further 123,743 individuals from surveys in 79 countries collected with the WHO STEPwise Approach to Surveillance.

Findings

This risk model derivation involved 376,177 individuals from 85 cohorts, and 19,333 incident cardiovascular events were recorded during 10 years of follow-up. The derived risk prediction models discriminated well in external validation cohorts (19 cohorts, 1,096,061 individuals, 25,950 cardiovascular disease events), with Harrell's C indices ranging from 0.685 (95% CI: 0.629–0.741) to 0.833 (0.783–0.882). For a given risk factor profile, there was substantial variation across global regions in the estimated 10-year predicted risk. For example, estimated cardiovascular disease risk for a 60-year-old male smoker without diabetes and with systolic blood pressure of 140 mmHg and total cholesterol of 5 mmol/L ranged from 11% in Andean Latin America to 30% in central Asia. When applied to data from 79 countries (mostly low-income and middle-income countries), the proportion of individuals aged 40–64 years estimated to be at greater than 20% risk ranged from less than 1% in Uganda to more than 16% in Egypt.

Discussion

These newly derived, calibrated, and validated WHO risk prediction models aimed to estimate cardiovascular disease risk in 21 Global Burden of Disease regions. The widespread use of these models could enhance the accuracy, practicability, and sustainability of efforts to reduce the burden of cardiovascular disease worldwide. Figure 1 describes the revised prediction models for CVD risk, adapted for low-income and middle-income countries, with the aim of their incorporation into the WHO HEARTS package. These models have been systematically adapted to contemporary risk factor levels and disease incidences across 21 global regions, for better identification of individuals at high risk of cardiovascular disease in different settings [19]. These models allow rapid revision of cardiovascular disease, for an easy future updating of models when new epidemiological data emerge about cardiovascular disease trends in different geographical areas, particularly the scale and geographical resolution of the datasets analyzed have improved the validity and generalizability of risk models for each sex-specific and disease-specific cardiovascular event. Another advantage of this approach is the easy recalibration, providing recalibrated calculators tailored to the sex-specific cardiovascular disease rates and risk factor levels of each region [21–23]. Considering that this approach allows the use of aggregate data on cardiovascular disease incidences and average risk factor values to screen any target population, new epidemiological data can be easily and quickly implemented to revise models according to country-specific cardiovascular disease incidence and new risk factor profiles. The statistical code used to calculate, validate, and recalibrate these models using updated population data is an open access one, useful for scheduled periodic revisions. Many attempts were just made in not-high-income countries [21–26]**.** The above-mentioned models have been derived for the sex-specific and age-specific rates of myocardial infarction and stroke in each region and recalibrated to the same, they avoid inaccuracies due to recalibration to overall cardiovascular disease rates [21], including softer endpoints, different than acute cardiovascular events. High-income countries evaluate cardiovascular events prevention by risk factors control in different setting of subjects as women [27] or postmenopausal women [28]. The strength of these models is that they do not assume availability of laboratory measurements, i.e., for serum lipid concentrations. This does not mean that, where economically sustainable, a stepwise approach cannot be used, performing target laboratory testing in people for the benefit of extra information [21], mostly when values for some risk factors are unavailable for individuals [11].

As expected, the new models illustrate the surveillance data from 79 countries, demonstrating that the proportion of individuals across different risk categories is very different across global regions. This finding, concerning risk estimation, is very useful for health authorities of each country to assume more appropriate decisions, linked to local needs for a better allocation of resources for cardiovascular prevention.

The format of the revised risk charts is not different form previous WHO–International Society of Hypertension (ISH) versions for a simple continuation of their use, only the colors have been changed to highlight the general lower estimated absolute risk levels compared with those of previous WHO–ISH models [29]. Orange sections now indicate 10-year risk greater than 10%, when it previously was > 20%, while red sections indicate a risk greater than 20%, when it previously was > 30%.

However, there is an important limitation in this new approach, already contained in the 2007 WHO cardiovascular disease risk prediction approach, namely the lack of large-scale prospective cohort data from several low-income and middle-income countries, with long-term follow-up and validated fatal and non-fatal endpoints [30–32]. The new risk prediction models are derived from 85 cohorts mostly from high-income countries in the ERFC. Data extrapolated from the GBD study and the NCD-RisC have been used for a recalibration of the models.

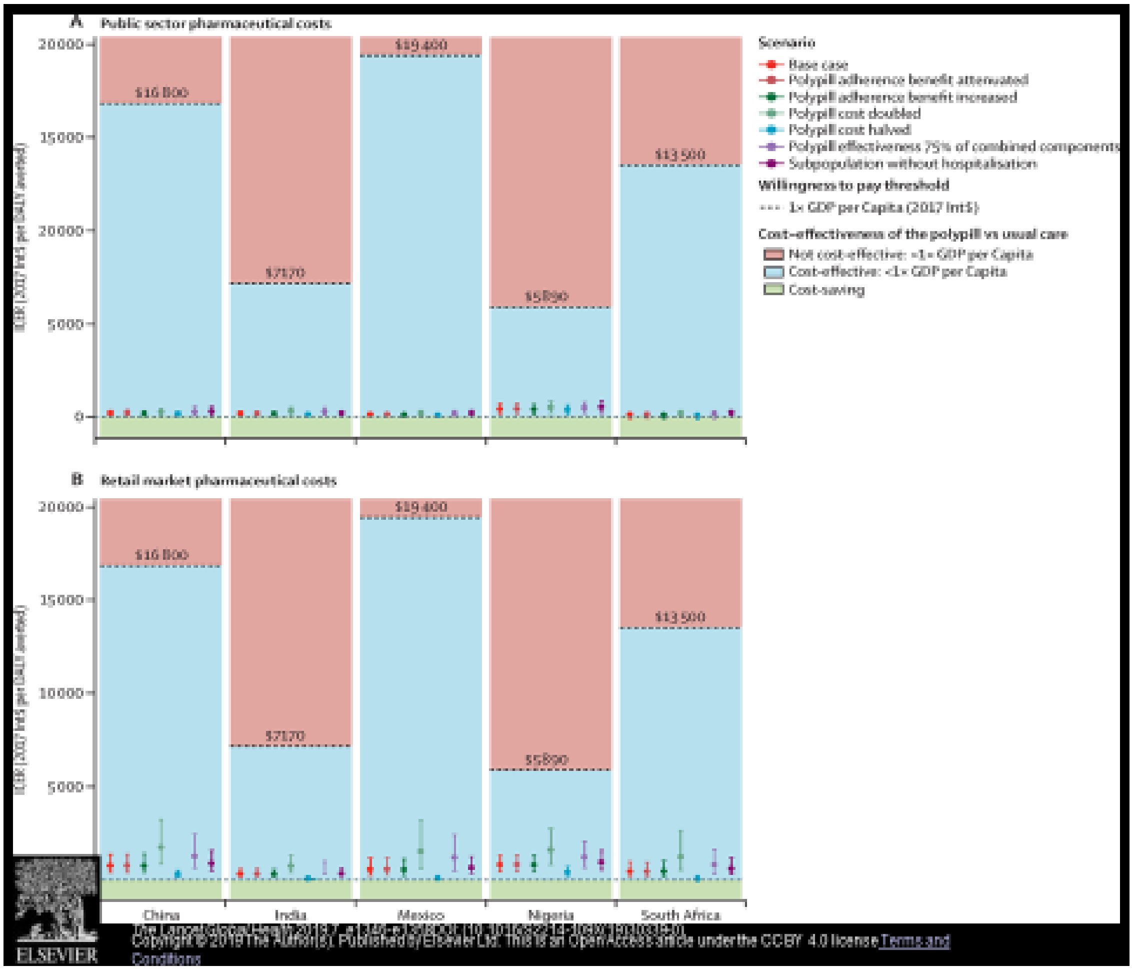

Figure 1. Cost-effectiveness of a fixed-dose combination pill for secondary prevention of cardiovascular disease in China, India, Mexico, Nigeria, and South Africa (Modified from WHO).

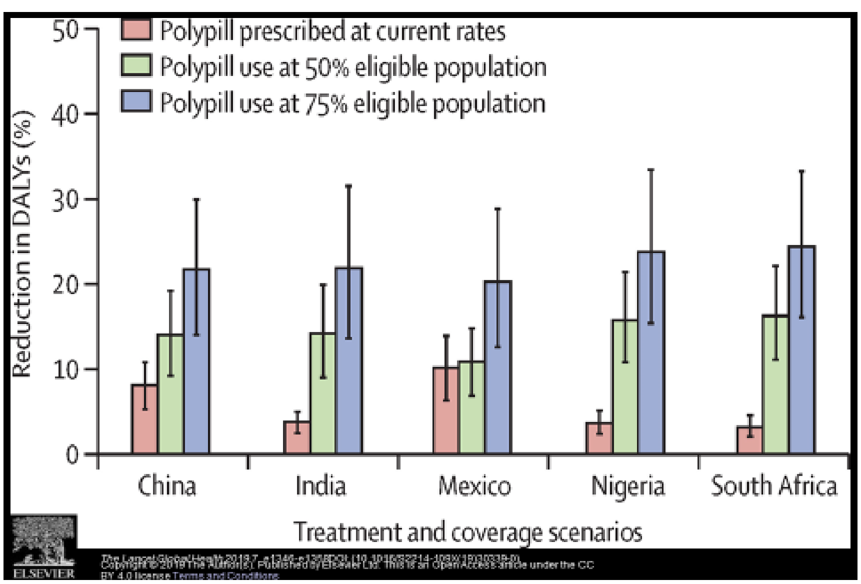

Figure 2. Disability-adjusted life years (DALYs) reduction at different rates of poly-pill prescription (modifird from WHO).

In an ideal world there a risk model should be validated for each country, but actually this is impossible because the necessary data do not yet exist for most low-income and middle-income countries [30–32]. Another limitation of this approach is the failure to identify some recurrent events from global regions used to recalibrate models, leading to an overestimation of cardiovascular disease risk for primary prevention [33]. On the other hand, there are population data used to estimate incidences derived from people already on cardiovascular disease prevention therapies, such as statins or anti-hypertensive medication, leading to an under-estimation of cardiovascular disease risk; therefore people on secondary prevention should be excluded from data collection. The performance of new risk models is not comparable with existing models used in specific high-income countries or regions because they use some variables that are not available, and hence are not practicably measurable in low-income and middle-income countries [6, 7, 11, 13, 15, 34].

Participants enrolled in new model derivation presented complete risk factor information, to avoid bias of recruitment and loss in efficiency. The collection of large sample size dataset increases the statistical power of the models and compensates for possible bias by the reasonable assumption that the probability of an individual having complete risk factor information is independent of cardiovascular disease, given the variables included in the prediction model [35].

In conclusion, the new WHO risk prediction models that estimated cardiovascular disease risk in 21 GBD regions are welcome. These risk prediction models have been adapted to the different features of many global regions and can be easily and quickly updated with routinely available information. Their implementation in clinical practice could improve the accuracy and sustainability of primary prevention to reduce the burden of cardiovascular disease worldwide.

Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Possible Worsening of Left Ventricular Diastolic Function in Patients with Statin-Associated Myopathy and the Role of Coenzyme Q10

Jan Fedacko[1, 2,*], Daniel Pella[1], Dominik Pella[1], and Ram B Singh[3]
[1]Faculty of Medicine, Pavol Jozef Safarik University, Kosice, Slovakia
[2]Cardio D&R Kosice, Cardiology Outpatient Clinic, Kosice, Slovakia
[3]Halberg Hospital and Research Institute, Moradabad, India

Abstract

Introduction: Inhibition of HMG-CoA reductase by statins leads to decreased synthesis of cholesterol and coenzyme Q10 (CoQ10). The aim of our double-blind, randomized, single-center 3-month study was to evaluate possible benefits of coenzyme Q10 supplementation in patients with possible statin-associated left ventricular diastolic dysfunction.
Subjects and Methods: 60 patients with possible statin adverse effects were enrolled in the study. Beside physical and laboratory examinations we focused on 2D echocardiography measurement of diastolic function at baseline and at the end of the study.
Results: We have observed increase of CoQ10 plasma concentrations in the active arm (from 0.81 ± 0.39 μg/ml to 3.31 ± 1.72 μg/ml at the 3-month visit ($p < 0.001$), compared to no change in the placebo arm. Doppler recordings of transmitral filling velocities show statistically significant normalization (CoQ10 group) in E/A ratio (ratio of the early (E) to late (A) ventricular filling velocities) and in IVRT (isovolumetric relaxation time). No changes were observed in deceleration time parameter.
Conclusions: Supplementation with Coenzyme Q10 resulted in normalization of Doppler recordings of transmitral filling velocities.

Keywords: coenzyme Q10, statins, statin adverse effects, left ventricular diastolic function

Introduction

Statins are very effective in reducing LDL cholesterol plasma lconcentrations. Although being safe for use, the most recent data have shown an increase in statins' adverse effects, and some of them are likely due to inhibition of geranyl pyrophosphate synthesis and subsequently dekaprenyl–4–benzoate, precursor of coenzyme Q10 (CoQ10) (Figure A). CoQ10 plays a role in mitochondrial

* *Corresponding author*: Jan Fedacko, MD, PhD, FICC. Faculty of Medicine, PJ Safarik University, Kosice, Slovakia. SLOVACRIN, Slovak Clinical Research Infrastructure Network, Kosice, Slovakia. Cardio D&R Kosice, Cardiology Outpatient Clinic, Kosice, Slovakia. Bioinformatics R&D Centre, TECHNICOM, Kosice, Slovakia. Tel. +421911315924.
Email: janfedacko@hotmail.com; jan.fedacko@upjs.sk

energy production, and CoQ10 deficiency could be responsible for statins' adverse effects, including statin myopathy and glucose intolerance [1, 2].

Reduced CoQ10 plasma concentrations by statins could lead to mitochondrial dysfunction (CoQ10 is one of the key substances in myocardial energetic metabolism and resulting in possible subclinical cardiomyopathy, reversed upon CoQ10 supplementation [3, 4]. In a recent study by Silver et al. [5], diastolic dysfunction with co-existing CoQ10 deficiency was observed among patients on long-term statin administration. Response to CoQ10 therapy suggested that diastolic dysfunction was attributable to statins.

In our previous trial [6], supplementation with CoQ10 led to a significant reduction of statin-associated myopathy in patient treated by statins.

The aim of the trial was to determine if treatment with coenzyme Q10 could have some positive effect on impaired diastolic function in statin-treated patients with mild to moderate statin-associated myopathy.

Subjects and Methods

Study Population

During our previous trial [6], we enrolled 60 patients with statin-associated myopathy for this echocardiography trial. Clinical data of the patients in the placebo and intervention (CoQ10) groups are shown in Table 1. No other lipid-lowering drugs were allowed. Concomitant use of non-statin medication by study participants is described in Table 3. There were 40 patients (66.7%) with arterial hypertension, 17 (21.7%) with coronary artery disease (4 of them after myocardial infarction), 7 with diabetes mellitus (11.7%), 5 with arrhythmias (8.3%), 4 with a history of gastroduodenal ulcer disease (6.7%) and 3 with bronchial asthma (5.0%). All patients gave informed consent to a protocol approved by the Pavol Jozef Safarik University ethics committee. Patients were consecutively randomized to either coenzyme Q10 or matching placebo.

We included statin-treated patients with associated myopathy (muscle pain, muscle weakness, muscle cramps, tiredness with or without elevated creatinine kinase). The exclusion criteria were the same as in previous trial [6].

Study Procedures

Physical examination, laboratory examination (Table 2) and echocardiography Doppler recordings of transmitral filling velocities (early diastolic velocity E, late diastolic velocity A, E/A ratio, isovolumetric relaxation time, deceleration time) were done at the beginningg and at the end of the trial.

Blood samples were collected at each phase of the trial for determination of CoQ10 plasma concentration. Plasma was separated and kept at -80°C until the analytical procedure was performed. Total coenzyme Q10 concentration was measured by the HPLC-UV method already described [11].

Echocardiography focused on diastolic function was performed at the start and at the end of the 3-month study. Doppler recordings of transmitral filling velocities were obtained in the apical 4-chamber view, with the sample volume positioned at the tip of the mitral leaflets. Doppler recordings of transmitral filling velocities (E/A ratio, deceleration time, and IVRT) were averaged from 3 cardiac cycles.

Presence and severity of adverse effects of statin therapy were checked at all study visits and always when needed during the whole study duration.

Study Intervention

All patients were randomized to coenzyme Q10 capsules (100 mg twice daily) (Bio-Quinon 100 mg B.I.D, Pharma Nord, Vejle, Denmark) or to corresponding placebo. Study medication was taken on top of regular medication. Patient compliance to study medication was performed as well. The coenzyme Q10 preparations have shown good absorption and efficacy in previous controlled trials [11, 12]; the capsules were identical to medicinal quality capsules registered for heart failure in a European Union Member State (Myoqinon®, authorization no. OGYI 11494-2010).

Statistical Analysis

Statistical analysis was performed with Student's t-test or ANOVA, and corresponding non-parametric tests when needed. For non-normal distributions, a Mann Whitney U or Wilcoxon test was used. For testing the equality of mean values of three or more groups, one-way ANOVA was used with post hoc LSD test. For testing of influence of different factors, two*way ANOVA was used. The data are expressed as mean ± SD.

Table 1. Clinical data in the placebo and intervention (CoQ10) groups

	Placebo			CoQ10		
	Baseline	After 3 months	P	Placebo	After 3 months	P
Number of patients	26	26	NS	34	34	NS
Men/women	7/19	7/19	NS	12/22	12/22	NS
Mean age	55.4 ± 12.4	56.3 ± 13.1	NS	59.6 ± 8.9	60.1 ± 9.7	NS
BMI (kg/m^2)	27.2 ± 4.1	27.3 ± 4.4	NS	29.0 ± 6.1	29.2 ± 6.5	NS
Systolic BP (mmHg)	130.12 ± 16.11	130.2 ± 11.7	NS	128.79 ± 13.39	128.2 ± 12.7	NS
Diastolic BP(mmHg)	82.04 ± 8.88	82.2 ± 9.1	NS	79.12 ± 6.69	79.2 ± 6.9	NS
Risk Factors						
Hypertension	17	17	NS	23	23	NS
Coronary artery disease (without myocardial infarction)	5	5	NS	8	8	NS
Myocardial infarction	2	2	NS	2	2	NS
Dysrhythmia	3	3	NS	2	2	NS
Diabetes mellitus type 2	4	4	NS	2	2	NS
Diabetes mellitus type 1	0	0	NS	1	1	NS
Thyroidal insufficiency	3	3	NS	2	2	NS
Bronchial asthma	2	2	NS	1	1	NS
Duodenal ulcer	2	2	NS	2	2	NS
Chronic heart failure	1	1	NS	0	0	NS
Adverse Events						
Muscle pain	18	15	NS	22	5	< 0.001
Muscle weakness	7	5	NS	13	1	0.011
Muscle fatigue	12	8	NS	10	0	< 0.001
Cramps	5	4	NS	13	2	0.024

BMI- body mass index, sBP- systolic blood pressure, dBP- diastolic blood pressure.

Table 2. Laboratory data in the placebo and intervention (CoQ10) groups

	Placebo			CoQ10		
	Baseline	After 3 months	P	Baseline	After 3 months	P
Plasma levels of CoQ10 (μmol/L)	0.74 ± 0.31	0.68 ± 0.31	NS	0.81 ± 0.39	3.31 ± 1.72	< 0.001
Total Chol (mmol/L)	5.15 ± 1.1	5.3 ± 1.2	NS	4.8 ± 1,.4	4.9 ± 1.2	NS
LDL-C (mmol/L)	2.97 ± 1.97	3.13 ± 1.14	NS	2.61 ± 0.97	2.60 ± 0.93	NS
HDL -C(mmol/L)	1.53 ± 0.36	1.5 ± 0.4	NS	1.43 ± 0.31	1.4 ± 0.3	NS
TG (mmol/L)	1.41 ± 0.67	1.5 ± 0.9	NS	1.62 ± 0.73	1.7 ± 1.0	NS
Glycaemia (mmol/L)	5.59 ± 1.67	5.63 ± 1.94	NS	5.55 ± 1.56	5.66 ± 1.89	NS
CK (μkat/L)	2.20 ± 1.21	2.15 ± 0.98	NS	3.51 ± 2.96	3.1 ± 4.77	0.04
AST (μkat/L)	0.43 ± 0.30	0.39 ± 0.17	NS	0.40 ± 0.11	0.39 ± 0.10	NS
ALT (μkat/L)	0.43 ± 0.28	0.45 ± 0.28	NS	0.40 ± 0.16	0.41 ± 0.22	NS
GMT (μkat/L)	0.49 ± 0.29	0.56 ± 0.39	NS	0.61 ± 0.61	0.60 ± 0.44	NS
ALP (μkat/L)	1.10 ± 0.30	1.10 ± 0.30	NS	1.18 ± 0.31	1.2 ± 0.4	NS
Urea (mmol/L)	4.87 ± 1.31	5.0 ± 1.41	NS	5.01 ± 1.27	5.27 ± 1.61	NS
Creatinine (μmol/L)	85.70 ± 13.52	86.08 ± 10.32	NS	86.47 ± 16.03	88.21 ± 15.78	NS

TChol- total cholesterol, LDL- LDL cholesterol, HDL- HDL cholesterol, TG-triglycerides, CK- creatine kinase, AST- aspartate aminotransferase, ALT- alanine transaminase, GMT- glutamyl transferase, ALP- Alkaline phosphatase.

Table 3. Concomitant statin therapy in the placebo and intervention (CoQ10) groups

	Placebo			CoQ10		
	Baseline	After 3 months	P	Baseline	After 3 months	P
Drug therapy						
ACE-i	11	11	NS	10	10	NS
Beta-blockers	7	7	NS	6	6	NS
Calcium channel blockers (Dihydropyridine)	5	5	NS	8	8	NS
AT1-inhibitors	2	2	NS	1	1	NS
Alpha-blockers	2	2	NS	3	3	NS
Nitrates	4	4	NS	5	5	NS
Aspirin	8	8	NS	8	8	NS
Trimetazidine	7	7	NS	4	4	NS
Ticlopidine	2	2	NS	2	2	NS
Metformine	2	2	NS	3	3	NS
Rosiglitazone	0	0	NS	1	1	NS
Insulin	0	0	NS	2	2	NS
L-thyroxine	3	3	NS	2	2	NS
PPI	1	1	NS	0	0	NS
Xantins	1	1	NS	0	0	NS
Analgesics	4	4	NS	3	3	NS

Results

60 eligible patients with statin-associated myopathy from our previous trial were enrolled for this echocardiography trial. The average doses of statins and number of patients of different statins are documented in Table 4. The study group consisted of 19 men and 41 women. The average age was 55.4 ± 12.4 years in the placebo group (n = 27) and 59.6 ± 8.9 years in the CoQ10 group (n = 33) (Table 1). The parameters of left ventricular diastolic function were evaluated in all 60 patients. Reference plasma concentrations of CoQ10 in the placebo group were 0.76 ± 0.31 µg/ml and did not change significantly during the whole study (0.68 ± 0.31µg/ml at 3-month visit). Plasma concentrations of CoQ10 in the active group increased from reference (0.81 ± 0.39 µg/ml) to 3-month visit (3.31 ± 1.72 µg/ml) (p = 0.001). A significant increase of CoQ10 plasma lconcentration was already observed at the 1-month visit in the active group of patients (plasma lconcentration of CoQ10 at 1 month was 3.16 ± 1.78 µg/ml; p = 0.001 compared to reference) (Table 4, Figure B).

Acetyl CoA
↓
HMG-CoA
↓ **HMG CoA reductase**
mevalonate
↓
isopentanyl – PP → selenoproteins
↓
geranyl-PP→dekaprenyl-PP→dekaprenyl 4-OH benzoate→ **CoQ10**
↓
squalene←——————farnesyl-PP———→farnesylated proteins
↓ ↓
cholesterol geranylated proteins

Figure A. Cholesterol (mevalonate) synthesis pathway.

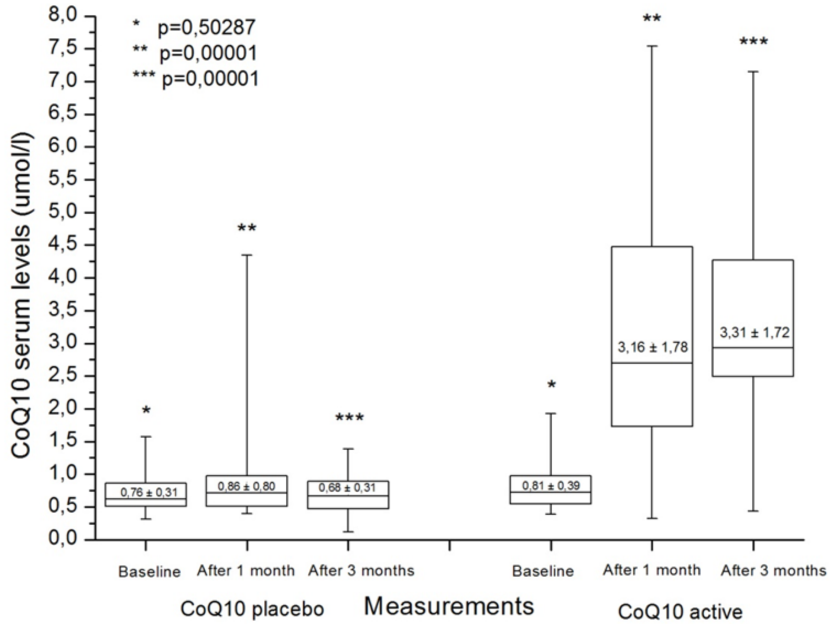

Figure B. CoQ10 serum levels in the CoQ10 active group of patients vs. the placebo group.

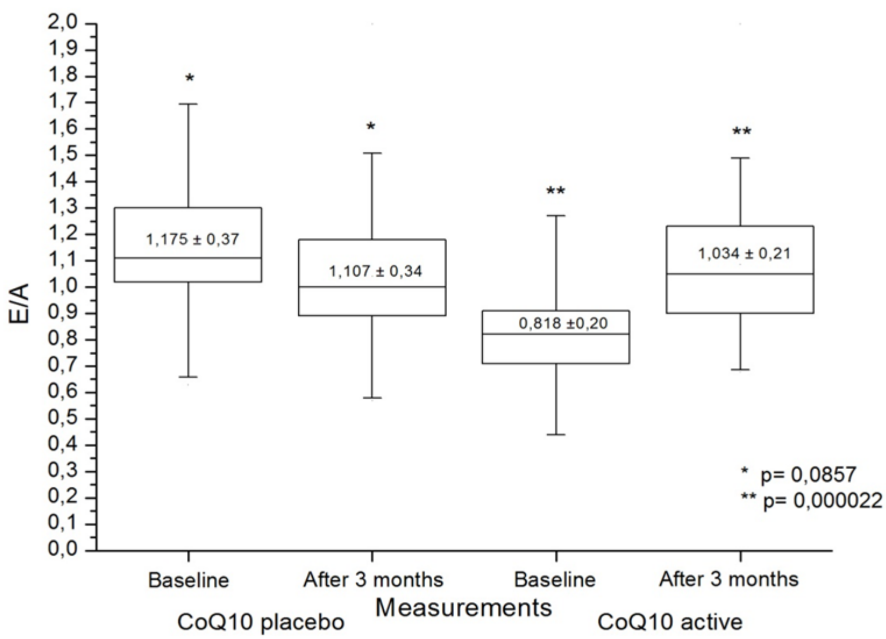

Figure C. E/A ratio in the CoQ10 active vs. placebo group of patients.

Doppler recordings of transmitral filling velocities of the diastolic function in both groups of patients between months 0 and 3 are listed in Table 5. We observed normalization of the E/A ratio from 0.818 ± 0.20 to 1.034 ± 0.21 ($p < 0.001$) (Figure C); IVRT decreased from 103.51 ± 16.35 ms to 86.40 ± 13.53 ms ($p < 0.001$) (Figure D) and deceleration time did not change statistically ($p = 0.263$) (Figure E) in the CoQ1O group of patients. In patients who did not take CoQ10, mild worsening of the E/A ratio was seen (from 1.175 ± 0.37 to 1.107 ± 0.34), but the difference was only of borderline statistical significance ($p = 0.085$). IVRT did not change after 3 months ($p = 0.275$). Deceleration time remained unchanged as well (from 162.62 ± 39.60 to 172.00 ± 39.27; $p = 0.281$).

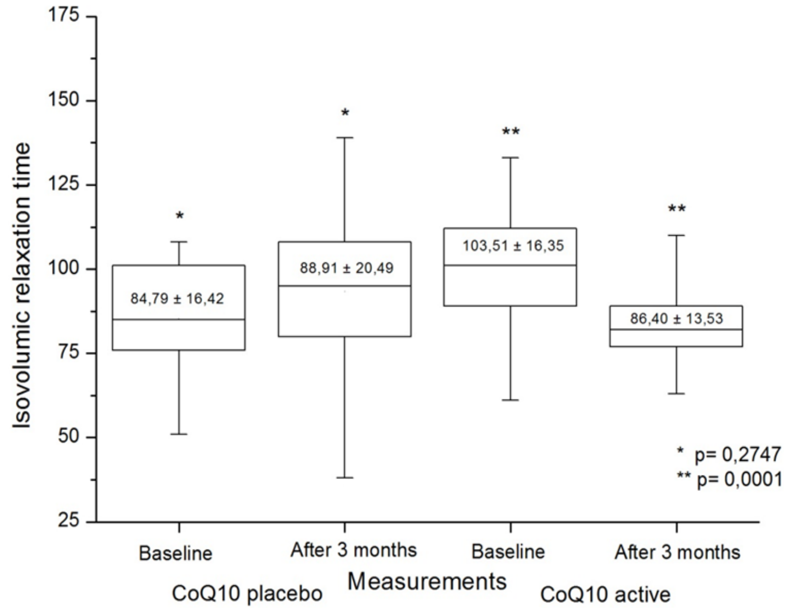

Figure D. Isovolumetric relaxation time in the CoQ10 active vs. the placebo group of patients.

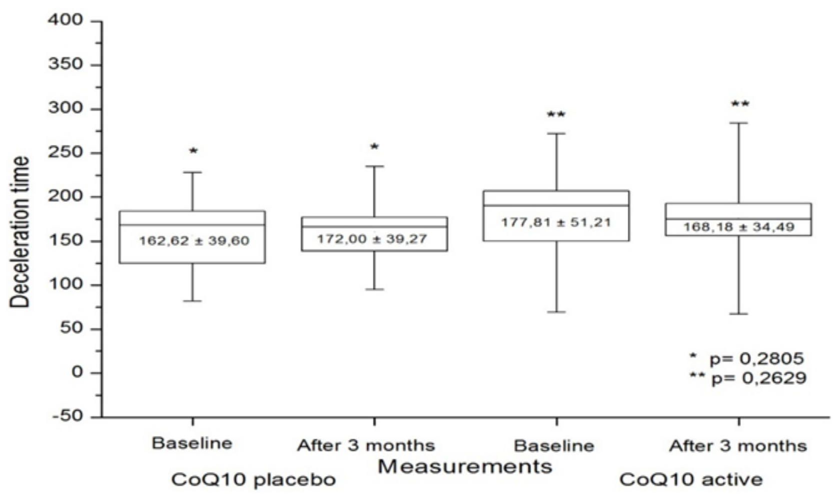

Figure E. Deceleration time in the CoQ10 active vs. the placebo group of patients.

Table 4. Statin use in the placebo and intervention (CoQ10) groups

	Placebo			CoQ10		
	Baseline	After 3 months	P	Baseline	After 3 months	P
Atorvastatin (n) Mean dose mg/day	11 16.7 ± 8.7	11 16.7 ± 7.4	NS	14 18.1 ± 9.1	14 18.1 ± 8.8	NS
Rosuvastatin (n) Mean dose mg/day	7 12.1 ± 5.3	7 12.1 ± 4.3	NS	9 10.4 ± 3.2	9 10.4 ± 2.8	NS
Simvastatin (n) Mean dose mg/day	6 26.9 ± 7.4	6 26.9 ± 8.2	NS	8 25.4 ± 6.5	8 25.4 ± 6.9	NS
Fluvastatin (n) Mean dose mg/day	2 80 ± 0	2 80 ± 0	NS	3 80 ± 0	3 80 ± 0	NS

Table 5. Echocardiographic diastolic dysfunction in the placebo and intervention (CoQ10) groups

	Placebo			CoQ10		
	Baseline	After 3 months	P	Baseline	After 3 months	P
Ratio E/A	1.175 ± 0.37	1.107 ± 0.34	0.085	0.818 ± 0.20	1.034 ± 0.21	< 0.001
IVRT	84.79 ± 16.42	88.91 ± 20.49	0.274	103.51 ± 16.35	86.40 ± 13.53	< 0.001
DT	162.62 ± 39.60	172.00 ± 39.27	0.805	177.81 ± 51.21	168.18 ± 34.49	0.2629

IVRT – isovolumetric relaxation time.
DT – deceleration time.

Discussion

We found that treatment with coenzyme Q10 (100 mg BID) in patients with statin-associated myopathy and with possible statin-associated left ventricular diastolic dysfunction decreased not only adverse effects of statins but also significantly improved left ventricular diastolic function. The importance of our findings is that treatment with CoQ10 allowed patients to continue treatment with statin drugs rather than withdraw due to adverse effects of statins.

A recent study by Silver et al. [5] examined 17 patients who met NTP III criteria for treatment with a statin (namely atorvastatin 20mg daily). The primary endpoint was left ventricular diastolic function, measured by Doppler echocardiography before and after statin therapy in patients with no prior indication of congestive heart failure, myocardial infarction and reduced left ventricular ejection fraction. The effect of atorvastatin on left ventricular diastolic function of was measured. Serum lipids and serum concentrations of coenzyme Q10 were determined at the beginning and end of the study. Patients with deterioration in at least one of the monitored parameters of left ventricular diastolic function during 3-6 months received replacement therapy with coenzyme Q10 at a dose of 100 mg three times daily for a further three months. During the follow-up, there was no development of heart failure, change in left ventricular dimensions or change in left ventricular systolic function, but statins for 3-6 months were associated with deterioration of diastolic function parameters in most patients. Ten patients had deterioration in at least one, 3 patients in two and 5 patients all three parameters. Subsequent substitution of coenzyme Q10 resulted in modification of monitored parameters [5].

Similar results were achieved by Kumar et al. [7] in more than 100 patients. In a recent double-blind trial, conducted on 32 patients, muscle pain and muscle interference with daily activities significantly decreased in the course of statin treatment when patients were supplemented with CoQ10 [8].

Statins are chemically and pharmacologically a varied group of drugs; the metabolic pathway may be important for their safety. Several of them are metabolized via cytochrome P 450 3A4 [10, 11], several are metabolized via cytochrome P450 2C9 and drug interactions associated with the use of these statins could play an important role in statin-related adverse effects, as confirmed in our study. Among 1,142 statin-treated patients, adverse effects were observed only in 1.46% cases on fluvastatin, versus 12.28% on simvastatin. Among the currently available statins, only two are hydrophilic (pravastatin, rosuvastatin). Hydrophilic ability could prevent state of “increased fluidity” of muscle cell membrane and the increased risk for myopathy or even rhabdomyolysis [12].

The cardioprotective role of statins by decreasing oxidation of LDL cholesterol or stabilization of myocyte membranes could be declared by antioxidant activity of statins, which differ between statins (from atorvastatin as best to rosuvastatin as lowest) [13]. Long-term effectiveness, utility and safety of statins have been confirmed in large multicenter trials.

Inhibition of HMG-CoA reductase and blocking the main metabolite of mevalonate not only blocks the synthesis of cholesterol, but also leads to the blockade of other metabolites, particularly coenzyme Q10. Coenzyme Q10 plays a dominant role during mitochondrial oxidative phosphorylation in myocardial cells. Since the diastolic myocardial function is highly ATP-dependent (more than systolic function), coenzyme Q10 deficiency may be responsible for early diastolic left ventricular dysfunction [15]. In the Anglo-Scandinavian Cardiac Outcomes Trial (ASCOT) [16], patients randomized to the atorvastatin arm were hospitalized more frequently for heart failure. This may have been due to deterioration in diastolic left ventricular function due to depletion of CoQ10.

The important role of CoQ10 in cellular energy and diastolic function was supported by initial studies by Oda [17] and Langsjoen [18]. Oda [17] in one of his studies examined the effects of coenzyme Q10 in patients with systolic and diastolic dysfunction on the basis of overload, where coenzyme Q10 supplementation led to normalization of both functions. In a study of stress, echocardiography in patients with mitral valve prolapse, CoQ10 supplementation improved left ventricular ejection fraction. Langsjoen [18] demonstrated a benefit of coenzyme Q10 in patients with chronic heart failure and symptomatology associated with diastolic dysfunction. Coenzyme Q10 supplementation led to a normalization of diastolic dysfunction, besides adjusting other parameters and symptoms [17, 18].

Possible mechanisms of statin-associated myopathy include possible decrease of intramito-chondrial coenzyme Q10 beside decreased sarcol-emmal cholesterol and small guanosine triphosphate–binding proteins, increased intracellular lipid production and lipid myopathy [19-22].

Based on the latest data, occurrence of both myopathy and rhabdomyolysis in clinical trials is rare; the prevalence of severe myopathy is around 0.1-0.5%, while the prevalence of rhabdomyolysis is 0.02-0.04% [23, 24]. Clinical trial protocols often exclude patients likely to develop some kind of myopathy (history of muscle pain, concomitant therapy, dual lipid-lowering therapy, previous history of elevated creatine kinase activity). On the other hand, mild muscular symptoms are often mistakenly overlooked by doctors, and their frequency is probably not accurate [25]. This possible under-estimated occurrence of muscular symptoms was declared in The PRIMO survey (observational study, hyperlipidemic patients treated by high-dose statin therapy) in France, which reported muscular symptoms by 832 patients (10.5%), with onset time of 1 month (median) after starting statin therapy [26]. Multivariate analysis showed that the strongest predictors for muscular symptoms are a personal history of muscle pain during lipid-lowering therapy, unexplained cramps, history of creatine kinase elevation, family history of muscular symptoms, hypothyroidism, duration and type of statin (the lowest rate of muscular symptoms observed in fluvastatin, the highest one in simvastatin – 5.1% versus 18.2 %). Further studies also reported beneficial effects of coenzyme Q10 in cardiac dysfunction [27-29].

The aims of the trial was to determine if treatment with coenzyme Q10 could have some positive effect on impaired diastolic function in statin-treated patients with mild to moderate statin-associated myopathy, and to determine any clinical benefit from CoQ10 supplementation.

Limitations

Although this was a randomized, prospective and placebo-controlled trial, we only involved a small number of patients in this pilot trial. No Tissue Doppler method was used (not available at the time) and the data must be qualified accordingly.

Conclusion

Our recent trial with coenzyme Q10 treatment (100 mg BID) in statin-associated myopathy with possible statin-associated left ventricular diastolic dysfunction led not only to decrease adverse effects of statins, but also to significantly improve left ventricular diastolic function.

Acknowledgments

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Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Development and Validation of a Questionnaire for Assessment of Happiness: The Happiness Rating Scale

Ram B Singh[1,†], Agnieszka Wilczynska[2,*], Jan Fedacko[3,§], Masaki Mogi[4], Rie Horiuchi[5,‡], Toru Takahashi[6,#], MA Niaz[7,|], Ghizal Fatima[8,°], Kumar Kartikey[9,‖], and Harpal S Buttar[10]

[1]Halberg Hospital and Research Institute, Moradabad, India
[2]The Tsim Tsoum Institute, Krakow, Poland
[3]PJ Safaric University, Kosice, Slovakia
[4]Department of Pharmacology. Ehime University, Graduate School of Medicine, Tohon, Ehime, Japan
[5]Faculty of Human Environmental Sciences, Nishinomia City, Mukogawa Women's University, Japan
[6]Koriyama Women's University, Koriyama, Japan
[7]Center of Nutrition Research, Halberg Hospital and Research Institute, Moradabad, India
[8]Department of Biotechnology, Era Medical College, Lucknow, India
[9]Halberg Hospital and Research Institute, Moradabad, India;
[10]Department of Pathology and Laboratory Medicine, University of Ottawa, Faculty of Medicine, Ottawa, Ontario, Canada;

Abstract

Background. Cohort studies indicate that chronic anxiety, aggression and depression may be associated with neuroendocrine dysfunctions and lack of happiness which may predispose to poor social and physical health, leading to cardio-metabolic diseases (CMDs). Dopamine, serotonin, oxytocin and endorphins are the quartet neurotransmitters responsible for the happiness of the mind, which may be blunted by stress hormones released due to psychological diseases. This study aims to develop and validate a questionnaire for assessment of happiness to find out the prevalence of happiness in relation to coronary artery disease (CAD).
Subjects and Methods. After written informed consent and approval from the hospital ethics committee, this cross-sectional survey was conducted at Halberg Hospital and Research Institute, Moradabad, India. All subjects, 980 urban (495 men and 485 women), 900 rural (510 men and 390 women) above 25 years of age were randomly selected and recruited from urban and rural populations. Clinical data as well as risk factors were recorded with the help of case record form and validated questionnaires. Assessment of happiness was made by a new questionnaire involving happiness of mind by including questions related to social, psychological and spiritual behaviors.
Results. The prevalence of happiness was 62.4% (n = 612) among urban and 63.2% (n = 569) among rural subjects, with overall prevalence of 62.8% (n = 1,181). The prevalence of happiness was significantly greater among subjects > 45 years among both urban and rural populations compared to subjects between 25 to 45 years and the trend was significant. The overall prevalence of happiness was also significantly greater among subjects above 45 years compared to subject below 45 years of age. The prevalence of happiness overall showed a graded increase with age from 25 to 84 years in both sexes among urban and rural populations and the trends were significant ($P < 0.02$). The overall prevalence of tobacco intake was 16.2%, (n = 304) and alcoholism 1.96%. The frequency of tobacco intake and alcoholism were significantly lower among subjects with happiness compared to those with no happiness. However, among subjects with happiness, the frequency of moderate alcohol intake was significantly greater compared to urban subjects with no happiness. The prevalence of CAD was significantly lower among subjects with happiness

† Email: rbs@tsimtsoum.net
* Correspondence: Dr Agnieszka Wilczynska, PhD. The Tsim Tsoum Institute, Krakow, Poland, Tel. +421911109836. Email: awk@tsimtsoum.net
§ Email: janfedacko@hotmail.com
‡ Email: mhor9496@mukogawa-u.ac.jp
# Email: tkhstr111@gmail.com
| Email: mohdarifniaz@yahoo.com
° Email: ghizalfatima8@gmail.com
‖ Email: drgarimadrkartikey@gmail.com

compared to those with no happiness, respectively both in urban (5.06 vs. 12.2%, P < 0.01) and rural populations (2.09% vs. 3.92%, P < 0.05), indicating that happiness may be protective against CAD. There were no gender differences in happiness.
*Conclusions***.** It is possible that this modified questionnaire can be successfully used for assessment of happiness rates in urban and rural areas of north India. The questionnaire allowed us to identify overall 63.2% (n = 569) subjects with happiness, including 62.4 % among urban and 63.2% (n = 569) among rural subjects. Moderate intake of alcohol may be in favor of happiness. Happiness may be protective against CAD. Further analysis is necessary to verify our findings.

Keywords: Western diet, sedentary behavior, mastication, cardiovascular diseases.

Introduction

Buddha said ages ago, "there is no path to happiness, happiness is the path," however, a new survey demystifies the true determinants of happiness [1]. A new global survey by Ipsos in India showed that happiness has ranked 9th in the pecking order on happiness, among 28 nations polled [2]. This study showed that the highest prevalence of happiness is found in developed nations: Australia and Canada tied at the top spot (at 86%) that have emerged as the happiest nations of the world, followed by China (83%), Great Britain (82%), France (80%), USA (79%), Saudi Arabia (78%) and Germany (78%) [2]. These countries are the eight markets preceding India, which is just below these countries (77%). The main criteria of happiness were personal safety, living conditions, physical wellbeing, and security. Friends, feeling in control of life emerged as top determinants of happiness in life. Lowest in happiness scale were Argentina (34%), Spain (46%) and Russia (47%), among the 28 markets. It is noteworthy that globally, happiness levels have decreased in 2019, as compared to 2018 in India, with a 6% drop (from 83% in 2018 to 77% in 2019.

It is clear that the criteria of happiness are not confined to economic development, personal safety and living conditions, because in India, all these attributes are very much lacking compared to Argentina, Spain and Russia, having lower levels of happiness [2]. It is important to find out the factors that make Indians happy. It seems that what Lord Buddha said, ages ago, may be correct for Indians and that "happiness is the path" [1]. This means that happiness may be an important component of total health, including physical, social, emotional and spiritual wellbeing of the subjects and of population health. Happiness appears to be important for health promotion as well as for prevention of cardio-metabolic diseases (CMDs) and neuro-psychiatric diseases [3-8]. Recent research has demonstrated that happiness may be associated with stress hormones and certain neuronal circuits regulating pleasure and happiness [8, 9]. It seems that unhealthy behaviors and attitudes as well as cultural factors involve progressive decline in physiological functions, immunological functions, neuroendocrine dysfunction, cognition and attitudes towards healthy behaviors and health [3-7]. These behaviors appear to be important in determining happiness and well-being of the mind that are important components of spiritual as well as psychological health which determine total health [7].

There is concurrent and temporal relationships between humility and emotional and psychological well-being that may be important for maintaining happiness [10]. Dreaming of a brighter future leading to anticipation of happiness may instill meaning in life, which may be protective for health promotion [11]. There is consistent evidence from cohort studies, that type A behavior, aggression, mood disorders and depression as well as chronic anxiety disorders can predispose to poor social and physical health, leading to non-communicable diseases (NCDs) [3-7]. However, active prayer, yoga therapy, moderate physical activity, mindfulness, Mediterranean-type diets, probiotics, mastication during eating and optimism can provide improved psychological health and wellbeing [3-7]. Dopamine, serotonin, oxytocin and endorphins are the quartet neurotransmitters responsible for the happiness of the mind [12]. It is hypothesized that chronic stress of short duration as well as depression and aggression may be reduced by inculcation of happiness, which prompts us to develop and validate a questionnaire for assessment of happiness. It is possible that this questionnaire could be used for assessment of happiness in epidemiological studies for assessment of happiness as well as in clinical trials to find out the effects of intervention on happiness. This study aims to develop and validate a questionnaire for assessment of quality of happiness, which may be used to find out its role in health promotion and in the prevention of diseases.

Subjects and Methods

After written informed consent and clearance from institutional ethics committee, recruitment of subjects was made at Halberg Hospital and Research Institute, Moradabad, India. Randomly selected samples of subjects, aged 25 years and above were obtained from apparently healthy urban and rural populations, based on voters list [13]. If the random number fell to a subject who was < 25 years or not available, it was assigned to the next person in the list. In this study, we contacted 2,422 urban and 1,769, rural subjects from a few villages. Among all subjects aged 25 years and above, approximately 9 % refused to participate and the rest gave their consent to be included in the study. Of all the subjects, 980 urban (495 men and 485 women), 900 rural (510 men and 390 women) above 25 years of age were randomly selected for assessment of happiness.

Collection of Data

A case record form was completed to record clinical data such as age, sex, body weight, height, waist circumference, past and family history of diseases in all the subjects. Validated questionnaires were used to examine behaviors, tobacco and alcohol intake, diet and lifestyle factors as well as mood disorders and depression [13-18].

Criteria of Happiness

Happiness was assessed by a new questionnaire modified from earlier questionnaires based on subjective feeling without consideration of other behavior [10, 19-22]. The happiness scale was also referred to as the Emotion Questionnaire as it assesses emotional wellbeing as an indication of perceived happiness [10]. It consists of two items: the first is a scale measuring happiness/ unhappiness by participants ranking descriptive phrases on a 0 – 10 scale. However, this questionnaire used in our study contained 12 items. These items making up the test require participants to give an approximate percentage of time or number of times that they feel about the attributes, indicating happy, unhappy and neutral, which are based on a scale of 0-10. The test shows adequate reliability and validity because many behavioral confounders have been included as attributes of happiness (APPENDIX I).

Criteria of diagnosis of other behaviors and risk factors and diseases were based on WHO criteria and on previous studies [13-18]. It is difficult to assess tobacco intake because it is consumed in various forms. Cigarettes, pipes, raw tobacco and chewing tobacco are all commonly consumed and some people use tobacco in more than one form [15]. We therefore categorized users of any form of tobacco as smokers, as was done in previous studies. Individuals who admitted to ingesting alcohol more than once a week were categorized as alcohol consuming. Moderate alcohol intake was considered if the intake was 10 drinks per week or less Alcoholism was considered if the intake was more than 10 drinks/week. Sedentary behavior was considered in presence of no spare time physical activity, walking less than 14.5 km/week, climbing fewer than 20 flights of stairs a week, or performing no moderate physical activity (< 300 K calories/day) on 5 days a week [15]. Validated questionnaires were used to assess type A behavior, depression and mood disorders, chronic anxiety disorders, mastication, and optimism, by the psychologist [13-18].

Physical Examination

Criteria of diagnosis of risk factors were based on WHO criteria for all the risk factors and diseases. Body weights and heights were measured in underclothes. Body weight was measured in kilograms up to minimum of 0.5 kg by a calibrated weighing machine. Height was measured in centimeters after removing shoes, asking the subject to stand on their back side, close to the measuring stand. Body mass index (BMI) was calculated and obesity was defined as a BMI of 30 kg/m^2 and above, overweight when body mass index 25 kg/m^2 to 29.9 kg/ m^2.

A 12 lead electrocardiogram was done in all subjects. The criteria for the diagnosis of coronary artery diseases (CAD) were based on WHO criteria: (a) history of angina or infarction and previously diagnosed disease, (b) affirmative response to Rose questionnaire, and (c) electrocardiographic findings, namely Minnesota codes 1-1, 4-1, 5-9, 5-2 or 9-2. Presence of all three criteria was taken as confirmation of the diagnosis of CAD. Individual clinical criteria included known CAD, affirmative response to Rose questionnaire and electrocardiographic changes (Q wave change codes 1-1 and 1-2, ST segment depression or elevation codes 4-1, 4-2 and 9-2 and T wave inversions, codes 5-1 and 5-2.) Prevalence of these electrocardiographic finding with and without clinical criteria for CAD is also given.

Statistical Analysis

The prevalence is given in percent and continuous variables as mean ± standard deviation. Subjects were classified based on grade of happiness; the association of various grades of happiness with other risk factors and protective factors were tested by the Chi square test. Only P-values < 0.05 with two -tailed t test are considered significant.

Results

This study included 980 urban (50.5% males) and 900 rural (53.6% males) subjects. Mean age among urban (44.5(10.5) vs. 42.6 (9.5) years) and rural subjects showed no significant difference, respectively. Mean body weights (61.8 (11.7) vs. 56.7 (8.9) kg) and body mass index (22.5 (2.3) vs. 21.6(1.9) kg/m^2) among urban subjects were significantly higher compared to rural subjects.

Table 1 shows the grades of happiness among urban and rural subjects. The prevalence of happiness was 62.4% (n = 612)* among urban and 63.2% (n = 569) among rural subjects. The overall prevalence of happiness was 62.8% (n = 1,181).

Table 2 shows the prevalence of happiness according to age groups. The prevalence of happiness was significantly greater among subjects > 45 years among both urban and rural populations compered to subjects between 25 to 45 years and the trend was significant. The overall prevalence of happiness above 45 years (n = 628 vs. 553 (65.7% vs. 48.2 %, P < 0.01) was also significantly greater among subjects above 45 years of age compared to subject below 45 years (Table 2).

Prevalence of happiness in relation to risk factors is shown in Table 3. The findings show that proportions of tobacco intake and alcoholism were significantly more common among men compared to women. No alcoholism was found among women and tobacco intake was also rare in women. The overall prevalence of tobacco intake was 16.2% (n = 304) and alcoholism 1.96%. Among subjects with happiness, the frequency of moderate alcohol intake was significantly greater compared to urban subjects with no happiness. The prevalence of CAD was significantly lower among subjects with happiness compared to those with no happiness, both in urban (5.06 vs.12.2%, P < 0.01) and rural populations (2.09% vs. 3.92%, P < 0.05), indicating that happiness may be protective against CAD. There were no gender differences.

Table 1. Grade of happiness based on scores, among urban and rural subjects

Grades and scores of happiness	Urban subjects, (n = 980)	Rural subjects(n = 900)	Total, n = 1,880
Very happy and satisfied, 81-120	208 (21.2)*	210 (23.3)*	418 (22.2)*
Moderately happy and satisfied,51-80	227 (23.1)*	201 (22.3)*	428 (22.7)*
Modestly happy and satisfied, 21-50	177 (18.0)	158 (17.5)	335 (17.8)
Total	612 (62.4)	569 (63.2)	1,181 (62.8)
Neutral, < 21	368 (37.5)	331 (36.8)	699 (37.2)

*P-values were obtained by Chi square test. Values are number (%).

Total score = 81-120; Very satisfied and very happy; Total score = 51-80; moderately satisfied and happy and Total score = 21-50; modestly satisfied and happy and Total score < 21 = Neutral. (Highlighted- proven).

Table 2. Prevalence of happiness by age among rural and urban populations in North India

		Urban (n = 980)	Rural (n = 900)		Total (n = 1,880)	
Age group	No	Happiness N (%)	No	Happiness N (%)	No	Dementia N (%)
25-34	301	115 (38.2)	333	156 (46.8)	634	271 (42,7)
35-44	281	147 (52.3)	232	135 (58.2)	513	282 (50.9)
45-54	172	148 (86.0)*	151	125 (82.2)*	323	273 (84.2)*
55-64	121	114 (94.2)*	110	94 (82.7)*	231	208 (90.4)*
65-74	60	53 (88.3)*	49	42 (85.7)*	109	95 (87.1)*
75-84	45	35 (77.7)	25	17 (68.0)	70	52 (74.3)
chi2 for trend P -value		92.7 0.01		60.5 0.02		104.4 0.01
Total	980	612 (62.4)*	900	569 (63.2)	1880	1,181 (62.8)

Values are N (%), P-value was obtained by Chi square test; * P < 0.02.

Table 3. Prevalence of tobacco and alcohol intake among urban and rural subjects with and without happiness

	Happiness (n = 1,181)		No Happiness (n = 699)	
	Urban (n = 612)	Rural (n = 669)	Urban (n = 368)	Rural (n = 331)
Tobacco intake (>once/week)	168 (27.45)	15 (2.24)	103 (27.9)	18 (5.44)
Alcoholism (>10 drinks/week)	12 (1.96)	2 (0.29)	22 (2.47)	1 (0.30)
Alcohol intake in moderation (10 drinks/week)	136 (22.2)*	28 (4.18)	28 (10.3)*	16 (4.83)
Coronary artery disease	31 (5.06)*	14 (2.09)*	45 (12.2)*	13 (3.92)

Values are n (%), *p < 0.01, P-values were obtained by Chi square test by comparing percentages of risk factors among subjects with happiness and no happiness.

Discussion

It is clear that this modified questionnaire can be successfully used for assessment of happiness rates in urban and rural areas of north India. Based on this questionnaire, it has been possible to identify the prevalence of happiness among 62.4% (n = 612) urban and 63.2% (n = 569) rural subjects, with overall prevalence of happiness among 62.8% (n = 1,181) subjects. There are only a few studies on prevalence of happiness among randomly selected subjects. In another survey from 28 countries, in which 1,000 subjects or more had online questionnaire from each country, conducted by non-physicians, the prevalence of happiness among Indians was 77% [2]. The factors considered as most important in triggering happiness were: living conditions (89%), physical well-being (88%), personal safety and security (88%), friends (87%), a feeling in control of my life (86%), feeling my life has meaning (86%), my hobbies/interests (85%), and well-being of the country (85%) [2]. In brief, most people are happy if they have comfortable living conditions, robust health, good financial conditions, friends and a social circle and a purpose in life, but such assessments are open to bias.

In the global happiness index, the highest determinants of happiness were: physical health/ wellbeing (88%), living conditions (86%), hobbies/ interests (85%), personal safety/security (85%), feeling that life has meaning (85%), feeling in control on life (84%), satisfaction (84%), having more money (84%) and friends (83%) [25, 26]. The World Happiness Report 2018 is a survey of the state of global happiness that includes ranking of 156 countries by their happiness levels, and 117 countries by the happiness of their immigrants [25]. This report shows that India has slipped 10 ranks; from 123 to 133 compared to last year. Finland tops the list of 156 countries ranked, pushing last year's winner Norway to second position. Pakistan (75) is above India, and so is China (86). While Bangladesh and Sri Lanka are at 115 and 116 positions, respectively, Nepal stands at 101 and Bhutan at 97 [25]. According to a 2019 report on happiness index, positive affect is generally falling in Western Europe, and falling and then rising in both Central and Eastern Europe [26]. It seems that the pattern of partial convergence of positive affect between the two parts of Europe leaves positive affect still significantly more frequent in Western Europe. The incidence of positive affect is falling in Americas, fairly stable and at similar levels in East and Southeast Asia, while starting lower and falling significantly in South Asia [26]. It is possible that rapid increase in risk factors and CMDs may be one important cause for the decline in the prevalence of happiness in South Asia [13-18, 27, 28].

A recent study reported the prevalence of perceived happiness and life satisfaction among 12,610 subjects aged 15–24 years in Malawi [29]. A descriptive and multivariable regression analysis showed that more than 80% of the men and women reported being satisfied about happiness, health, and life. Ethnic disparities in perceived happiness and status of health were greater among men, whereas that of life satisfaction was more pronounced in women. Housing in highest wealth quintile was positively associated with health and life satisfaction, but not with happiness. The results highlight the need for priority in relation to psychosocial needs of the populations in designing health and social policy in Malawi. It is clear that economic status alone in the form of luxurious housing is not the sole criterion of happiness [29]. In another cross-sectional survey from Finland, Poland and Spain, among 10,800 adults, in whom experienced well-being was measured using the Day Reconstruction Method and evaluative well-being was measured with the Cantril Self-Anchoring Striving Scale. The results showed that respondents with younger age (effect size, $\beta = 0.19$), higher levels of education ($\beta = -0.12$), a history of depression ($\beta = -0.17$), poor health status ($\beta = 0.29$) or poor cognitive functioning ($\beta = 0.09$) had worse well-being experience. Male sex ($\beta = -0.03$), not living with a partner ($\beta = 0.07$), and lower occupational ($\beta = -0.07$) or income levels ($\beta = 0.08$) were also associated with poor wellbeing. After controlling for age, depression, income and other sociodemographic variables, status of health was most strongly correlated with wellbeing indicating that strategies to improve population health may improve people's wellbeing [19].

In previously collected samples from Poland (1991-2015) and Russia (1994-2014), Brzezinski investigated the microeconomic determinants of declines in unhappiness rates observed in the studied periods in Poland (a 56% fall in unhappiness) and Russia (a drop in the range from 46 to 75% depending on the unhappiness threshold chosen) [20]. The results indicate that unhappiness reductions in both countries were mostly driven by coefficient effects, while characteristics played a smaller role. In both countries, financial gains accounted for about 15% of the total unhappiness reduction. These effects were doubled by growing return to income with unemployment, as unhappiness-increasing factor in Russia. However, in Poland income has been losing protecting power and in overall, income showed no effect on increase in unhappiness. In case of Poland, good self-rated health and having children explains additional 15–20% of the happiness observed in this country [20].

It should be noted that our questionnaire is different from existing questionnaires because it includes 12 attributes related to behavioral factors: optimism, prayer, yoga practice, occupational physical activity, no shift work, meditation, Mediterranean diet, mastication, no night time eating, life satisfaction and wellbeing, as well as perceived happiness with quantity and duration of happiness in a day on a scale of 0-10 (Appendix I). In other questionnaires, satisfaction in life, position in society, health status and economic status are major attributes of happiness, which do not cover emotional and spiritual wellbeing [1, 2, 19, 20, 29]. Therefore, our questionnaire appears to be much better for the assessment of multiple dimensions of happiness attributed by emotional, physical, social and spiritual behaviors that are indicators of self-health, humanity and harmony.

Our study also showed that among subjects with happiness, the frequency of moderate alcohol intake was significantly greater compared to urbansubjects with no happiness among. The prevalence of CAD was significantly lower among subjects with happiness compared to those with no happiness, both in urban (5.06 vs. 12.2%, $P < 0.01$) and rural populations (2.09% vs. 3.92%, $P < 0.05$), indicating that happiness may be protective against CAD without any gender differences (Table 3). Clinical experimental studies indicate that behavioral factors such as laughing, feeling of pleasure, sexual activity, contentment and satisfaction may be associated with increase in neuroendocrine hormones [7-12, 16]. However, aggression, depression and humiliation may be associated with increase in cortisol and decline in serotonin, resulting in insulin resistance and endothelial dysfunction [9, 10, 16, 30]. Emotional stress that begins in the brain may induce increase in stress hormones, which may have adverse effects on happiness and health and may trigger emotional diseases as well as other NCDs [9, 12, 16, 30]. Therefore, it is important to understand the adaptive and damaging effects of stress and stress mediators because they inhibit the inculcation of happiness and may predispose CMDs [9-12, 16, 30]. A positive role of lifestyle and behaviors, subjective health, and quality of life have also been observed among Chinese men living with type 2 diabetes [31].

Previous studies indicate that acute stress responses promote adaptation and survival via responses of neural, cardiovascular, autonomic, immune and metabolic systems which may become chronic due to other environmental factors [7, 9]. Other studies indicate that apart from happiness, other protective behaviors: prayer, religious service attendance, yoga therapy may be protective against psychological disorders as well as CMDs [32-37]. Thus, evidence from neuroscience, paired with evidence from the measurement of subjective well-being, or happiness, suggests that a scientific explanation of happiness is, in fact, possible [35]. The effects of happiness on brain and of brain dysfunction induced psychological disorders and CMDs appear to be bidirectional. This means that happiness may inhibit brain dysfunction, leading to decline in psychological disorders and CMDs, whereas increase in brain dysfunction may predispose to lack of perceived happiness resulting in depression, anxiety and aggression leading to CMDs. Therefore, we may agree with the quotation that “there is no path to happiness, happiness is the path.”

Life satisfaction as a component of happiness also favors reproduction [38]. Western-style diets, eating late in the night, short sleep, sedentary behavior, low cognitive work and depression as well as tobacco intake and alcoholism are behavioral risk factors [7, 9]. These risk factors may influence many of the attributes of happiness and adversely affect catecholamines and cortisol release that are damaging to target organs, resulting in the neurodegenerative diseases and CMD [7, 9, 39-44]. The hypothesis is that holistic manipulations, such as Mediterranean-style diets with probiotics, mastication during eating, intermittent fasting and caloric restriction, physical activity, yoga therapy, meditation and prayer along with other positive behaviors like optimism and social support may be an important complement to happiness [7, 45-50]. This scale needs to be reproduced by other investigators, for assessment of happiness (Appendix I) in other populations. There is a need to study the role of biomarkers: endorphins, serotonin, oxytocin, dopamine and melatonin as underlying mechanisms of happiness [45-50]. While glucocorticoids and cortisol associated with unhappiness are pro-inflammatory and predispose to cellular damage, the biomarkers associated with happiness may be anti-inflammatory and protective to cells [51-56]. Happiness may also be under the influence of social determinants of health and life outcomes have been gaining widespread attention in public health including psychological health, and sociology, environment and international development. These factors are broadly classified to encompass the cultural, behavioral, social and environmental aspects of health, quality of life, and well-being, especially among the adult population, because happiness of mind depends on these factors [7, 9, 19, 20, 29].

Neuroimaging studies by functional Magnetic Resonance imaging (fMRI) and PET have demonstrated that brain regions such as hippocampus, prefrontal cortex and amygdala and hypothalamus respond to acute and chronic stress [57-65]. These changes in morphology and chemistry of the neurons may possibly be reversed by inculcation of happiness, if the chronic stress lasts for weeks [7, 9]. However, it is not clear whether prolonged stress for many months or years may have irreversible effects on the brain. The adaptive plasticity of chronic stress involves many mediators, including glucocorticoids, excitatory amino acids, endogenous factors such as brain-derived neurotrophic factor (BDNF), polysialated neural cell adhesion molecule (PSA-NCAM) and tissue plasminogen activator (tPA) [12]. The role of this stress-induced remodeling of neural circuitry is not well known in relation to psychiatric illnesses, as well as chronic stress and the concept of top-down regulation of cognitive, autonomic and neuroendocrine function. It seems that policies of government and the private sector play an important role in this top-down view of minimizing the burden of chronic stress and lifestyle factors, sedentary, tobacco, alcohol, aggression etc., and related NCDs to achieve sustainable development goals of the UNO.

An experimental study in monkeys via linear systems analysis on a trial-by-trial basis showed that the impulse response of the neurovascular system to a stimuli is both animal- and site-specific, and that local field potential yield a better estimate of blood oxygen dependent fMRI responses than the multi-unit responses [60]. These findings suggest that the blood oxygen dependent contrast mechanism of fMRI reflects the input and intracortical processing of a given area, rather than its spiking output. The brain is the central organ of the stress response and determines what is stressful, as well as the behavioral and physiological responses to potential and actual stressors. The brain is also a target of stress and it changes structurally and chemically in response to both acute and chronic stressors [12, 57-70]. Glucocorticoids play a role in these changes, but there are other mediators; endorphins, serotonin, dopamine and oxytocin as well [8-12]. Although glucocorticoids and catecholamines are the two defining hormones of the “fight or flight” stress response, there are many other mediators, such as pro- and anti-inflammatory cytokines and the parasympathetic nervous system that are also involved in the istress by releasing acetylecholine and nitric oxide that provide anti-infammatory effects.

Limitations

In our study, more time should have been given to find out exact details of all the behaviors. Multivariate regression analysis should be conducted to find out the association of risk factors and protective factors with happiness.

In existing studies based on online surveys, results cannot be taken as representative of the general adult population under the age of 75 in most countries of the world [2]. Samples in Brazil, Chile, China (mainland), Colombia, India, Malaysia, Mexico, Peru, Russia, Saudi Arabia, Serbia, South Africa and Turkey are more urban, more educated and/or more affluent than the general population and the results should be viewed as reflecting the views of a more “connected” population. Sample surveys and polls may be subject to other sources of error, including, but not limited to coverage error, and measurement error. The precision of the Ipsos online polls is measured using a Bayesian Credibility Interval. The credibility interval around percentages based on single-month data is of +/-3.5 percentage points for markets where the monthly sample is 1,000+ and +/-4.8 points for markets where the monthly sample is 500+. National surveys in various countries are in progress to evaluate well-being as an indicator of societal progress that goes beyond traditional indices, such as gross domestic product (GDP) [20]. These surveys educate the policy-makers about the factors that can affect the well-being of populations. Health and wellbeing are interconnected because each can influence the other. Good health may be associated with wellbeing and happiness, whereas presence of diseases or disability may provide negative effects, leading to unhappiness and depression [55-60]. The Commission on the Measurement of Economic Performance and Social Progress recommended shifting emphasis from measuring economic production to measuring people’s well-being and that this measurement be done at a country level. The Better Life Initiative, launched by the Organisation for Economic Co-operation and Development, has a major objective of measuring the progress of people across eleven attributes of well-being, such as life satisfaction, health, education and environment. Efforts are also being made at the national level in many countries.

Clinical studies may distinguish differences in ways of assessment of well-being and happiness, which may include asking people to evaluate their life. It may be called evaluative well-being. Another method may be to ask subjects to narrate the positive and negative emotions that they experience day-to-day which may be called experienced happiness. Evaluation of happiness may refer to overall evaluation of the subject indicating the quality of his or her life. Experiencing happiness captures the positive and negative emotions that people experience from one moment to another moment, which are relevant, because these do not necessarily have the same correlates. High-income subjects report more satisfaction with their lives when their evaluative happiness is assessed but the same subjects may not express better experienced happiness due to various social and occupational problems. Some of the life circumstances, such as occupation, marital status and education may also be more strongly correlated with evaluative than experienced happiness. However, presence of diseases, caring for an adult, loneliness and smoking have been reported to be strong predictors of low experienced happiness as observed in our study (Table 3). The correlation between health status and evaluative and experienced happiness indicate that CAD was associated with reduced perceived happiness in this population. Public health research and policy in the developing world are generally disease-oriented, with the focus being largely confined within the biological determinants of health. So far, little attention has been given to developing a more health-oriented approach by emphasizing the psychosocial dimensions of health, especially among the younger population.

In brief, our study showed that our questionnaire can accurately identify happiness. The prevalence of happiness is quite high in India, despite it is a lower middle-income coutry. This modified questionnaire may be successfully applied for assessment of happiness rates in urban and rural areas. The questionnaire allowed us to identify overall 63.2% (n = 569) subjects with happiness, including 62.4 % among urban and 63.2% (n = 569) among rural subjects. Moderate intake of alcohol may be in favor of happiness and happiness may be protective against CAD. Further analysis and further studies are necessary to verify our findings.

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Authors Contributions

All the data were collected by Dr RBS and MAN others Dr JF, VM, AI, NR, MD, SW helped in writing of the manuscript.

Appendix I. Study of Healthy Lifestyle, Life Satisfaction, Wellbeing and Behaviors as Attributes of Happiness, Based on References 5-9

Attributes of happiness: Scores	0	1	2	3	4	5	6	7	8	9	10
1. Right, Understanding (Think for universal benefits)											
2. Right Intent, (Good intention for people)											
3. Right Speech (Speak truth, not unpleasant to others)											
4. Right Action (Do right things without harm to others), healthy behaviors: prayer, yoga, physical activity.											
5. Right Livelihood (Occupation with no harm to others, and self, satisfaction with income, status, no night shift)											
6. Right Effort (Make efforts without harming others)											
7. Right Mindfulness (Speak with full attention and feeling of goodness) optimism.											
8. Right Concentration (Concentrate, pray and meditate to allay bad thoughts, optimism).											
9. Right eating (Eat healthy foods with mastication).											
10. Right time of eating (Eat before sun set).											
11. Life satisfaction and wellbeing, optimism.											
12. Perceived quantity of happiness and duration of happiness including sexual activity, cuddling etc. in 24 hours on a scale of 0-10.											

Total score = 81-120; Very satisfied and very happy; Total score = 51-80; moderately satisfied and happy and Total score = 21-50; modestly satisfied and happy and Total score < 21 = Neutral. (Highlighted- proven).

Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Correlation between 2D Speckle Tracking Echocardiography and Exercise Stress Echocardiography in Asymptomatic Patients with Severe Rheumatic Mitral Regurgitation

Mohamed Khalfallah[1,*], MD,
Randa Abdelmageed[2], MD,
and Ibtsam Khairat[1], MD
[1]Lecturer of Cardiology, Tanta University, Egypt
[2]Assistant Professor of Cardiology, Tanta University, Egypt

Abstract

Background: Mitral regurgitation (MR) is a common valvular heart disease. Data available about the proper diagnosis and management of asymptomatic patients with severe rheumatic MR with good systolic function are still controversial. Our objective was to assess the role of Two D-Speckle Tracking Echocardiography (2DSTE) in patients with severe rheumatic MR with correlation to exercise stress echocardiography.
Patients and methods: The present study was conducted on 187 participants (90 patients and 97 healthy controls). Patients were asymptomatic and had severe rheumatic MR. 2DSTE was done to all participants and correlated with the results of exercise stress echocardiography.
Results: Global longitudinal strain (GLS) of the left ventricle (LV) by 2DSTE was significantly lower in rheumatic patients (-16.95 ± 2.12%) than in controls (-23.01 ± 0.71%) (P=0.001). Controversially, global circumferential and radial strain of LV (GCS, GRS) showed no statistically significant difference between the two groups. GLS had a significant negative correlation with exercise duration and percentage of age-gender predicted METs. Interestingly, GLS ≤ -17 had better exercise parameters, longer exercise duration and better percentage of age-gender predicted METs.
Conclusion: 2DSTE has incremental value beyond conventional measures of LV function. Patients with GLS ≤-17 have a good functional capacity and better exercise parameters than those with GLS >-17. A cut off value of GLS >-17 could identify patients at high risk of cardiovascular morbidity.

Keywords: Correlation; 2D speckle tracking echocardiography; exercise stress echocardiography; rheumatic; mitral regurgitation

Introduction

Valvular heart disease (VHD) is a one of the most common causes of morbidity and mortality

* *Corresponding author:* Dr. Mohamed Khalfallah, MD. Department of Cardiology, Faculty of Medicine, Tanta University, Egypt. E-mail: khalfallah@yahoo.com

worldwide. Rheumatic heart disease is considered one of the most important causes of VHD in developing countries. Mitral regurgitation (MR) is a common type of rheumatic VHD [1]. Left ventricular dysfunction is frequently subclinical despite normal LV ejection fraction (LVEF) and may precede the onset of symptoms. Current guidelines [2] recommend valve replacement/repair in cases of severe MR that cause symptoms or reduced LVEF, as surgery before development of irreversible myocardial damage prevents progressive post-operative LV dysfunction and improves clinical outcome [3].

Echocardiography and cardiac magnetic resonance are the best available current modalities for assessment of LV function; they allow dynamic imaging of the heart and assess regional and global LV function. But cardiac magnetic resonance is costly, not easily available in developing countries, and cannot be used for routine clinical application in all patients [4]. Routine echocardiographic parameters such as volume and LVEF are often unable to disclose initial LV dysfunction; LV function may remain compensated despite the changes in myocardial deformation properties. Tissue Doppler imaging (TDI) allows measurement of tissue velocities within the myocardium and assessment of LV function and filling pressures [5]. However, TDI depends on the angle and measures only a single component of regional velocity vector along the scan line [5, 6]. 2D Speckle tracking echocardiography (2DSTE) is a newer technique that provides non-Doppler, relatively angle independent measurement of myocardial deformation [6]. 2DSTE evaluates LV mechanics, rotation, and torsion and allows for semi-automated quantification of myocardial deformation (strain and strain rate) in the three spatial directions (longitudinal, radial, and circumferential), which is useful to assess LV long and short axis function as the chronic LV remodeling process may have a different effect on LV short and long axis function [7-9]. Although strain parameters are all partially load-dependent, the strain rate is relatively load-independent [10].

Exercise testing plays an important role in primary asymptomatic MR; it allows better assessment of symptoms and its relation to valve disease severity, and safe deferral of surgery in patients with preserved exercise capacity and helps in individual risk stratification. The objective of the present study was to assess the role of 2DSTE and its incremental value beyond conventional measures of LV function in patients with severe rheumatic MR with correlation to exercise stress echocardiography. Results suggest that the application of this new modality for identification of early abnormalities in LV function, particularly in the case of preserved LVEF and before the onset of clinical symptoms.

Patients and Methods

The present study was conducted in the Cardiology Department, Faculty of Medicine, Tanta University, during the period from October 2017 to March 2019 on 90 asymptomatic patients with severe rheumatic mitral incompetence (group I) who were selected randomly during this period and 97 age- and sex-matched healthy controls (group II). The mean age of group I was 33.02 ± 6.12 years with 51 male patients (56.7%), while the mean age of group II was 33.94 ± 6.18 years with 45 male patients (46.4%). All participants gave written informed consent to participate in this study and the study was approved by the local ethics committee.

Inclusion and Exclusion Criteria

Patients were asymptomatic with severe rheumatic mitral incompetence, LV systolic function ≥ 60%, LVESD < 4.5 cm, while patients with evidence of double mitral valve lesion (N = 205), multivalvular affection (N = 350), ischemic heart disease (N = 20), LV function <60% (N = 150), atrial or ventricular arrhythmia (N = 88), renal disorders, hepatic disorders, diabetes mellitus, hypertension, congenital heart diseases, hematological and neoplastic diseases (N = 185) were excluded.

All patients and healthy controls were subjected to history taking and general and local cardiac examination. Routine 12-lead ECG was performed for the studied populations to detect changes suggestive of coronary artery disease and exclude arrhythmia and conduction abnormalities.

Data were collected by a case record form, which included routine laboratory investigations including kidney function tests, random blood sugar and liver function tests. All participants were subjected to 2D, Doppler, and color Doppler echocardiography for the assessment of the mitral valve. Echocardiographic examinations were carried out with the Vivid E9 dimension (General Electric Medical Systems, Horten, Norway) equipped with a 2.5-MHz variable-frequency transducer S probe for 2D imaging. Standard views (parasternal long and short axis, apical four, five and two chamber, subcostal and suprasternal views) with ECG tracing were obtained in 2D modes, with assessment of LV diameter in end diastole, and end systole by M mode with estimation automatically of LVEF, assessment of LA diameter in parasternal long axis view by M mode at end systole [11] and assessment of other cardiac valves to exclude any associated valvular lesion. Assessment of the mitral valve was done first by 2D, then assessment by color flow mapping to asses origin and direction of jet, width of jet at the level of the orifice averaged from all views (vena contracta) [12]; continuous wave signal and dense forward flow ascertained the degree of mitral incompetence [13]. The right side of the heart and right ventricular systolic pressure (RVSP) were assessed by estimating maximum velocity of tricuspid incompetence by continuous Doppler flow [14]. In all patients, RVSP was < 50mmHg. In apical 4 chamber view pulsed wave TDI sample volumes were placed at the level of the lateral mitral annulus and at the level of the septal mitral annulus, with proper alignment to minimize the incident angle. Peak myocardial velocities during systole (s') were measured in 3 consecutive cycles and averaged [15].

Two D-Speckle Tracking Echocardiography and Exercise Stress Echocardiography

Gated images with high quality are obtained during end expiratory breath hold with stable ECG tracing. Three consecutive heart cycles were recorded and averaged at frame rate adjusted at 60 - 80 frames/second. Three, four and two apical chambers are necessary for estimation of global longitudinal strain; however for circumferential and radial strain, parasternal views were obtained at basal, papillary muscle and apical levels [16, 17]. Measurement of the aortic valve closure was performed by acquiring a clear M-Mode signal where the valve closure was clearly visible or from a proper Doppler signal including both mitral and aortic valves. Recordings were processed by software (Echo Pac, GE Vivid E9 echocardiography system version 113) for off-line measurements of speckle-based strain [16, 17]. The region of interest was outlined manually by tracing the endocardium and then the epicardium and automatically outlined; the peak systolic global longitudinal, circumferential and radial strain were calculated [16, 17]. All strain measurements were performed independently by two separate investigators experienced in analysis and blinded to clinical and other echocardiographic patient's data. Exercise stress echocardiography was performed for all patients (group I) using a treadmill with a modified Bruce protocol in which an expected exercise level for each gender and age group was expressed as a functional aerobic capacity, and also expected maximum heart rate achieved. Rapid image acquisition was performed at peak stress to estimate LV systolic function and RVSP by estimating maximal velocity of tricuspid incompetence for correlation with the results of 2DSTE. The patients were followed-up for 6 months for cardiovascular symptom development in the form of dyspnea and palpitation, signs of heart failure and cardiac hospitalization.

Statistical Analysis

Statistical analysis was dperformed using SPSS 20 (IBM, Armonk, NY, USA). Quantitative data were expressed as mean ± standard deviation (SD). Qualitative data were expressed as frequency and percentage. The Student's t test was used for the comparison between two groups in quantitative data and the chi-square (χ^2) test in proportions. A P-value <0.05 was considered to indicate statistical significance. Correlation analysis (using Spearman's method) assessed the strength of association between 2DSTE results and exercise stress echocardiography results. The correlation coefficient denoted symbolically "r" defines the strength and direction of the linear relationship between two variables.

Results

The present study included 187 participants: 90 patients with severe rheumatic mitral incompetence with preserved systolic function (group I) and 97 healthy controls with no history of any disease (group II). The patients' ages ranged from 27-39 years with a mean of 33.02 ± 6.12 years including 51 male patients and 39 female patients with no statistically significant difference between the two groups. The heart rate was significantly higher among patients with severe rheumatic MR compared to control subjects (71.26 ± 7.77 vs. 68.74 ± 8.12 bpm, P=0.032), Table 1.

Table 1. Baseline demographic, clinical and echocardiographic data of the two groups

	Group I (N= 90) (patients with severe MR)	Group II (N= 97) (control group)	P-value
Age, years	33.02 ± 6.12	33.94 ± 6.18	0.310
Male gender, n (%)	51 (56.7%)	45 (46.4%)	0.160
Smoking, n (%)	31 (34.4%)	29 (29.9%)	0.506
Dyslipidemia, n (%)	16 (17.8)	24 (24.7%)	0.246
Height (cm)	166.97 ± 5.21	165.94 ± 11.31	0.431
Weight (kg)	69.31 ± 7.20	69.35 ± 7.52	0.971
BMI (kg/m^2)	24.92 ± 2.86	26.17 ± 12.36	0.348
Heart rate (bpm)	71.26 ± 7.77	68.74 ± 8.12	0.032*
Systolic BP, (mmHg)	117.39 ± 9.66	119.18 ± 9.23	0.198
Diastolic BP, (mmHg)	79.13 ± 4.90	77.65 ± 6.18	0.072
LVEDD, (cm)	6.20 ± 0.20	5.25 ± 0.18	0.001*
LVESD, (cm)	4.23 ± 0.14	3.43 ± 0.22	0.001*
LVEF, (%)	63.57 ± 2.90	62.95 ± 2.58	0.124
LA diameter, (cm)	4.97 ± 0.19	3.39 ± 0.39	0.001*
S' lateral, (cm/s)	7.96 ± 0.30	9.35 ± 1.10	0.001*
S' septal, (cm/s)	8.00 ± 0.31	9.15 ± 1.11	0.001*
GLS, (%)	-16.95 ± 2.12	-23.01 ± 0.71	0.001*
GCS, (%)	-22.58 ± 0.73	-22.67 ± 0.68	0.407
GRS, (%)	45.31 ± 0.86	45.37 ± 0.71	0.623

MR: mitral regurgitation; BMI: body mass index; BP: blood pressure; LVEDD: left ventricular end diastolic dimension; LVESD: left ventricular end systolic dimension; LVEF: left ventricular ejection fraction; LA: left atrium; S' lateral: peak systolic lateral mitral annular velocity; S' septal: peak systolic septal mitral annular velocity; GLS: global longitudinal strain; GCS: global circumferential strain; GRS: global radial strain.

All included participants' LVEF was more than 60% with no significant difference between the two groups, but left ventricular dimensions in diastole and systole (LVEDD, LVESD) showed statistically significant differences with more dilatation in the rheumatic group (6.20 ± 0.20 cm and 4.23 ± 0.14 cm, respectively than in the control group (5.25 ± 0.18 cm and 3.43 ± 0.22 cm, respectively) (P = 0.001). The left atrium (LA) was also significantly dilated in group I (4.97 ± 0.19 vs. 3.39 ± 0.39 cm in the control group) (P = 0.001). Despite LVEF being higher than 60% in all participants, peak systolic S' wave of LV at septal and lateral mitral annulus was significantly lower in rheumatic patients (8.00 ± 0.31, 7.96 ± 0.30 cm/s, respectively) as compared to the control group (9.15 ± 1.11, 9.35 ± 1.10 cm/s, respectively) (P=0.001). Global longitudinal strain (GLS) of LV by two-dimensional strain was significantly lower in rheumatic patients (-16.95 ± 2.12%) than in the control group (-23.01 ± 0.71%) (P=0.001). Controversially, global circumferential and radial strain of LV (GCS, GRS) showed no statistically significant difference between the two groups.

Furthermore, group I was subjected to exercise stress echocardiography to assess functional capacity of the patients with severe rheumatic mitral regurgitation. Correlation between 2DSTE parameters and exercise stress echocardiographic parameters showed that GLS had a significant negative correlation with exercise duration and percentage of age-gender predicted METs (P=0.001). GCS correlated negatively with peak stress LVEF (P=0.002). GRS did not correlate with exercise stress echocardiographic parameters, as shown in Table 2 and Figures 1 and 2.

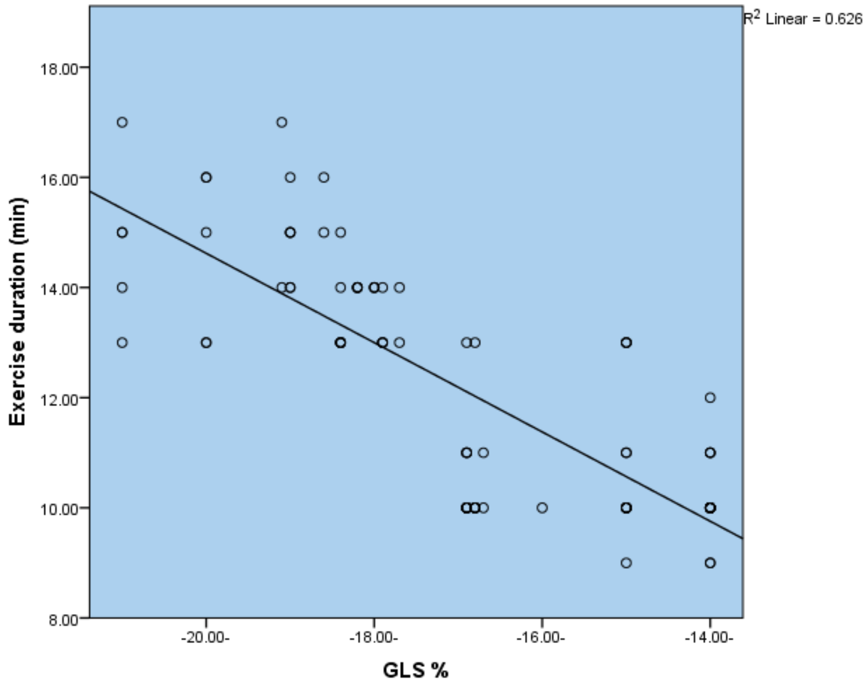

Figure 1. The correlation between global longitudinal strain and exercise duration.

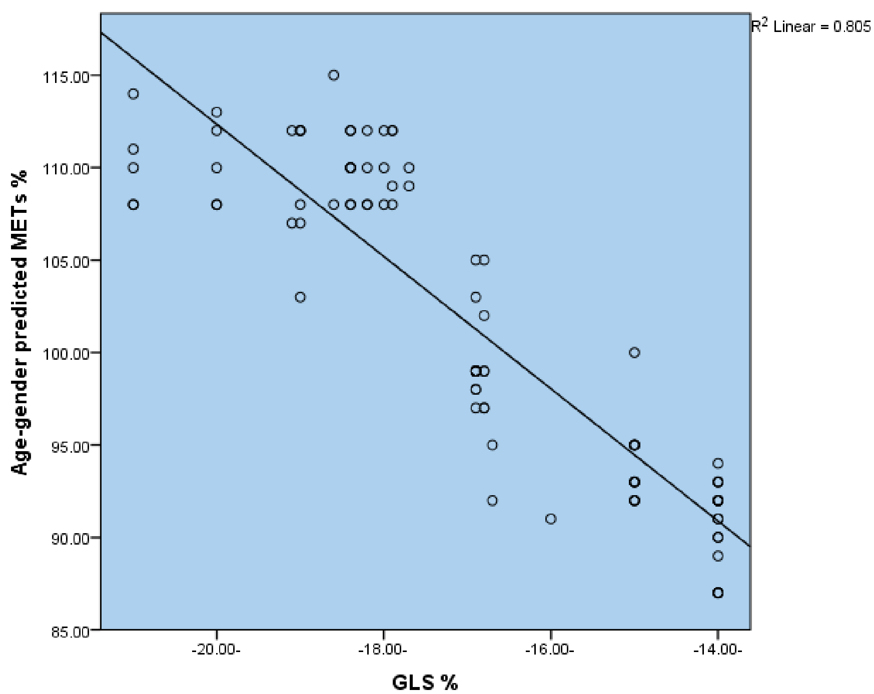

Figure 2. The correlation between global longitudinal strain and age-gender predicted METs.

Table 2. The correlation between 2D speckle tracking echocardiographic results and exercise stress echocardiographic results

	Exercise duration (minutes)		Maximum predicted HR, (%)		% Age-gender predicted METs		Peak stress RVSP, (mm Hg)		Peak stress LVEF, (%)	
	r	P value	r	P value	r	P value	r	P value	r	P value
GLS, (%)	-0.791	0.001*	-0.085	0.427	-0.897	0.001*	0.146	0.170	-0.001	0.991
GCS, (%)	-0.107	0.317	0.082	0.443	0.104	0.330	0.115	0.283	-0.323	0.002*
GRS, (%)	-0.201	0.057	0.165	0.121	0.002	0.987	0.095	0.372	-0.193	0.068

HR: heart rate; METs: metabolic equivalents; RVSP: right ventricular systolic pressure; LVEF: left ventricular ejection fraction; GLS: global longitudinal strain; GCS: global circumferential strain; GRS: global radial strain.

Table 3. Comparison between the two subgroups of global longitudinal strain

	Group IA GLS > -17 (N = 50)	Group IB GLS ≤ -17 (N = 40)	P-value
Exercise duration (minutes)	10.46 ± 1.01	14.25 ± 1.15	0.001*
Maximum predicted heart rate, (%)	92.82 ± 6.06	94.27 ± 7.37	0.307
% Age-gender predicted METs	94.64 ± 4.49	109.90 ± 3.32	0.001*
Peak stress RVSP, (mm Hg)	34.66 ± 6.05	32.37 ± 3.42	0.036*
Peak stress LVEF, (%)	66.72 ± 3.36	67.05 ± 2.82	0.621

METs: metabolic equivalents; RVSP: right ventricular systolic pressure; LVEF: left ventricular ejection fraction; GLS: global longitudinal strain.

Furthermore, group I was divided into two subgroups according to the median value of GLS, which was -17; Group IA had GLS >-17 (N = 50) and Group IB had GLS ≤ -17 (N = 40). As compared to patients with GLS >-17, patients with GLS ≤-17 had better exercise parameters, longer exercise duration, more percentage of age-gender predicted METs and lower peak stress RVSP, Table 3. During follow-up of the patients for 6 months, we noticed that 4 patients of Group IA (GLS >-17) developed symptoms in the form of dyspnea and palpitation and one patient was admitted to hospital with atrial fibrillation.

Discussion

The present study emphasized the role of 2DSTE and its incremental value beyond other conventional measures of LV function. Moreover, we correlated it to exercise stress echocardiography and to the best of our knowledge we were the first to put a cut-off value of GLS>-17 to determine the patients with severe MR at high risk of cardiovascular morbidity. Severe MR is associated with increased risk of morbidity and mortality; recent guidelines recommend surgery in symptomatic patients with severe MR and in asymptomatic patients with signs of LV dysfunction to confer a better outcome [18, 19]. The LVEF is often overestimated in the presence of MR and may remain in the normal or supernormal range for a long time, even if alterations in contractility develop. Thus, it is important to detect LV contractile dysfunction very early to favor better surgical outcome and life expectancy after surgery for severe MR [20, 21]. Conventional echocardiographic parameters cannot detect subclinical LV dysfunction; recently 2DSTE is used for quantitative assessment of global and segmental LV function [22, 23]. In the present study GLS of LV by two-dimension strain was significantly lower in rheumatic patients with severe MR in comparison with the control group, while GCS and GRS were similar in two groups. In agreement with the present study, Kim et al. [24] studied 59 patients with severe MR and LVEF ≥50% and 34 healthy controls. Speckle tracking imaging was performed to measure peak systolic radial, circumferential and longitudinal strain rates. In all patients, GLS was significantly depressed when compared with a control group. Preoperative speckle tracking–derived GLS and longitudinal strain and strain rate values strongly predicted a postoperative LVEF decrease of > 10%. The authors reported that long axis function (longitudinal strain) becomes depressed earlier than short axis function (circumferential and radial strain) in remodeling process [24]. Lee et al. and Agricola et al. also showed previously that the deterioration in LV GLS is a predictor of latent LV dysfunction and a predictor of postoperative LV dysfunction [25, 26]. In disagreement to our study, Marciniak et al. [8] showed that circumferential strain has a biphasic pattern, being enhanced in moderate MR and then significantly reduced with severe MR in comparison with a control group, but this was studied in a small group of patients [8].

When assessing patients with severe rheumatic MR, there is a diagnostic and management dilemma. Exercise testing plays an important role in the assessment of the clinical status of those patients and in sorting out some of these clinical challenges. In asymptomatic patients with severe MR, exercise testing allows symptom assessment and determines the link of symptoms to valve disease severity, deferral of surgery in patients with preserved exercise capacity and helps in individual risk stratification [27]. In the current study, rheumatic patients with severe MR were subjected to exercise stress echocardiography to assess functional capacity, RVSP and LVEF at peak stress and correlation between 2DSTE parameters and exercise parameters was performed. GLS had a significant negative correlation with exercise duration and percentage of age-gender predicted METs. In agreement with the present study Supino et al. [28] prospectively followed 38 patients with chronic severe non ischemic MR who underwent modified Bruce exercise Treadmill testing; all patients lacked surgical indications at study entry. Their baseline exercise parameters were also compared with those from 46 patients with severe MR who, at entry, had already reached surgical indications. Their esults from univariate analysis showed that exercise duration ($p = 0.004$) predicted peak heart rate ($p = 0.01$) and heart rate recovery ($p < 0.02$). In multivariate analysis, only exercise duration was predictive ($p < 0.02$). Average annual event risk was 5-fold lower (4.62%) with exercise duration ≥15 minutes vs. <15 minutes (average annual risk=23.48%, P=0.004). The authors concluded that among asymptomatic patients with chronic severe non-ischemic MR and no objective criteria for operation, progression to surgical indications is rapid. However, those patients with excellent exercise tolerance hadve a relatively benign course. Mentias et al. [29] studied a group of patients to determine whether resting LV GLS and exercise test provide incremental prognostic utility in asymptomatic patients with severe MR and preserved ejection fraction. The conclusion was that reduced exercise capacity and worsening resting LV GLS were associated with increased mortality and providing additive prognostic utility. In the same issue Mentias et al. [30] sought to evaluate the incremental impact of LV GLS on exercise capacity and studied 660 patients who underwent rest-stress echocardiography, Standard strain data were obtained and their conclusion was that in asymptomatic patients with ≥3+ primary MR, non-dilated LV and preserved LVEF, worsening resting LV GLS is significantly associated with reduced functional capacity, independently and incrementally to other known predictors [30].

In the current study, group I was divided into two subgroups according to the median value of GLS. Patients with GLS ≤ -17 were found to have better exercise parameters, longer exercise duration, more percentage of age-gender predicted METs and lower peak stress RVSP.

Conclusion

The role of 2DSTE and its incremental value beyond conventional measures of LV function was studied herein with correlation to exercise stress echocardiography. The application of this new modality for identification of early abnormalities in LV function, particularly in the case of preserved LVEF and before the onset of clinical symptoms, rheumatic patients with severe MR (with GLS ≤-17) had a good functional capacity and better exercise parameters than those with GLS>-17. Using GLS>-17 as a cut-off value could be an important criterion in the future to identify such high risk group of patients at high risk of cardiovascular morbidity.

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Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Echocardiographic Evaluation of the Right Ventricular Function in a Sample of Iraqi Patients with Left Ventricular Dysfunction

Marwah Qasim Mohammed[1], Muataz Fawzi Hussein[2], Samar I. Essa[3,*], Yaarub Shawkat Dahaam[4], and Ammar A. Ismaeel[5]

[1]Specialist of Echocardiography, Baghdad Teaching Hospital, Baghdad, Iraq
[2]Department of Medicine, College of Medicine, University of Baghdad, Baghdad, Iraq
[3]Department of Physics, College of Science, University of Baghdad, Baghdad, Iraq
[4]Department of Medicine, Kirkuk General Hospital, Iraq
[5]Interventional Cardiologist, Iraqi Center for Heart Diseases, Iraq

Abstract

Background: Assessment of function of the right side of the heart in cases of left ventricular dysfunction has been widely studied but the sensitive and specific echocardiographic parameter to be tested is still a matter of controversy. Right ventricular function is related to left ventricular function by ventricular independence so function of both should be assessed carefully.
The objective of this study was to evaluate the effects of left ventricular systolic dysfunction on right ventricular systolic and diastolic functions and pulmonary pressure using conventional and tissue Doppler echocardiography.
Patients and Methods: Sixty patients (39 males and 21 females) with heart failure due to left ventricular systolic dysfunction diagnosed by echocardiography presented to the Baghdad Teaching Hospital and the Iraqi Centre of Heart Diseases were included in this study. The mean age of study patients was 57±11years. Ethical approval was obtained from the ethical committee of college of Medicine/ University of Baghdad. Informed consent was obtained from all patients included in the study. Sinus rhythm and mild to severe left ventricular systolic dysfunction were the inclusion criteria. All patients underwent a standard 2-dimensional and tissue Doppler echocardiography.
Results: Of the study patients, 63.3% had abnormal right ventricle myocardial performance index. 35% of the sample have abnormal tricuspid annular plane systolic excursion. Pulmonary artery systolic pressure was elevated in 28.3% of patients. The right ventricular performance index was the most sensitive parameter of right ventricular dysfunction with a sensitivity 100% but the specificity was about 52%, while tricuspid annular plane systolic excursion was less sensitive and more specific than right ventricle myocardial performance index with a sensitivity of 38% and specificity of 84%.
Conclusion: The right ventricular function is affected in patients with left ventricular dysfunction.
To study right ventricular function; right ventricular myocardial performance index which measure both systolic and diastolic functions is more sensitive than TAPSE which is a measure of only the systolic function of right ventricle.

* Corresponding Author's Email: samaroem78@yahoo.com

Keywords: Right ventricular dysfunction, left ventricular dysfunction, myocardial performance index, tricuspid annular plane systolic excursion.

Introduction

The term congestive heart failure (CHF) is usually used to refer to heart inability to pump sufficient amount of blood to maintain the body's needs. Signs and symptoms commonly include shortness of breath, excessive tiredness, and leg swelling [1].

A previous myocardial infarction is the most common cause with or without high blood pressure that is by itself an important common cause. Other causes include diabetes mellitus, atrial fibrillation, valvular heart diseases, excess alcohol use, infection, and cardiomyopathy. There are two main types of heart failure either heart failure with low ejection fraction or heart failure with preserved ejection fraction depending on whether the ability of the left ventricle to contract or to relax is predominantly affected [1].

It is well known that the left side of the heart is affected in CHF as observed by chamber enlargement as the left side pumps blood into the systemic circulation. However, the role of the right ventricle (RV) in CHF is still relatively overlooked in part because of the perception that it is somewhat a passive conduit. However, accurate assessment of the RV by echocardiography yields early detection of cardiac diseases, enhances risk stratification, and allows timely initiation of appropriate therapy [2].

Records showed that both pulmonary hypertension (PH) and CHF are becoming increasingly common. As such, PH due to left ventricular dysfunction (LVD) is now the most common form of PH. The backward transmission of an increased left ventricular filling pressures into the pulmonary circulation and, ultimately, RV strain/dysfunction is the pathophysiology of the condition and these processes are both the effect and cause of left heart disease progression [3].

Conversely, primary right ventricular pressure overload, as may occur with pulmonary hypertensive states, may compromise left ventricular function and lead to coincident evidence of left ventricular failure, such as pulmonary oedema or effusion. Furthermore, when the right ventricle fails in the setting of left ventricular failure, it may be unable to maintain the flow volume required to maintain adequate left ventricular preload [4]. Because of the multiple influences affecting right ventricular function due to left ventricular failure, right ventricular status may constitute a “common final pathway” in the progression of congestive heart failure and therefore may be a sensitive indicator of impending decompensating or poor prognosis [5].

The geometry of the chamber is complex, consisting of an inlet (sinus) portion and an outlet section separated by the crista supraventricular is. Longitudinal shortening is a greater contributor to right ventricular stroke volume than short-axis circumferential shortening [6].

In this study; the consequences of left ventricular systolic dysfunction on the function of the right ventricle were evaluated using conventional and tissue Doppler echocardiography.

Patients and Methods

Sixty patients (39 males and 21 females) with heart failure due to left ventricular systolic dysfunction were included in this study. The mean age of study patients was 57 ± 11years. Ethical approval was obtained from the ethical committee of College of Medicine/ University of Baghdad. Informed consent was obtained from all patients included in the study and they were informed that their information will be used only for research purposes.

Recruitment of Study Patients

The study patients were recruited from patients presented to the echocardiography units of Baghdad Teaching Hospital and the Iraqi Centre of Heart Diseases during the period from January 2019 to April 2019 (two-three days/week). The total number of patients examined in the echocardiography units in each of these two centers is about 20-30 patients daily. In this study, only adult patients with LV systolic dysfunction were included.

Inclusion Criteria

Sinus rhythm and mild to severe LV systolic dysfunction due to chronic ischemic heart diseases (IHD) or idiopathic dilated cardiomyopathy (IDCM) were the inclusion criteria.

Exclusion Criteria

Patients with any valvular heart disease, cardiac amyloidosis, hypertrophic cardiomyopathy, active alcoholism, chronic obstructive pulmonary disease, malignancy, advanced liver or renal diseases, recent myocardial infarction (<6 months) and any other disease leading to pulmonary hypertension were excluded. Additionally, any patient who did not accept to be included in the study was excluded.

Collection of Data

Full history was obtained from patients and complete physical examination was performed and classification of patients according to New York Heart Association (NYHA) functional class [7].

In addition, resting electrocardiography (ECG) and chest radiography were obtained in all patients.

A standard 2-dimensional and tissue Doppler echocardiographic examination was done using Philips CX-50 echocardiographic equipment with phase-array transducer of 2.5 MHz frequency. The echocardiographic examinations, measurements and grading of different echocardiographic parameters of LV and RV functions were done according to recommendations of the American Society of Echocardiography [8]. Left ventricular systolic dysfunction was defined as left ventricular ejection fraction of less than 52% in males and less than 54% in females.

Statistical Analysis

Statistical analysis was done using statistical package for social sciences (SPSS) version 17. Descriptive statistics were presented as (mean ± standard deviation) and frequencies as percentages. Level of significance (p value) was set at ≤ 0.05.

Results

The mean age of study patients was 57 ± 11 years, 50% of patients were ≥60 years old.

Males were more than females with male to female ratio of 1.8:1. All these findings were shown in Table 1.

Most of patients had NYHA class II (36.7%). Most of patients had etiology of IHD and 16.7% of them had etiology of dilated cardiomyopathy (DCM). Hypertension (HT) was present in 29 patients and diabetes mellitus (DM) was present among 32 patients as shown in Table 1.

LVIDd was abnormal in 68.3% and LVIDs was abnormal in 93.3% of patients. 48.3% of patients had mild LV systolic dysfunction, 33.3% had moderate and 18.4% had severe LV systolic dysfunction as shown in Table 2.

Table 1. Demographic and clinical characteristics of patients with left ventricular dysfunction

Variable	No.	%
Age		
<40 years	4	6.7
40-49 years	5	8.3
50-59 years	21	35.0
≥60 years	30	50.0
Gender		
Male	39	65.0
Female	21	35.0
NYHA Class		
I	11	18.3
II	22	36.7
III	11	18.3
IV	16	26.7
Etiology		
Ischemic heart disease	50	83.3
Idiopathic dilated cardiomyopathy	10	16.7
Hypertension		
Present	29	48.3
Absent	31	51.7
Diabetes mellitus		
Present	32	53.3
Absent	28	46.7

Table 2. Left and right ventricular function parameters in patients with left ventricular dysfunction

Variable	No.	%
LVIDd		
Normal	19	31.7
Abnormal	41	68.3
LVIDs		
Normal	4	6.7
Abnormal	56	93.3
LV systolic function		
Mild dysfunction	29	48.3
Moderate dysfunction	20	33.3
Severe dysfunction	11	18.4
RVMPI		
Normal	22	36.7
Impaired	38	63.3
PASP (mmHg)		
< 45	55	91.7
45-64	2	3.3
≥ 65	3	5.0
TAPSE		
Normal	39	65.0
Abnormal	21	35.0
RVS′		
Normal	42	70.0
Abnormal	18	30.0
RV dimensions		
Normal	39	65.0
Dilated	21	35.0
PVR		
Normal	16	72.7
Abnormal	6	27.3
RV dysfunction		
No	46	76.7
Yes	14	23.3

Impaired RV myocardial performance index (RVMPI) was present in 63.3% of patients. According to the level of pulmonary artery systolic pressure (PASP), normal or mildly elevated PASP was present in 91.7% of patients, 3.3% had moderate pulmonary HT and 5% had severe pulmonary HT. Tricuspid annular plane systolic excursion (TAPSE) was abnormal in 35% of patients, and systolic velocity of right ventricular free wall (RVS′) was abnormal in 30% of patients. RV dimensions were increased in 35% of patients. 27.3% of patients had elevated pulmonary vascular resistance (PVR).

RV dysfunction was defined as at least impaired three RV function parameters. It was present in 14 patients (23.3%). These findings are presented in Table 2.

ANOVA analysis revealed significant increase in means of LVIDd and LVIDs with severe LV systolic dysfunction ($p < 0.05$) as shown in Table 3.

No significant differences were observed in age, gender, NYHA class, etiology, and presence of HT or DM between patients with RV dysfunction and patients with no RV dysfunction ($p > 0.05$) as shown in Table 4.

A significant association was observed between dilated RV dimensions and RV dysfunction ($p < 0.05$) as shown in Table 5.

No significant differences were observed between patients with RV dysfunction and those without regarding LVIDd, LVIDs, PASP and PVR means ($p > 0.05$). There was a significantly lower mean of LVEF among patients with RV dysfunction ($p<0.05$) as shown in Table 6.

Binary logistic regression analysis for parameters associated with RV dysfunction revealed that RVMPI and TAPSE were significant predictors of RV dysfunction ($p < 0.05$). Odds ratio OR (probability) of RVMPI was higher than OR for TAPSE (16.6 versus 0.2) as shown in Table 7.

ROC analysis revealed that area under curve (AUC) for RVMPI was significant predictor for RV dysfunction (>0.5). On other hand, AUC for TAPSE and RVS′ were not significant (<0.5) as shown in Figure 1.

Table 3. ANOVA analysis of LV dimensions according to severity of LV systolic dysfunction

LV systolic dysfunction	LVIDd	LVIDs
	Mean ± SD	Mean ± SD
Mild	56.3 ± 5.5	42.8 ± 5.3
Moderate	61 ± 8.4	49.5 ± 7.9
Severe	61.3 ± 9.2	54.9 ± 8.2
p- value	0.04	<0.001

Table 4. Distribution of demographic and clinical characteristics according to RV dysfunction

Variable	No RV dysfunction		RV dysfunction		χ^2	P
	No.	%	No.	%		
Age (years)						
<40	3	75.0	1	25.0		
40-49	3	60.0	2	40.0	3.5*	0.2
50-59	19	90.5	2	9.5		
≥60	21	70.0	9	30.0		
Gender						
Male	32	82.1	7	17.9	1.8*	0.1
Female	14	66.7	7	33.3		
NHYA-class						
I	10	90.9	1	9.1		
II	17	77.3	5	22.7	2.3	0.5
III	7	63.6	4	36.4		
IV	12	75.0	4	25.0		
Etiology						
IHD	39	78.0	11	22.0	0.2	0.5
DCM	7	70.0	3	30.0		
HT						
Present	21	72.4	8	27.6	0.5	0.4
Absent	25	80.6	6	19.4		
DM						
Present	23	71.9	9	28.1	0.8	0.3
Absent	23	82.1	5	17.9		

Table 5. Distribution of RV dimensions according to RV function

Variable	No dysfunction		RV dysfunction		χ^2	p- value
	No.	%	No.	%		
Normal	34	87.2	5	12.8	2.3*	0.009
Dilated	12	57.1	9	42.9		

Table 6. Distribution of LV and RV function parameters according to RV function

Variables	No dysfunction	RV dysfunction	p- value
	Mean ± SD	Mean ± SD	
LVIDd(mm)	59.6 ± 7.6	56 ± 7.1	>0.05
LVIDs (mm)	47.7 ± 8.1	45.7 ± 8.4	>0.05
LVEF (%)	40.1 ± 8.3	36.1 ± 7.4	<0.05
SPAP (mmHg)	27.2 ± 15	33 ± 17.8	>0.05
PVR (wood)	1.9 ± 0.8	2.3 ± 1.2	>0.05

Table 7. Binary logistic regression analysis for variables correlating with RV dysfunction

Variable	B	SE	p-value	OR
Constant	17.1	6.4	<0.05	-
RVMPI	7.3	3.3	<0.05	16.6
TAPSE	-1.4	0.5	<0.05	0.2
RVS′	0.03	0.2	>0.05	1.02

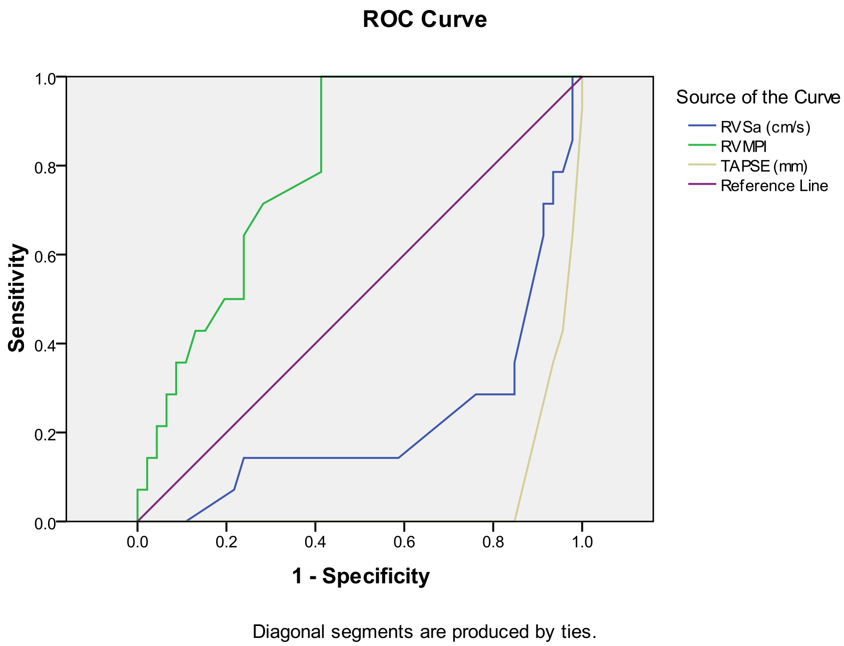

Figure 1: ROC curve analysis for prediction of RV dysfunction.

RVMPI at 0.54: Sensitivity = 100%, Specificity = 52%.

TAPSE at 17: Sensitivity = 38%, Specificity = 84%.

RVS′ at 9.5: Sensitivity = 29%, Specificity = 82%.

Discussion

RV function is related to LV function by the phenomenon of ventricular independence. Although the right ventricle pumps the same stroke volume as the left ventricle, it does so with 25% of the stroke work due to low resistance of the pulmonary circulation which is the reason of the thinner walls and more compliance of the pulmonary circulation according to Laplace’s law which explain the right ventricular function impairment in patient with left ventricular dysfunction found in the present study [9].

According to Kukulski et al., it is difficult to make an evaluation of the right side of the heart using methods that depend on the visual assessment due to geometric complexity that could not be explained by mathematic models as its shape changes during the cardiac cycle [10]. Therefore, many studies were conducted to find an accepted method to assess right side function.

In the present study, the mean age of study population was 57±11 years, 50% of patients were ≥60 years old. Males constituted the majority of patients with male to female ratio of 1.8:1. These results are in concordance with those of other studies [11] and can be explained by the prevalence of ischemic heart disease as an underlying cause of heart failure in these studies including this study and the risk for development of ischemic heart diseases increases with increasing age and in male sex [11].

Hypertension (HT) was present in 53.3% of patients and diabetes mellitus (DM) was present in 46.7% of patients. It is well known that HT and DM are independent risk factors for the development of ischemic heart disease which is in turn the most common etiology for heart failure [12].

LVIDd and LVIDs were increased in most patients in this study. This is expected as left ventricular remodelling including progressive dilatation occurs well before the onset of systolic dysfunction and symptomatic heart failure [13].

The majority of patients in this study had mild-moderate LV systolic dysfunction. This may be due to early presentation of patients and early treatment and also because patients were recruited from cardiology outpatients and not from inpatient medical services.

Impaired RV myocardial performance index (RVMPI) was the most commonly affected parameter of RV function (63.3%) much more than abnormalities in TAPSE, RVS' and increased RV dimensions.

Using Binary logistic regression analysis and ROC analysis, it was found that RVMPI was the most significant predictors of RV dysfunction followed by TAPSE and lastly RVS.' These results agree with Vizzardi et al. [6]. It is well known that TAPSE is evaluating the systolic function and it is widely used in RV systolic assessment but the results of the current study showed that in spite of the decrease of TAPSE in LV systolic dysfunction, there was a lower sensitivity but higher specificity as compared with RV MPI which was also in harmony with the findings of Vizzardi et al. [6].

Miller and his co-workers in 2004 compared the RV EF with the TAPSE using radionuclide angiography and showed a significant correlation [14]. For years TAPSE simplicity and reproducibility made it an attractive tool in clinical practice and has been considered as a rough estimate of RV longitudinal systolic function but as it is strongly influenced by overall heart motion and load condition it may fail to reflect RV function accurately [6] and the current study is supporting this hypothesis.

In this study, the majority of patients had normal or mildly elevated pulmonary artery systolic pressure (PASP) (91.7%) of patients, while 27.3% of patients had elevated pulmonary vascular resistance (PVR).

The present study showed that the pulmonary artery systolic pressure (PASP) is not significantly associated with left ventricular systolic dysfunction. This is in agreement with Enriquez-Sarano et al. [15] who found that the pulmonary hypertension is highly variable in patients with left ventricular dysfunction with independency to the degree of left ventricular systolic dysfunction unlike the LV diastolic dysfunction in which pulmonary HT is more commonly elevated than in LV systolic dysfunction.

In this study, there was no significant association between pulmonary vascular resistance and left ventricular systolic dysfunction but it was highly associated with the severity of pulmonary artery systolic pressure. This was also shown by Mayet and Hughes who stated that the degree of change in vessel structure and pulmonary vascular resistance in response to venous hypertension varies widely among patients of HF [16].

Both "passive" and "reactive" secondary PH might be observed in patients of CHF due to LV systolic dysfunction, while pulmonary venous hypertension and post-capillary PH is due to an increase in LV filling pressures. Fortunately, the passive component of PH is reversible when LV filling pressure is improved [10]. However, functional and structural changes in the pulmonary vascular organization are expected after a sustained and excessive exposure to high pulmonary venous pressure ubiquitous among patients with advanced HF that initially damages the capillaries and later the damage is observed in the arterioles and arteries [17]. Endothelial dysfunction is the key factor for the functional alterations of the pulmonary vasculature through an impaired pulmonary vascular smooth muscle relaxation. The local control of vascular tone in the pulmonary vasculature is endothelium-mediated which is briefly based on a balanced release of nitric oxide and endothelin (ET), and pulmonary vascular resistance (PVR) is very sensitive to any changes or disturbance of the balance of these two opposing systems [18].

In this study a significant association was found between right ventricular dilatation and right ventricular dysfunction which is in agreement with Haddad et al who showed that right ventricle adapted better to volume overload than to pressure overload [16]. In contrast to volume overload, moderate to severe acquired pulmonary hypertension often leads to right ventricular dilatation and failure. In addition; pressure overload also might lead to right ventricular ischemia which may further aggravate ventricular dysfunction [19].

The RV afterload is affected by changes in the pulmonary vasculature and the associated increase of its impedance which activate complex adaptive mechanisms. Hypertrophy is the initial response of the right ventricle, which however, decreases RV sub-endocardial perfusion. Eventually, the progressive dilatation will increase wall stress and leads to functional tricuspid regurgitation due to altered geometry of the tricuspid annulus. It also shifts the septum towards the left side leading to compression of the LV and impairment of the global ventricular function through ventricular interdependence [8].

The left ventricular dimensions were not significantly associated with RV dysfunction. Also, pulmonary artery systolic pressure and pulmonary vascular resistance were not significantly associated with RV function. On the other hand, there was a significantly lower mean of LVEF among patients with RV dysfunction [20].

Conclusions

The right ventricular function is affected in patients with left ventricular dysfunction.

To study right ventricular function; right ventricular myocardial performance index which measure both systolic and diastolic functions is more sensitive than TAPSE which is a measure of only the systolic function of right ventricle.

Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Clinical Outcomes of Intracoronary (IC) Tirofiban Administration in Patients with Large Anterior STEMI Undergoing Primary PCI

Ahmed Basuoni[1,*], Wael El-Naggar[1,2], Mohamed El-Mahdy[1,2], and Sameh El-Kaffas[1,2]
[1]Cardiology Department, Dar Al-Fouad Hospital, Cairo, Egypt
[2]Cardiology Department, Faculty of Medicine, Cairo University, Cairo, Egypt

Abstract

Background/Aim. Thrombus embolization during percutaneous coronary intervention (PCI) in STEMI results in sub-optimal myocardial perfusion and increased infarct size. This study aimed to evaluate the clinical outcomes of intra-coronary (IC) delivery of bolus Tirofiban following aspiration thrombectomy in patients with large anterior STEMI undergoing primary PCI.
Patients and Methods. A prospective single-blinded randomized controlled trial of 100 patients with large anterior STEMI was screened at 2 sites in one country (Egypt). Aspiration thrombectomy was performed in all patients using a 6F-aspiration catheter. Patients were randomized to IC Tirofiban (Study group) and no IC Tirofiban (control group). To ensure high intra-thrombus drug concentrations, Tirofiban was administered locally at the site of the infarct lesion via the aspiration catheter after flushing of the aspiration catheter well.
Results. There is no significant difference in major adverse cardiac and cerebrovascular events (MACCE) at 90 days between patients receiving or not receiving bolus IC Tirofiban ($P = 0.723$). Bleeding risk (TIMI major and minor) and thrombocytopenia showed no difference between the 2 groups.
Conclusions. In patients with large anterior STEMI presenting early after symptom onset and undergoing primary PCI, there is no significant difference in clinical outcomes between patients receiving or not receiving intracoronary Tirofiban.

Keywords: MACCE; Tirofiban; aspiration thrombectomy

Introduction

Infarct size is one of the main predictors of mortality after myocardial infarction [1]. Thrombus embolization results in ineffective microvascular perfusion and myocardial recovery despite of TIMI 3 flow after primary PCI [2]. Different parameters can measure infarct size, including biomarkers such as

* Correspondence: ahmedmos179@gmail.com

Creatinine Phosphokinase (CK), Lactate Dehydrogenase (LDH), and Troponin (Tpn). While these tests are promptly available and can be obtained in an acute setting, they are unreliable in comparison to infarct size assessed by cardiac imaging [3]. Furthermore, myocardial Blush grade is an independent predictor for outcome in acute myocardial infarction patients after primary PCI [4]. For measuring the infarct size in humans, the reference investigation is cardiac magnetic resonance (cMR) [1]. Although all mentioned methods can determine the effect of intracoronary Tirofiban, previous studies showed conflicting data about whether (IC) glycoprotein IIb/IIIa or manual aspiration can reduce infarct size or improve clinical outcome [5, 6]. No such data are available among Egyptians and other Asian populations. In view of the above reports, we performed this pilot study to evaluate the clinical outcomes of intracoronary delivery of bolus Tirofiban following aspiration thrombectomy in patients with large anterior STEMI undergoing primary PCI.

Patients and Methods

A prospective randomized controlled study of patients with large anterior STEMI was undertaken. Patients were assigned either to the Study group (n = 50) receiving intra-coronary Tirofiban (IC) or to the untreated Control group (n = 50). Randomization was carried out using computer-generated random numbers. All patients included in the study had presented with acute coronary syndrome (STEMI) and all patients were admitted through emergency room in CCU. Of the 118 patients originally entered in the study, 18 were excluded, as shown in Figure 1.

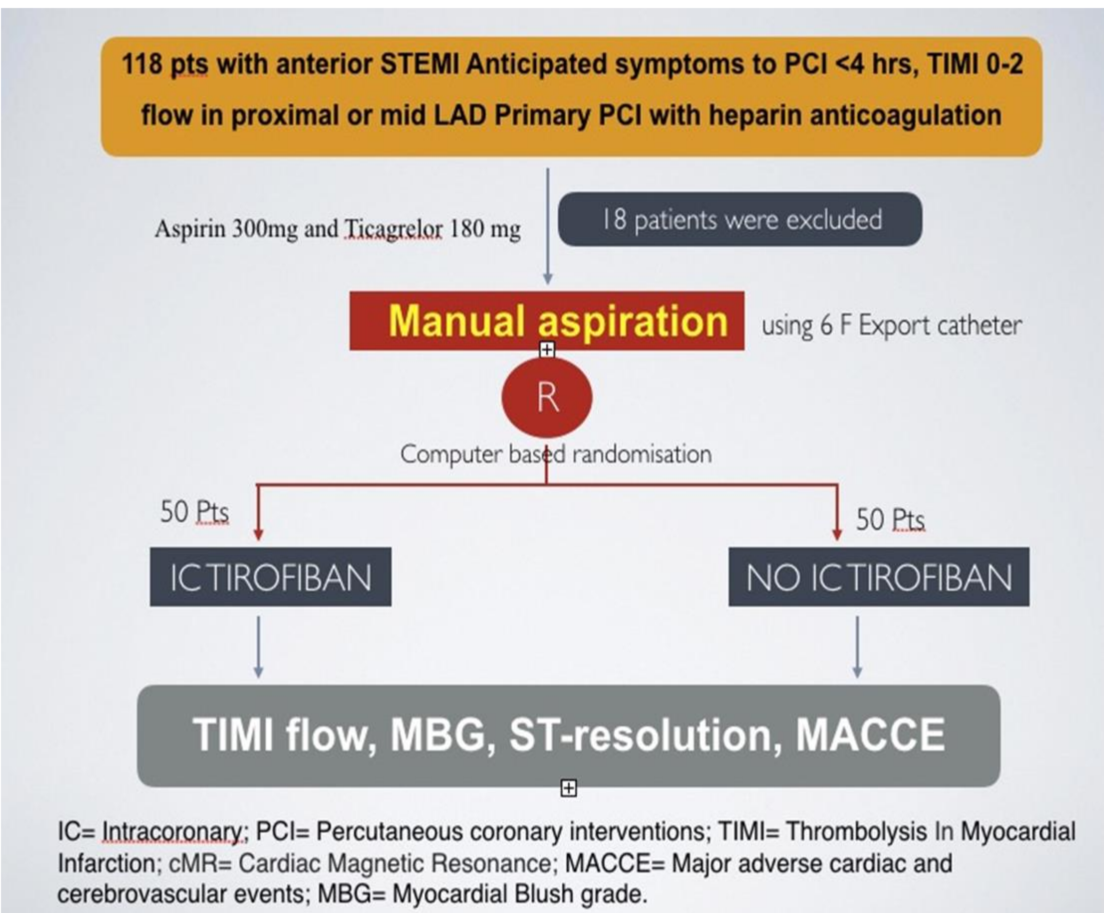

Figure 1.

Inclusion Criteria

The trial was approved by each participating center; all patients included in the study signed a written consent. Patients between 18 and 80 years of both genders with symptoms of myocardial infarction (ECG showing 1-mm or greater ST-segment elevation in 2 or more contiguous leads in V1-V4, or new left bundle-branch block, with anticipated symptom-onset-to-device time of 6 hours or less) were eligible to be included in the trial. The coronary anatomy of all patients included should have proximal or mid-grade III or VI thrombus in left anterior descending artery (LAD) of TIMI 0/1/2.

Exclusion Criteria

The exclusion criteria included prior myocardial infarction, prior systolic dysfunction (ejection fraction <35%) (n = 3), prior Coronary Artery Bypass Graft (CABG) (n = 2), and previously inserted stents in LAD (n = 4). Patients who had severe calcification, diffuse disease and severe vessel tortuosity (n = 5) were excluded, as were patients with contraindication to cardiac magnetic resonance (cMR) (n = 4).

Aspirin (300 mg) and Ticagrelor (180 mg) were given upon admission before the coronary angiography. The anticoagulant of choice was IV un-fractionated heparin guided by anti-coagulation time (ACT). A guiding catheter was placed; then a PTCA wire was placed distal to the culprit lesion. A 6F-thrombus aspiration catheter was used in all patients. By the same catheter (export catheter), after flushing it very well and to ensure high intra-thrombus drug concentrations, a 25 mcq/kg intra-coronary bolus of Tirofiban was injected locally at the place of the high thrombus burden through the bore of aspiration catheter. Percutaneous coronary intervention was performed by using standard techniques, with drug-eluting stents implantation. Assessment of final TIMI flow and myocardial blush were carried out by a blinded observer. Clinical follow-ups were scheduled for all patients at 90 days.

End Points and Definitions

Death, reinfarction, stroke, heart failure or target vessel revascularization (TVR) were defined as major adverse cardiac and cerebrovascular events (MACCE). Bleeding risks were calculated by using TIMI minor and major risk scores.

Statistical Analysis

Statistical analysis used the SPSS software. All continuous variables were tested for normality. Data were expressed as percentage for discrete variables and mean values ± SD for normally distributed and medians [(with interquartile range (P25-P75)] for skewed data. Numerical differences between the two study groups were tested using non-parametric Mann-Whitney tests in skewed variables and Student’s t test for normally distributed variables. Chi square and Fisher’s exact tests were used for comparison between categorical variables. All p-values were calculated using two-sided tests. Differences were considered statistically significant at $p < 0.05$.

Results

The baseline characteristics of randomized groups were all matched (Table 1). Procedural data for the patients assigned to the study group who received intracoronary Tirofiban vs. the control group are shown in Table 2. Thrombus aspiration was performed in all patients. Discharge medications included aspirin, ticagrelor, statins, beta blockers and ACEIs in 100% of patients with no difference between the 2 groups.

Myocardial Perfusion and ST-Segment Resolution

Post-PCI Thrombolysis In Myocardial Infarction (TIMI) 3 flow, an myocardial blush grade (MBG) of 2 or 3, and complete ST-segment resolution (STR) at 60 minutes were achieved in 100% of patients.

Table 1. Baseline characteristics of the randomized groups

Variables	Aspiration + IC Tirofiban n = 50	Aspiration + no IC Tirofiban n = 50	p-value
Age	52.2 ± 6.9	47.32 ± 7.4	0.02
Sex	38 (76%)	42 (84%)	0.48
Smoking	36 (72%)	34 (68%)	0.76
Diabetes Mellitus	22 (44%)	20 (40%)	0.77
Hypertension	8 (16%)	16 (32%)	0.19
Dyslipidemia	6 (12%)	6 (12%)	1
FH of CAD	4 (16%)	4 (16%)	1
Obesity	10 (20%)	4 (8%)	0.42
CKD	0	0	
LEAD	0	0	
Kilip class II	6 (12%)	6 (12%)	1
Prior MI or CABG	0	0	
Anterior MI	50 (100%)	50 (100%)	
Pain to door (hours)	median 4, IQR 2	median 4, IQR 2	0.99*
Door to balloon (minutes	median 30, IQR 30	median 30, IQR 15	0.75*

Data provided as mean ± standard deviation, median [interquartile range], or number [%].

IC = Intracoronary; FH = Family history; CAD = Coronary artery disease; LEAD = Lower extremities arterial disease; MI = Myocardial infarction; CABG = Coronary artery bypass graft.

Table 2. Procedural data for the patients

Variables	Aspiration + IC Tirofiban n = 50	Aspiration + no IC Tirofiban n = 50	p-value
Proximal LAD artery	30 (60%)	28 (56%)	0.77
Mid LAD artery	20 (41.7%)	22 (45.8%)	0.77
Thrombus grade III	8 (16%)	4 (8%)	0.42
Thrombus grade IV	0	0	
Thrombus grade V	40 (80%)	46 (92%)	
Drug Eluting Stents	50 (100%)	50 (100%)	1
Number of stents	median 1, IQR 0	median 1, IQR 0	0.5*
stent length >30 mm	10 (41.7%)	7 (29.2%)	0.37
Radial approach	30 (60%)	29 (58%)	

Data provided as mean ± standard deviation, median [interquartile range], or number [%].

IC = Intracoronary; LAD = Left anterior descending.

Table 3. Myocardial perfusion and infarct size assessment

Variables	Aspiration + IC Tirofiban n = 50	Aspiration + no IC Tirofiban n = 50	p-value
Cardiac enzymes CKMB (TIME TO PEAK)	median 13.5, IQR 7	median 12, IQR 8	0.58*
ECG st. segment resolution post PCI	50 (100%)	50 (100%)	
ECHO (EF%)	median 46, IQR 13	median 40.5, IQR 16	0.13*
TIMI FLOW 3	44 (88%)	46 (92%)	1
Myocardial Blush grade 2/3	42 (84%)	46 (92%)	0.67

Data provided as mean ± standard deviation, median [interquartile range], or number [%].

IC = Intracoronary; CKMB = Creatine kinase-muscle/brain; PCI = Percutaneous coronary interventions; EF = Ejection fraction; TIMI = Thrombolysis In Myocardial Infarction; cMR = Cardiac Magnetic Resonance

No significant differences in these measures were present between patients randomized to intracoronary tirofiban vs. no IC tirofiban (Table 3).

Clinical Outcomes

Patients randomized to intracoronary Tirofiban compared with no intracoronary Tirofiban had no significant reduction in MACCE results at 90 days (P = 0.723), Table 4. Bleeding risk (TIMI major and minor) and thrombocytopenia showed no difference between the 2 groups (Table 4).

Table 4. Clinical outcomes

Variables	Aspiration + IC Tirofiban n=50	Aspiration + no IC Tirofiban n=50	p-value
MACCE	4 (8 %)	6 (12 %)	0.723
Heart failure	3	5	
Stroke	0	0	
Reinfarction	1	1	
Death	0	0	
Bleeding			
TIMI major	0	0	
TIMI minor	6 (12 %)	4 (8 %)	0.48
Thrombocytopenia	0	0	

Data provided as mean ± standard deviation, median [interquartile range], or number [%].
IC = Intracoronary; MACCE = Major adverse cardiac and cerebrovascular events; TIMI = Thrombolysis In Myocardial Infarction.

Discussion

Two of solid infarct size determinatives are the location of myocardial infarction and the abnormal TIMI flow [7].

The main findings from this study are as follows: (A) STR, MBG, TIMI flow and 90-day clinical outcomes showed no significant difference; (B) Bleeding risk (TIMI major and minor) and thrombocytopenia showed no significant difference between the 2 groups.

Therefore, we enrolled patients with proximal lesions in LAD, grade 3 or 4 thrombus burden (without previous myocardial infarction) and TIMI 0-2 flow. Furthermore, we enrolled patients who came early, in the effective time window [8]. The average time from the beginning of symptoms to doing the ECG was 4 hours and average door-to-needle time was 30 minutes. Thus, the study population expressed a markedly selected group of patients with the highest clinical need who came to the hospital in their first hours when early intervention was achievable to decrease the infarct size.

Myocardial reperfusion was assessed by several complementary parameters, including post-PCI TIMI flow, MBG, and STR [9]. Despite using heparin as the procedural anticoagulant in addition to intracoronary Tirofiban, there was no significant difference in TIMI bleeding risks. These results must be settled in relation to prior studies. Two earlier studies showed infarct size reduction with intracoronary glycoprotein IIb/IIIa antagonists compared with intravenous glycoprotein IIb/IIIa antagonists (despite recruitment of patients with non-anterior MI presenting up to 12 hours after symptoms) [10, 11]. A meta-analysis of 6 studies (1,246 patients) described improving survival with intracoronary abciximab [12]. However, the AIDA STEMI trial (2,065 randomized patients) showed closely similar rates of MACE (and biomarker-assessed infarct size) with bolus intracoronary and intravenous abciximab. In contrast to INFUSE, AMI trial which used bivilurdin as the procedural anticoagulant [13], we used heparin in our trial as many trials have advocated that infarct size might be decreased by adding both intravenous glycoprotein IIb/IIIa antagonist and heparin [14, 15]. However, in addition to recruiting patients with anterior STEMI presenting early, all prior trials (including AIDA-STEMI and except INFUSE AMI trial) inject intracoronary glycoprotein IIb/IIIa antagonist proximally through the guiding catheter, restricting its invasion into the occlusive thrombus and favoring drug flow to lower resistance pathways (such as the left circumflex artery) and return to the aorta. By contrast, the local drug injection through the aspiration device used in our present study precisely attains high intra-thrombus concentrations of glycoprotein IIb/IIIa antagonist at the site of LAD occlusion, which may inhibit platelet aggregation and thrombus formation [16, 17].

Regarding thrombus aspiration, in the TAPAS trial, 1,071 patients (with different types of STEMI) were randomized to a group using thrombus aspiration vs. no aspiration during primary PCI; using thrombus aspiration showed moderate enhancement in MBG and STR but a great improvement in 1-year mortality [6, 1]. Other trials of manual thrombus aspiration showed opposing results [19, 20]. And contrary to these trials, multi-center large trials (TASTE 2013 and TOTAL 2015) have largely been negative [21]. Moreover, in TAPAS, aspiration did not reduce infarct size as measured by cardiac biomarkers [6], calling into question the mechanism underlying the survival benefit. The fact that manual thrombus aspiration did not reduce infarct size in our study makes a substantial clinical benefit unlikely, questioning its routine use in STEMI.

Although infarct size was decreased in the intracoronary tirofiban group, early indices of microcirculatory reperfusion (MBG and STR) did not differ significantly. This discordance may show that different ascertainment times give infarct progress over 30 days (especially as edema is substantially improved during this period) and certainty of different biomarkers. However, the comparable 90-day clinical outcomes between groups are consistent with MBG and STR results [22].

Limitations

The current study has various limitations. First, the physician who did the operation knew the randomization assignment which made the trial single-blinded. Thus, while some bias cannot be excluded, the patient and follow-up personnel did not know the treatments supplied, and the study used multiple core laboratories and clinical outcomes committee blinded to treatment assignment. Second, using an aspiration device for local drug delivery may have a risk of distal embolization but there was no information about its hazardous before. Third, manual aspiration catheters with a larger inner diameter are now obtainable. However, trials have not greatly shown improvement in thrombus retrieval or myocardial perfusion with larger inner-diameter catheters. Fourth, further trials are required to determine whether the great infarct size diminution at 30 days resulted from aspiration thrombectomy followed by intracoronary tirofiban in the current study that translates improvement in late clinical outcomes without increase in bleeding risks. Finally, the trial was conducted with a small number of patients which need to be supported with further large trials.

Conclusion

In patients presented early with a large anterior STEMI, clinical outcomes at 90 days were not significantly reduced by bolus intracoronary Tirofiban delivered to the infarct lesion site following thrombus aspiration. Large trials may be necessary to verify these findings.

Ethical Compliance

The authors have stated all possible conflicts of interest within this work. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.and written consent.

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In Silico Analysis on mRNA, lncRNA, and miRNA Expression Profiles in Ischemic Cardiomyopathy Left Ventricular Tissues and Differentiated Cardiomyocytes

Akram Gholipour[1], Ali Sharifi-Zarchi[2], Farshad Shakerian[3,4], Ali zahedmehr[4], Mehrdad Oveisi[3,5], Maedeh Arabian[3], Majid Maleki[3], Elham Taheri Bajgan[6], Seyed Javad Mowla[6], and Mahshid Malakootian[3,*]

[1]Department of Biology, Science and Research Branch, Islamic Azad University, Tehran, Iran
[2]Department of Computer Engineering, Sharif University of Technology, Tehran, Iran
[3]Cardiogenetic Research Center, Rajaie Cardiovascular Medical and Research Center, Iran University of Medical Sciences, Tehran, Iran
[4]Cardiovascular Intervention Research Center, Rajaie Cardiovascular Medical and Research Center, Iran University of Medical Sciences, Tehran, Iran
[5] Department of Computer Sciences, The University of British Columbia, Canada
[6]Department of Molecular Genetics, Faculty of Biological Sciences, Tarbiat Modares University, Tehran, Iran

Abstract

Ischemic cardiomyopathy (ICM) is a common form of myocardial injury. For all investigations on molecular diagnostic tests for ICM, a set of biomarkers is missing. This study aimed to identify efficient diagnostic biomarkers and also subscriptions of the expressions of mRNAs, lncRNAs and miRNAs through an integrated analysis of the expression profile. Differentially expressed mRNAs, lncRNAs, and miRNAs were performed between the ischemic tissues and normal samples. Moreover, the percentage of the transcriptome similarity in the expression profiling matrices between the normal samples and the differentiated cardiomyocytes as in vivo and in vitro models was analyzed. Pathway enrichment and molecular function analyses of those models that shared mRNAs were accomplished. The data showed that miR-5683 and the *COMP* gene were upregulated whereas TCNOS_00014011, an lncRNA, was downregulated in the ischemic cardiomyopathy tissues. Accordingly, they can be introduced as ICM biomarkers owing to their highest differential expression. Furthermore, 89% of the lncRNAs, 55.2% of the mRNAs, and 38.6% of the miRNAs were shared between the non-failing heart tissues and the differentiated cardiomyocytes. Furthermore, the shared mRNAs were involved in adrenergic signaling in cardiomyocytes, glycolysis/gluconeogenesis, calcium signaling pathway, PI3K/Akt pathway, MAPK signaling pathway, TGF-beta signaling pathway, Wnt signaling pathway, and BMP receptor binding. In conclusion, this study provides foundations for new ICM biomarkers and suggests an important role for lncRNAs among in vivo and in vitro models that would shed more light on molecular processes in ICM tissues.

Keywords: Ischemic cardiomyopathy; differentially expressed mRNAs; lncRNAs and miRNAs; cardiomyocytes

* Corresponding Author's Email: Malakootian@rhc.ac.ir

Introduction

Ischemic heart disease (IHD) continues to be the leading cause of death worldwide [1]. Ischemic cardiomyopathy (ICM) is defined as left ventricular (LV) systolic dysfunction due to a history of prior myocardial revascularization and/or myocardial infarction [2]. The heart is a very structured tissue that contains ventricular or atrial cardiomyocytes, pacemaker cells, Purkinje cells, vasculature, and connective tissue. Between 2 and 4 billion ventricular cardiomyocytes compose the human LV, and ventricular cardiomyocytes account for about 50% of heart weight. Additionally, myocardial infarctions are associated with LV systolic dysfunction in nearly 40% of cases. Most recent evidence-based therapies for ICM focus on LV remodeling [3, 4]. The damage to the cardiac tissue caused by ischemia sets in motion the creation of metabolic waste products, the failure in the maintenance of cell membranes, and the release of proteolytic enzymes into the cell and its surrounding tissues [5].

Genome-wide analyses of the eukaryotic transcriptome have shown that most parts of the human genome are transcribed, only 2% of the human genome is encoded for protein synthesis, and the bulk of the transcripts remain as noncoding RNAs (ncRNAs) [6]. Currently, a wide variety of ncRNAs have been introduced that assume an equally broad range of regulatory functions. MicroRNAs (miRNAs), with a length of 18–22 nucleotides, constitute a group of ncRNAs—including miR-1 and miR-133—that play essential roles in many physiological and pathophysiological aspects of the heart [7]. Long noncoding RNAs (lncRNAs), a class of ncRNAs longer than 200 nucleotides that regulate many physiological and pathological processes, are still largely unknown despite evidence on their role in heart diseases. It has been previously shown that lncRNAs regulate gene expression by interfering in the splicing, transcription, and translation of messenger RNAs (mRNAs) [8, 9].

Gene expression profiling is a powerful method with diagnostic and prognostic accuracy for studying biological processes, particularly in tissues and organs at the molecular level [10]. Since the heart is an organ with a limited capacity to regenerate and repair [11], it is of interest to determine the expression profile of its physiological and pathophysiological states. In the present study, we utilized bioinformatics and comprehensive analyses to examine the expressions of mRNAs, lncRNAs, and miRNAs in non-failing and ischemic samples of the heart with a view to introducing novel ICM biomarkers. Moreover, we aimed to determine the similarity of the transcriptome (expression profiles of mRNAs, lncRNAs, and miRNAs) between non-failing tissues (LV) and cardiomyocytes as in vivo and in vitro models, respectively.

Materials and methods

Data Sets and Samples

All the analyzed data sets in the present study were downloaded from the NCBI Gene Expression Omnibus (GEO). ICM and non-failing samples of the LV apex tissue were obtained from the GSE46224 Data Set. These data comprised a complete transcriptome profiling in failing and non-failing human hearts [12]. Samples of differentiated cardiomyocytes were obtained from the GSE76523 Data Set (for RNA-Seq data) [13], which contains information on cardiomyocyte differentiation from human embryonic stem cells (H7). The GSE108021 Data Set was used to profile miRNA expression. These data relate to human embryonic stem cells differentiated toward mesoderm lineage and cardiomyocytes (for miRNA-Seq data) [14].

Bioinformatics Analysis

The downloaded data sets were analyzed via the removal of the adapter sequences using Trimmomatic, version 0.36, and the alignment of the sequence reads to the human genome hg38 using HISAT2, version 2.1.0. NONCODE, version 5.0 (hg38), was applied to analyze the expression pattern of the lncRNAs; and RefSeq, obtained from the UCSC databases, was chosen as annotation references for mRNA analyses. The read counts of each transcript were analyzed by HTSeq, version 0.9.1. Additionally, miARma, version 1.7.2, was utilized to check the miRNAs.

Identification of Differentially Expressed mRNAs, lncRNAs, and miRNAs

Eight ICM samples and 8 non-failing heart samples were scrutinized to identify differentially expressed mRNAs, lncRNAs, and miRNAs. With the use of the DESeq2 Package in the R program with log2FoldChanges ≠ 1 and adjusted *P* values < 0.05, mRNAs, lncRNAs, and miRNAs with statistically significant differential expressions were identified. (The *P*-values were adjusted for multiple comparisons).

The expression levels of the candidate mRNAs, lncRNAs, and miRNAs were also calculated in the samples by analyzing the expression profiles of each sample in the R program. The expression graphs of the mRNAs, lncRNAs, and miRNAs were depicted by GraphPad Prism, version 7. With the aid of miRWalk, version 2.0, the targets of the candidate miRNA were determined and the expression levels of the targets were analyzed as well. One-way ANOVA was employed to examine the significance of the observed miRNA expression and its targets in ICM. Adjusted *P*-values < 0.05 were considered to indicate statistical significance.

Studying the similarity in the expression profile of the non-failing heart samples and the cardiomyocyte samples as in vivo and in vitro models, respectively.

The similarity percentage in the expression profiling matrix between the non-failing and the cardiomyocyte samples was determined using Venn diagrams (Venny 2.1). The non-failing heart tissues were considered the in vivo pattern, and the differentiated cardiomyocytes were considered the in vitro pattern.

Molecular Function Analysis of the Similar mRNAs between the Non-Failing Heart Samples and Cardiomyocytes

The Enrichr-Database was utilized for molecular function and KEGG (Kyoto Encyclopedia of Genes and Genomes) analyses to determine the important biological pathways probably shared between the mRNAs.

Results

Differential Expressions of the Transcriptome (miRNAs, mRNAs, and lncRNAs) in the Failing and Non-Failing Samples

The expression profiles of the miRNAs, mRNAs, and lncRNAs were determined using the DESeq2 Package in the R program. The differentiated expressions of the transcriptome were determined by log2FoldChanges ≠ 1 and adjusted *P*-values < 0.05 between the ICM and non-failing heart samples. In this study, log2FoldChanges ≥ 4 were considered to be the highest upregulation and log2FoldChanges ≤ -4 were regarded as the highest downregulation.

Our result showed that 5 miRNAs (miR-125b-1-3p, miR-708-5p, miR-141-3p, miR-34c-5p, and miR-5683) were upregulated and 4 miRNAs (miR-221-3p, miR-222-5p, miR-483-3p, and miR-543) were downregulated in the ischemic samples by comparison with the non-failing ones. Among these miRNAs in the ischemic tissues, miR-5683 demonstrated the highest upregulation with a log2FoldChange = 4.09. The targets of this miRNA were predicted using the miRWalk Database (identified by scores ≥ 3) (Supplementary Table S1). The results demonstrated that miR-5683 might be able to downregulate bone morphogenetic protein 7 (*BMP7*, NM_001719), FK506 binding protein 5 (*FKBP5*, NM_001145777), periplakin (*PPL*, NM_014692), and especially ribonuclease A family member 2 (*RNASE2*, NM_002934) genes (adjusted *P*-values = 0.0016) via interaction with their 3´-UTRs. Thereafter, the expressions of the candidate genes were analyzed by reviewing the expression profile of each sample in the R program. Interestingly, the results exhibited a reciprocal expression pattern between this miRNA and the candidate targets. All candidate genes exhibited downregulation in the ischemic samples (Figure 1*a*). The analysis showed differential expressions in 294 genes between the failing and non-failing heart samples. Of this total, 101 genes were downregulated and 193 genes were upregulated (Supplementary Table S2). Moreover, natriuretic peptide C (*NPPC*, NM_024409) and cystic fibrosis transmembrane conductance regulator (*CFTR*, NM_000492) had the highest downregulation (log2FoldChanges ≤ - 4) and secreted frizzled-related protein 4 (*SFRP4*, NM_003014), cartilage oligomeric matrix protein (*COMP*, NM_000095), hemoglobin subunit alpha 1 (*HBA1*, NM_000558), hemoglobin subunit beta (*HBB*, NM_000518), proenkephalin (*PENK*, NM_001135690), natriuretic peptide B (*NPPB*, NM_002521), hemoglobin subunit alpha 2 (*HBA2*, NM_000517), family with sequence similarity 180 member B (FAM180B, NM_001164 379), secreted frizzled-related protein 2 (*SFRP2*, NM_003013), phosphoenolpyruvate carboxykinase 1 (*PCK1*, NM_002591), and tachykinin receptor 1 (*TACR1*, NM_001058) had the highest upregulation (log2FoldChanges ≥ 4) in the ICM samples. The *COMP* gene with a log2FoldChange = 6.28 was selected as the gene with maximum differential expression in the failing samples (Figure 1b).

In addition, the lncRNAs with differential expressions among the ischemic and non-failing heart samples were studied. The results showed that 393 lncRNAs had differential expressions: 228 lncRNAs were upregulated and 165 lncRNAs were downregulated (Supplementary Table S3). These lncRNAs were predicted by NONCODE, version 5.0. The highest downregulation was observed in NONHSAG034276.2, NONHSAG049216.2, NONHS AG091293.1, NONHSAG002747.2, and NONHSAG 044173.2 (log2FoldChanges ≤ -4) and the highest upregulation was detected in NONHSAG047367.2, NONHSAG076427.1, NONHSAG018159.2, NONHS AG007503.2, NONHSAG057509.1, NONHSAG05 0281.2, NONHSAG017590.2, NONHSAG079370.1, and NONHSAG053940.3 (log2FoldChanges ≥ 4) (Figure 1*c*). Among them, NONHSAG049216.2 (TCNOS_00014011) lncRNA was downregulated among the ICM samples with a log2FoldChange = -4.69. This predicted an lncRNA located on chromosome 7 on the minus DNA strand with a length of 819 bp and categorized in a class of long intervening/intergenic ncRNAs (lincRNAs).

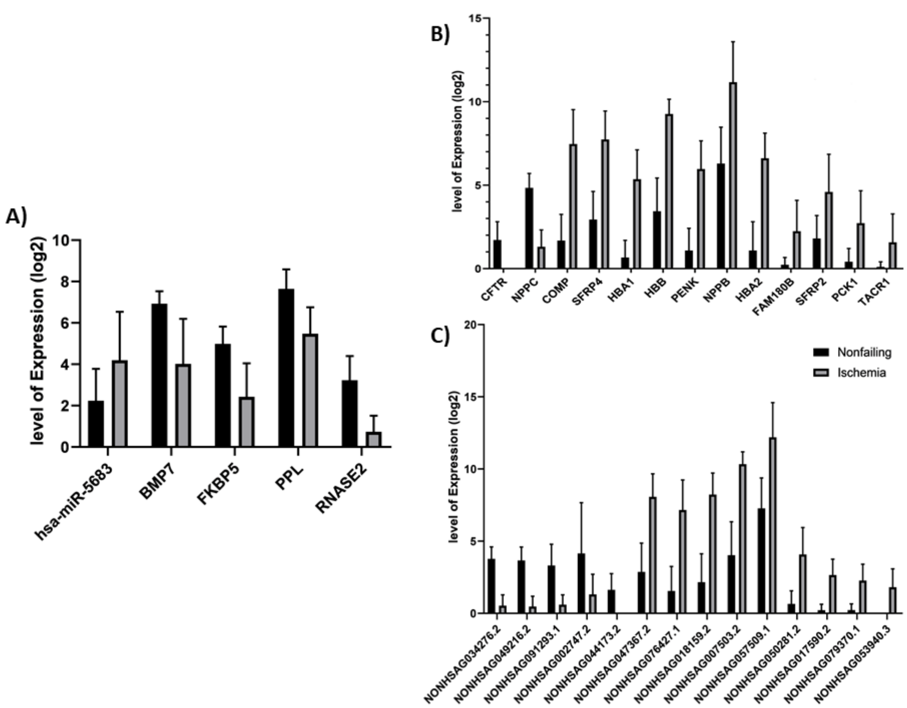

Figure 1. Expression levels of the highest upregulation and the highest downregulation of miRNAs, mRNAs, and lncRNAs a) Expression of miR-5683 and its target genes between the ischemic and non-failing tissues, indicating that miR-5683 might be able to downregulate the RNASE2 gene (adjusted *P*-value = 0.0016); b) Expression of the COMP gene and also the other up/downregulated genes among the ischemic and non-failing heart tissues; c) Expression of NONHSAG049216.2 lncRNA and the other up/downregulated genes between the ischemic and non-failing heart tissues.

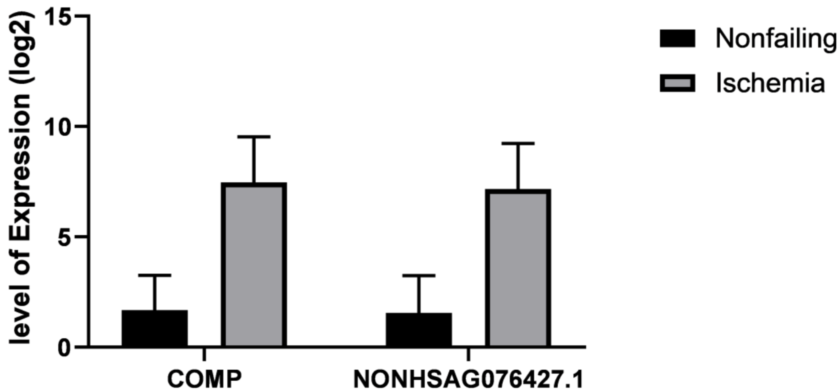

Figure 2. Level of the COMP gene expression and NONHSAG076427.1, predicting the expression of lncRNAs in the ischemic and non-failing heart tissues.

Another candidate lncRNA—NONHSAG076 427.1-upregulated in the ischemic tissues with a log2FoldChange = 6.03, was mapped to the 8–13 exons of the *COMP* gene. The upregulation of this lncRNA might have been in consequence of the elevated expression of the *COMP* gene in the ICM samples (Figure 2).

Taken together, by studying transcriptome expression profiling between ICM and non-failing tissues, we introduced the upregulation of miR-5683 due to a rise in the expression of the *COMP* gene and the downregulation of TCNOS 00014011 lncRNA as biomarkers for ischemic tissues.

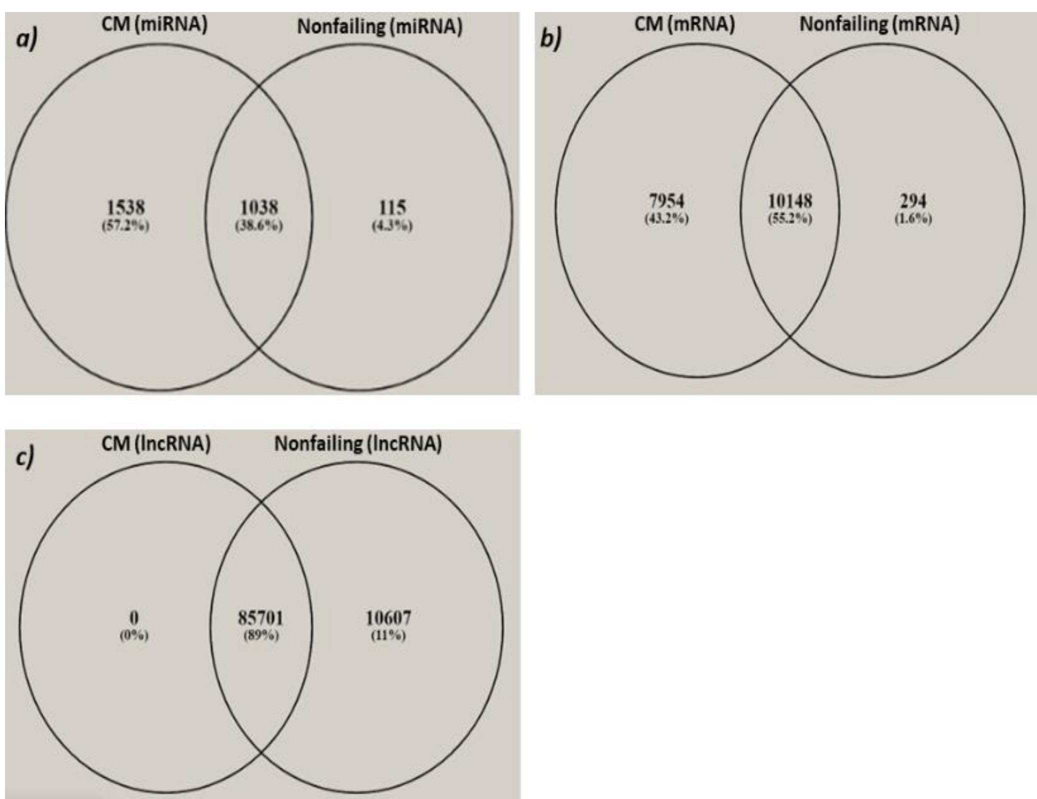

Figure 3. Venn diagram of the similarity in the miRNAs, RNAs, and lncRNAs between the non-failing and differentiated cardiomyocyte (CM). a) Similarity percentage of the miRNAs: 38.6%. b) Similarity percentage of the mRNAs: 55.2%; c) Similarity percentage of the lncRNAs: 89%.

In Vivo and In Vitro Analysis Results

The expression profiles of the normal samples and the differentiated cardiomyocytes as in vivo and in vitro models, correspondingly, were examined by Venn diagrams. The results showed that 89% of the lncRNAs, 55.2% of the mRNAs, and 38.6% of the miRNAs were shared between the non-failing heart tissues and the differentiated cardiomyocytes (Figure 3*a*-*c*). Additionally, our results indicated that among all the RNAs, the lncRNAs had the most similarity between the normal samples and the differentiated cardiomyocytes.

The functional roles of the mRNAs shared between these 2 models (in vivo and in vitro) were examined by molecular function analysis. Subsequently, the signaling pathway was analyzed using the KEGG pathway and the results showed that the shared mRNAs were involved in some important cardiac regulation pathways—including adrenergic signaling in cardiomyocytes, glycolysis/ gluconeogenesis, the calcium signaling pathway, cardiac muscle contraction, the regulation of actin cytoskeleton, the PI3K/Akt pathway, the MAPK signaling pathway, signaling pathways regulating the pluripotency of stem cells, the TGF-beta signaling pathway, and the Wnt signaling pathway. Inter-estingly, the *COMP* gene played a role in the PI3K/ Akt pathway.

The molecular function of the mRNAs shared between these 2 samples was analyzed in the Enrichr Database. The results showed that the shared mRNAs could have the following functions: Wnt-activated receptor activity, myosin heavy chain binding, MAP kinase tyrosine/serine/threonine phosphatase activity, adrenergic receptor binding, BMP receptor binding, muscle alpha-actinin binding, and calcium channel regulator activity.

Discussion

Ischemic injury to the myocardium begotten by coronary occlusion is one of the common types of cardiac tissue damage and accounts for a huge proportion of all hospital admissions and all causes of death in Western societies. ICM occurs when the blood supply to the heart muscle is interrupted by coronary artery disease or heart attacks [4]. Recent years have witnessed several advances in the treatment of ICM; the magnitude of myocardial damage, however, still thwarts the success of the therapeutic modalities. What exacerbates the situation is that these approaches are often applied too late to prevent irreversible damage to the myocardium. Consequently, given the complications of ischemic injury, a better understanding of the cellular and molecular mechanisms involved in myocardial ischemia may further boost clinical care [15-17].

In the current study, we primarily sought to compare the expression profiles of mRNAs, miRNAs, and lncRNAs between non-failing heart tissues (considered normal samples) and ischemic heart tissues with the intention of introducing novel ICM biomarkers. Our results indicated miR-5683, TCNOS_00014011 lncRNA, and the *COMP* gene as the probable biomarkers for the diagnosis of ischemic cardiac tissues. Needless to say, these findings should be validated by future experimental analyses.

Yang et al. [12] in 2014 compared the expression profiles of cardiac mRNAs, miRNAs, and lncRNAs between non-failing and failing human hearts before and after LV assist device support. The authors used Cuffdiff, version 2.0 (mRNA, RefSeq, and ENSEMBL transcript databases as the annotation references vs lncRNA and NONCODE, version 3.0, as the annotation files) and miRanalyzer (miRNA, miRBase v.18) for differential expression analyses. In addition, they applied alignment to the human genome hg19. In contrast, in our analysis, we aligned the sequence reads to the human genome hg38 with HISAT2 (version 2.1.0) and applied NONCODE (version 5.0) and the UCSC RefSeq as the annotation references for lncRNAs and mRNAs, respectively. Additionally, we performed miARma, version 1.7.2, for the analysis of miRNAs and used the DESeq2 package for the differential expression analysis. Yang et al. [12] concluded that lncRNAs were more suitable for the discrimination of non-failing from failing human hearts. They also reported that 37% of the genes and 71% of the lncRNAs had mitochondrial origins. Another salient finding in their study was that the expression profile of lncRNAs was able to distinguish between ischemic and non-ICM tissues as well as cardiomyopathy samples before and after LV assist device support, whereas the expression profiles of mRNAs and miRNAs could not. Their results also showed that lncRNA n340651 and the neighboring mRNA, retinoic acid receptor alpha (RARA), were upregulated in heart failure. In our analysis, the expression of lncRNA n340651 (RARA-AS1, NON HSAG021742.2) showed no differences between the ICM and non-failing samples.

Li et al. [18] studied the role of miRNAs in the network of regulating gene expressions between ischemic and non-ischemic heart tissues and reported that while the expression of miRNA-21 was increased in the ischemic tissues, the expression of miR-17 was downregulated in the non-ischemic samples. They mapped small RNA-Seq sequence reads to miRBase, version 18, with miRanalyzer and Bowtie and analyzed the differential expression of miRNAs using the edgeR package. The authors suggested that miRNAs were striking clinical biomarkers. The results of their study do not chime in with our results insofar as we demonstrated that miRNA5683 with the highest upregulation in ischemic tissues could be an attractive clinical biomarker for ICM.

In 2016, Pang et al. [19] studied the expression of lincRNAs in heart failure by applying the same data set (GSE46224) but categorizing the samples in 2 groups: heart failure tissues (16 samples including ischemic and non-ischemic) and non-failing tissues (considered to be normal). The investigators reported that WNT9A, FZD7, PRKACB, beta-actin, LEF1, and CCND1 were upregulated in heart failure and ICAT was downregulated (all are involved in the Wnt signaling pathway). These authors also suggested several lincRNAs—including ENSG00000260000, ENSG00000233137, XLOC_052037, ENSG00000 260091, XLOC_002393, and ENSG00000232044—as the critical regulators of the Wnt signaling pathway in heart failure. In their study, RNA-Seq data were mapped against the human genome by TopHat; Cufflinks and Cuffmerge were used for transcript merging; read counts were computed by GENCODE, version 19; and differential expression analysis was applied using DESeq2, edgeR, and voom-limma.

In line with our results, Pang et al. [19] reported that the WNT9A gene and ENSG00000260000 lncRNA were upregulated with a log2FoldChange = 3.01 and a log2FoldChange = 1.29, respectively, and ENSG00000232044 was downregulated with a log2FoldChange = -3.07. Nonetheless, the expressions of FZD7, CCND1, and ENSG00000260091 did not exhibit any significant differences between their samples. This discrepancy with our results may be because Pang et al. [19] categorized ICM and non-ICM tissues as heart failure. Based on our comparisons between ICM and non-failing heart samples, we introduced the COMP gene and TCNOS_00014011 lncRNA, which had the highest differential expression, as ICM biomarkers.

The analyses of the signaling pathway of COMP mRNA, which is introduced as a biomarker in our study, have shown that this gene plays a role in the PI3K/Akt pathway. This pathway has important molecular functions in the cardiac tissue such as apoptosis, proliferation, and cell survival [20, 21]. Negoro et al. [22] reported that the PI3K/Akt pathway transduced signals that inhibited apoptosis in cardiomyocytes in response to myocardial ischemia/ reperfusion injury. A large number of studies have concluded that the PI3K/Akt signaling pathway lessens myocardial ischemia/reperfusion injury by inhibiting cardiomyocyte apoptosis and enhancing myocardial viability [23-25].

The *COMP* gene, a pentameric glycoprotein with a molecular weight of 524 kDa, is a matrix cellular protein. Recent studies on the role of *COMP* in the cardiovascular system have shown that this gene is an inhibitor of post-injury neointima formation. Additionally, bone marrow-derived *COMP* can play an important role in atherosclerotic calcification [26].

Wang et al. [27] suggested the serum level of this gene was associated with the severity of coronary heart disease and artery calcification. Elsewhere, the presence of this gene was analyzed in different tissues, where the aorta was the only tissue outside the skeletal system in which *COMP* expression was observed [28]. Additionally, the level of *COMP* expression in the serum is deemed a biomarker for rheumatoid arthritis [29].

Qin et al. [30] indicated that the level of the COMP protein in patients with aortic aneurysms was significantly downregulated by comparison with a control group and concluded that the *COMP* gene could be used as a therapeutic target for cardiovascular disease. Our results demonstrated upregulation in the *COMP* gene in the ICM samples, which shows the importance of this gene.

Heart failure is known to commonly result from cardiomyocyte deficiency; hence, a thorough therapeutic knowledge of heart failure that would confer the regeneration of the lost cardiomyocytes and the recovery of cardiac function may become a future therapeutic target [3].

In the current study, we also studied the similarity in the expression profile between non-failing tissues (LV apex tissue) and differentiated cardiomyocytes as in vivo and in vitro models. Correspondingly, we found that 89% of the lncRNAs, 55.2% of the mRNAs, and 38.6% of the miRNAs were shared between the non-failing tissues and the differentiated cardiomyocytes, with the lncRNAs exhibiting the most similarity.

Given the low similarity between genes and miRNAs, more precise studies are required to examine the ischemic disease. Indeed, meticulously designed in vivo models are needed to validate the biomarkers suggested by in vitro models.

According to the results of the present bioinformatics study, miR-5683, the *COMP* gene, and TCNOS_00014011 lncRNA are probable ICM biomarkers. In addition, our in vivo and in vitro analyses indicated that lncRNAs expression had the most similarity between these two models. Altogether, it is important to pay sufficient attention to transcriptome profiling—especially lncRNAs—and the origin of the damaged tissue in myocardial ischemia to find accurate biomarkers. Indubitably, experimental validation is necessary to obtain results that are more robust.

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Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Left Atrial Enlargement Prevalence in Normotensive, Normoglycemic, Obese, Postmenopausal Women and Its Correlation with Left Ventricular Hypertrophy and Diastolic Function. An Echocardiographic Real World Observational Study

Maria Maiello[1], Annagrazia Cecere[2], Marco Matteo Ciccone[2], and Pasquale Palmiero[1,*]

[1]ASL BRINDISI, Cardiology Equipe,
District of Brindisi, Brindisi, Italy
[2]Cardiovascular Diseases Section, Department of Emergency and Organ Transplantation (DETO), University of Bari, Italy

Abstract

Background. Left atrial (LA) enlargement, as increased LA indexed volume, is a strong predictor of cardiovascular outcomes, mainly atrial fibrillation (AF), mostly in obese populations. However its prevalence among obese postmenopausal women (PMW) without diabetes and/or hypertension and its correlation with left ventricular (LV) hypertrophy and/or LV diastolic function is not well known. The purpose of the study was to detect prevalence and correlation of LA enlargement among obese PMW.
Methods. We enrolled 125 consecutive obese PMW, with a body mass index (BMI) greater than 30, from 45 to 55 years of age; 150 normal-weight PMW (BMI lower than 25), comparable for age, served as a control group. All patients underwent M-B mode echocardiography, PW-Doppler and tissue-Doppler evaluation. LA indexed volume, LV mass and LV diastolic function were assessed according to current ASE guidelines.
Exclusion criteria were diabetes mellitus and arterial hypertension.
Results. LA enlargement rate was higher in obese than in normal weight PMW: 38 out of 125 (30.4%) versus 10 out of 150 (6.6%), $p < 0.0001$. The association between LA enlargement and increased LV hypertrophy was more common among obese than in normal weight PMW: 22 (17.6%) versus 3 (2%), $p < 0.0001$. The association between LA enlargement and LV diastolic dysfunction was more common among obese than normal weight PMW: 13 (10.4%) versus 8 (5.4%), $p < 0.07$, which was not statistically significant.
Conclusions. LA enlargement is common in normotensive, non-diabetic, obese PMW and correlates well with LV hypertrophy. LV diastolic function indices were little affected in this study, however, likely reflecting an adaptation of the heart to the obese status. Further studies are recommended to determine whether LA enlargement in normotensive, healthy, obese PMW is associated with adverse long-term outcomes.

* *Corresponding Author*: pasqualepalmiero@iciscu.org;
Tel: +39 0831 536556

Keywords: left atrial enlargement, postmenopausal women, left ventricular mass, diastolic function, obesity

Introduction

Left atrial (LA) enlargement is a strong predictor of cardiovascular outcomes and correlates with atrial fibrillation (AF) [1-3], stroke [4-5], heart failure [6-7] and mortality [8-9].

Echocardiographic estimation of LA size is of great relevance because of its prognostic clinical implications. Left atrium volume (LAV) estimated by 2D echocardiography is more accurate than LA diameter for the evaluation of LA size [10] according to the American Society of Echocardiography (ASE) recommendations for cardiac chamber quantification [11].

LA enlargement (LAE) is an expression of LA pressure and/or volumetric overload, but many other factors contribute to increase LA size [12]. There are significant differences concerning LAV between genders, mainly due to variation in body size [13]. Therefore, body size, based on weight, represents the major determinant for LA size. In patients with preserved left ventricular (LV) systolic function, LAV increase is also associated with severity of LV diastolic dysfunction (LVDD) and LV mass increase [14-15].

Post-menopausal women (PMW) develop visceral obesity, with a remarkable increase of body fat mass, due to reduced concentration of circulating estrogen [16]. Although the correlation between body size, based on weight, and LAE has been widely mentioned in the literature, the true prevalence of LAE in this population of obese PMW is lacking.

The main goal of our study was to evaluate the prevalence of LAE in obese PMW without hypertension and/or diabetes mellitus (DM). Another goal was to evaluate the correlation of LAE with left ventricular (LV) hypertrophy and/or LV diastolic function.

Methods

Patients

We screened 125 consecutive obese PMW according to body mass index (BMI) greater than 30, recruited in a population of 4,982 consecutive women sent to our Heart Station in Brindisi (Italy) by general practitioners, from January 2018 to June 2019. All women provided an informed consent to be enrolled. Menopause was assessed when the last period occurred at least 12 months before our medical examination. 150 consecutive PMW with BMI lower than 25 were enrolled as a control group. All women on therapy for hypertension and/or diabetes were excluded; other exclusion criteria were: severe to moderate mitral regurgitation and all types of atrial fibrillation. We collected age, body mass index, most recent blood pressure measurement, comorbid conditions, number of visits, free medications dispensed, smoking status, and HbA1c on all women. EKG, Pulse Wave (PW) Tissue Doppler and Trans Thoracic Echocardiogram (TTE) were performed on all enrolled women. Demographic characteristics and clinical conditions at the start of study are listed in Table 1.

Table 1. Baseline characteristics (demographics, diseases and clinical conditions)

Demographic variables	All	Obese		C G		
Age (%)		57 ± 11		56±8		ns
Postmenopausal women	275	125		150		
BMI		> 30 kg/m^2		< 25 kg/m^2		
HbA1c		4.8-6.0%		4.6-5.8%		ns
Abnormal ECG	173	54	43.2%	48	32.0%	ns

ns: not statistically significant (P > 0.05).

Echocardiography

Transthoracic echocardiography was performed with the patient in left lateral decubitus, after 10-min rest, with the exam table elevated by 30°. The examination was carried out with a 3.5 MHz probe, with ECG trigger. We used an echo-Doppler system equipped with a multi-frequency transducer (Philips, Epiq 7, Utrasound System for Cardiology, Healthcare, viale Sarca 235, Milan, Italy).

We estimated: intraventricular septum thickness in diastole (IVSd), left ventricular diastolic diameter (LVDd), left ventricular posterior wall thickness during diastole (LVPWd), intraventricular septum thickness in systole (IVSs), left ventricular systolic diameter (LVSd), and left ventricular posterior wall thickness during systole (LVPWs). LV mass was determined according to the formula of Devereux et al. [17] and indexed to body surface area (BSA) to obtain LV mass index (LVMI).

The prolate ellipsoid left atrial volume indexed by Body Surface Area (BSA) was assessed from both apical four- and two-chamber views. Reference values of echocardiographic parameters are in accordance with the latest guidelines of the American Society of Echocardiography [11].

Peak velocities of early (E wave) and late (A wave) trans-mitral flow and deceleration time (DT) were determined, and E/A ratio was calculated. According to current ASE guidelines, LVDD was assessed by at least two expert cardiologists by PW Doppler of mitral inflow and Doppler tissue imaging of the mitral annulus [18]. LVDD was assessed as mild or grade I (impaired relaxation pattern), moderate or grade II (pseudonormal pattern), or severe or grade III (restrictive filling). Mild (impaired relaxation pattern) LVDD corresponded to mitral E/A ratio ≤0.8, average E/E' ratio < 10 (septal and lateral), and peak tricuspid regurgitation (TR) velocity <2.8 m/sec. Moderate (pseudonormal pattern) LVDD corresponded to mitral E/A ratio from 0.8 to 2, average E/E' ratio from 10 to 14, and peak TR velocity >2.8 m/sec. Severe (restrictive filling) LVDD corresponded to E/A ratio >2, average E/E' ratio> 14, peak TR velocity >2.8 m/sec [18].

Statistical Analysis

Statistical analysis was performed by IBM SPSS version 25.0 (Chicago, IL, USA). Results are described as means with 95% confidence intervals (95% CI). Chi-squared test was used for categorical variables. For contingency tables, odds ratios were calculated. A P-value <0.05 was considered to indicate statistical significance.

Results

Results of the study are described in Table 2.

LA volume indexed by BSA > 34 mL/ m^2 was diagnosed as LAE, LV mass > 95 g/m^2 was diagnosed as LVH. All women presented a grade I on LVDD: impaired relaxation pattern.

LA enlargement was more common among obese than in normal-weight PMW: 38/125 (30.4%) versus 10/150 (6.6%) (Chi-squared = 26, Odds ratio = 6.1, P < 0.0001). The association between LA enlargement and LV hypertrophy was more common among obese than in normal weight PMW: 22/125 (17.6%) versus 3/150 (2%) (Chi-squared = 20, Odds ratio = 10, P < 0.0001). Finally, the association between LA enlargement and LVDD tended to be more common among obese than in normal weight PMW: 13/125 (10.4%) versus 8/150 (5.4%) (Chi-squared = 3, Odds ratio = 2.2, P < 0.07).

Table 2. Results of the study

	Obese post-menopausal women	Post-menopausal women	Chi Squared	Odds Ratio	P-value
Doppler echocardio findings					
Left atrial enlargement	38/125 (30.4%)	13/125 (10.4%)	26	6.1	p < 0.0001
Left ventricular hypertrophy	22/125 (17.6%)	3/150 (2.0%)	20	10	p < 0.0001
Left ventricular diastolic dysfunction	13/125 (10.4%)	8/150 (5.4%)	3	2.2	ns

ns: not statistically significant (P >0.05).

Discussion

LA Function

LA plays an important hemodynamic role. It is a reservoir collecting pulmonary venous return during ventricular systole; then allows the passage of blood into LV during ventricular diastole [19]. For these reasons, diastolic filling and global systolic function of LV are affected by LA size. Many factors can contribute to an adaptive remodeling of LA. Obesity causes an adaptive dilation of the LV associated with an increased LV mass with development of eccentric hypertrophy. Similarly, arterial hypertension is also responsible for an adaptive LV remodeling, resulting in concentric hypertrophy. Both eccentric hypertrophy, mediated by obesity, and concentric hypertrophy, related to arterial hypertension, compromise normal LV filling, with consequent LVDD, and, therefore, LAE [15]. So LAE is due to a chronic volume overload related to obesity and to chronic pressure overload linked to arterial hypertension.

LAE and Obesity

The main aim of our study was to evaluate the prevalence of LAE in a population of obese PMW without arterial hypertension and/or DM. To our knowledge, this is the first study in the literature that evaluated the prevalence of LAE in a population of obese PMW, in the absence of confounders such as arterial hypertension and DM.

In our obese PMW, we found a higher prevalence of LAE, compared to a control group ($P < 0.0001$). This can be explained by the remarkable hormonal changes that occur in the post-menopausal period, promoting a cardiac remodeling. In the post-menopausal period, the distribution of fatty tissue increases especially in the trunk and abdominal regions. This expansion of fatty tissue is due to the reduction in circulating levels of estrogen, resulting from the decline in ovarian function, with a relative increase in androgen levels [20-21]. Therefore, hormonal changes typical of PMW are responsible for a greater predisposition to visceral obesity [16]. Oliver et al. [22] demonstrated that the association between Δ-LAV and BMI can only be explained by increasing of the visceral fat mass. For this reason, in obese PMW, the increase of visceral obesity promotes an adaptive cardiac remodeling with development of LAE.

The result of our study agrees with evidence by other authors. The MONIKA/KORA prospective study evaluated the association between obesity and arterial hypertension with LAE in 10 years of observation. The study concluded that obesity is the strongest predictor of LAE. In a linear regression model, the effects on cardiac remodeling mediated by obesity have an almost doubled role compared to arterial hypertension [23]. Similarly, Wang et al. [24] observed that obesity is a substantial risk factor for the new-onset of AF. The correlation between obesity and the new-onset of AF appears to be mediated by LAE and not by systolic blood pressure [24].

For a correct correlation between LA dimensions and body size, the evaluation of LAV needs to be normalized for BSA, as suggested by current American echocardiographic recommendations [17]. However, BSA is assessed by weight/squared height ratio; LAV normalized for BSA could be underestimated in obese patients [13]. On the basis of this “bias,” the best method for measuring normalized LAV is still under discussion.

LAE and LV Hypertrophy

The secondary aim of our study was to evaluate the correlation between LAE and LV hypertrophy. Our study showed a significant association between LAE and LV hypertrophy in obese PMW ($P < 0.0001$). This result agrees with authors who assessed a direct correlation between systolic blood pressure and LA size. In patients with arterial hypertension, LAE seems to be the echocardiographic finding that better correlates with LV mass [25]. Cioffi et al. [26] demonstrated that LV concentric ventricular hypertrophy, common in hypertensive patients, is closely related to LA size. Tsioufis et al. [27] assessed in hypertensive patients a positive correlation between LAV and systolic blood pressure, LV mass and BNP. Nevertheless, in multiple linear regression analysis only the LV mass showed a significant association with LAV, emphasizing that LAE is a crucial adaptive result to concentric hypertrophy [27]. In hypertensive patients, LV stiffness greatly influences left atrial performance. Impaired LV filling, common among patients with LV hypertrophy, increases atrial pressure promoting LAE [28].

Since the adaptive mechanisms concerning LAE appear to be different in obesity and in arterial hypertension, we chose to exclude hypertensive PMW from enrollment in our study to evaluate if a greater body size may determinate LA remodeling, confirming that obesity alone is an important predictor of LAE.

LAE and LVDD

A clear prevalence of LVDD in PMW has just been assessed [29-30]. It could be explained by reduction of circulating estrogen causing endothelial dysfunction, atherosclerotic plaque progression, activation of renin-angiotensin-aldosterone system and the consequent reduction of ventricular compliance [31]. According to these observations, our study identified a higher prevalence of LVDD in obese PMW; however, the correlation between LAE and LVDD was not statistically significant ($P < 0.07$).

Many studies supported that arterial hypertension, left ventricular hypertrophy and LVDD promote cardiac remodeling and, finally, contribute to LAE [32]. Naito et al. [33] demonstrated that chronic pressure overload is responsible for active atrial remodeling, while chronic volume overload causes a passive atrial remodeling. Particularly, different patterns of LA chronic overload determine distinct LA geometrical remodeling and, consequently, a different cardiovascular prognosis [33]. Based on this consideration, our research group evaluated the association between LVDD, mitral regurgitation and LAV. In our study, patients with LVDD had a more elongated atrial shape, while patients with mitral regurgitation presented a spherical atrial morphology, which was associated with a less effective atrial contractility with subsequent development of AF, which leads to a poor prognosis, demonstrating that LAV is superior to the LA diameter as parameter of LAE [34].

LAV has been proposed as a marker of LVDD grading, but its prognostic value is still debated [14]. Nishimura et al. [35] assessed that increased atrial pressures positively correlated with the degree of LVDD in a group of patients with cardiomyopathy and LV ejection fraction less than 40%. Matsuda and Matsuda [36] found that LAV increases proportionally with severity of LVDD.

Pritchett et al. [15] evaluated the association between LVDD and LAV and assessed that LAV is a specific and sensitive marker for identifying patients with severe grade of LVDD (grade III or IV). This finding justifies the lack of statistical significance in the correlation between LAE and LVDD in our patients with mild-moderate LVDD.

Future Perspectives

Results of our study raise some questions concerning the prognosis and management of obese PMW with LAE.

Body size and obesity are the strongest predictors of LAE. Because of the high prevalence of obesity among PMW, preventive strategies to avoid cardiac remodeling and LAE need to be implemented. However, data on a beneficial effect of weight loss on LAE are still limited. Garza et al. [37] assessed long-term effects of weight loss on LAV in 57 obese patients undergoing gastric bypass surgery, observing that body weight reduction is associated with a favorable change in LAV, regardless of comorbidities related to obesity. Often, visceral obesity is the first sign of metabolic syndrome and united to insulin resistance leads to LA enlargement and dysfunction [38].

There is a correlation between LAE and onset of many adverse cardiovascular events, mainly atrial fibrillation and heart failure [39-40]. Although there is much evidence assessing that LA dimensions can be reduced by medical therapy [41-42], further studies are needed to evaluate long-term clinical benefits from a reduction in LA dimensions.

Conclusion

LA enlargement is common in normotensive, normoglycemic, obese PMW, and correlates well with LV hypertrophy. It seems to be less due to LV diastolic function impairment and likely reflects an adaptive response of the heart to increased fatty tissue of the obese status. LAV is superior to the LA diameter measurement and it should be included into routine echocardiographic evaluation of PMW because of its clinical and prognostic implications.

Further studies will assess if LA enlargement in normotensive, normoglycemic, obese PMW is associated with adverse long-term outcomes.

Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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YouTube Videos as a Source of Information on Blood Pressure Measurement

Saif Aldeen AlRyalat[1,*], Muayad Azzam[2,†], Esraa Al-salamin[2,‡], Ghaida Khraisat[2,§], and Salam Alkhreasha[2,#]

[1]Department of Ophthalmology, The University of Jordan, Amman, Jordan
[2]Medical Student, The University of Jordan, Amman, Jordan

Abstract

Background: With the increasing use and dependence on the internet to obtain medical information and the effectiveness and attractiveness of video-based learning linked with the stressed importance on proper blood pressure measurement, the purpose of this study is to assess the scientific quality of YouTube videos that present information on blood pressure measurement.
Methods: In the study, we included 99 relevant videos out of the first 500 videos returned by YouTube's search engine in response to 'Blood pressure measurement' sorted by view count. We measured the scientific quality of these videos based on an 11-element skillset of blood pressure measurement.
Results: The sample was found to have a mean scientific score of 4.87 (±2.28) with the most common step mentioned of the 11 being placing the cuff over a bare arm. A significant difference in mean scientific quality with source upload ($p = 0.015$) was found, with the highest quality for health information websites (8.6 ± 2.3). A significant correlation was found between scientific quality and duration of video ($p < 0.001$); however, no significant difference was found between scientific quality and view count.
Conclusions: Although the YouTube content on blood pressure measurement is scientifically weak at its current state, it has the potential to greatly benefit its viewers if properly utilized. To make better use of the YouTube platform we recommend the evaluation of medical information obtained from YouTube by health professionals and organizations as well as developing and uploading medical videos themselves.

Keywords: Blood Pressure Measurement; YouTube; Content Analysis; Medical videos

Introduction

Previous studies suggested that there is an increasing growth in the use of the internet as a medical information resource by both patients and physicians [1–5], as found in a recent survey that out

* *Corresponding Author*: Department of Ophthalmology, The University of Jordan. Queen Rania Street, Amman, Jordan. Phone: +962798914594 ORCID: https://orcid.org/0000-0001-5588-9458. Email: saifryalat@yahoo.com
† Email: azzam.muayad@outlook.com
‡ Email: esraaodah0@gmail.com
§ Email: Ghaida.khraisat@yahoo.com
# Email: Salamalkhre@gmail.com

of 10 Internet users, 8 have accessed health information online [1]. This is specifically significant for patients with chronic illnesses; Madathil et al. [1] have reported an increase in chronic patients' dependence on Internet-based resources to manage their illnesses; various surveys conducted by the Pew Research Center have found that online health information influenced how 75% of chronic patients managed their condition [1]. Video-sharing sites, the most well-known of which being YouTube, are particularly popular resources for healthcare information with the potential to enhance the learning experience if appropriately used [1, 3, 6]. However, concerns about the accuracy and quality of the information shared, because of the minimal guidelines and interventions regulating the uploaded material, have been expressed by healthcare providers and government agencies [1, 2, 5, 7].

Hypertension is one of the leading causes of morbidity and mortality worldwide with 29 percent of adults in the United States affected [7–9]. According to the literature, blood pressure (BP) measurement is the most common and important office test performed in clinical practice [9–12]. Burgess. S et al. [13] have proposed that significant differences exist between BP readings obtained using recommended techniques and readings without following the recommended techniques; therefore, failure to follow measurement guidelines may have significant clinical consequences.

With the deleterious effects hypertension imposes and the stressed importance on proper blood pressure measurement, linked with the status and popularity of video-based learning and its effectiveness, the aim of this study is to assess the quality of YouTube videos on blood pressure measurement and provide recommendations on how to find the most appropriate video to learn or recall the essentials of blood pressure measurement.

Methods

On February 10, 2019, at The University of Jordan, we searched the YouTube website using the keywords 'Blood pressure measurement'. We sorted the results according to view count, and watched the first 500 videos for videos teaching how to measure blood pressure. We included the following videos:

- English language videos.
- Videos providing information on the steps of blood pressure measurement.

We excluded the following videos:

- Duplicates.
- Videos measuring blood pressure in sites other than the arms.
- Blood Pressure Measurement device advertisements.
- Videos without a demonstration on blood pressure measurement
- Videos unrelated to blood pressure measurement

Of the 500 videos, 100 met the set criteria. The videos were watched independently by two raters, trained previously on the 11 steps of blood pressure measurement suggested by Rakotz et al. [11]. In cases of differing advice scores, the videos were re-watched by both raters simultaneously, and re-scoring was settled by discussion.

Outcome Measures

- Scientific quality based on the 11 steps listed below.
- Max video quality in pixels (e.g., 144, 240...).
- Duration, in seconds.
- Source upload
- Presence of advertisement
- Views, likes, dislikes, and number of comments.

Blood Pressure Measurement Scientific Quality Score

The performance score was based on 11 skills: resting the patient for 5 minutes prior to the measurement or expressing intent to do so; legs uncrossed; feet on floor; arm supported; correct cuff size; cuff placed over bare arm; no talking; no mobile phone use or reading; BP measurement taken in both arms; correctly identifying BP from the arm with the higher reading as being clinically more important when asked; correctly identifying which arm to use for future readings (the arm with higher BP). Participants received pass or fail scores for each of the 11 skills tested, where the videos should have either performed or mentioned each step. These scores were combined into an overall performance score [11].

Upload Source

All videos were also categorized according to source into 4 groups:

- Health information websites (for example, AnswersTV.com);
- University channels/professional organizations (for example, Mayo Clinic, Arthritis Foundation);
- Medical advertisements/for-profit companies (for example, Immunogizerfatreducer.com, Dr. McDougall's Health and Medical Center)
- Independent users.

Statistical Analysis

We used SPSS version 21.0 (Chicago, USA) in our analysis. We used mean (± standard deviation) to describe continuous variables (e.g., views). We used count (frequency) to describe other nominal variables (i.e., electronic vs. manual device).

We performed independent sample t-test to analyze if there is a difference in mean scientific quality between videos with or without a reference, and also for videos with and without ads, and we presented data as mean (standard deviation). We used one-way ANOVA to analyze if there is a difference in mean scientific quality for source upload, type of device used, year published, and maximal video quality. We analyzed the correlation between scientific quality and each of views, like, dislikes, number of comments, and duration, using Pearson correlation.

All underlying assumptions were met, unless otherwise indicated. We adopted a P-value of <0.05 to indicate statistical significance.

Table 1. Details the characteristics of the included sample

		Count	Column N %	Mean	Standard Deviation
Source	Health information website	5	5.1%		
	Governmental or university	39	39.4%		
	Companies	16	16.2%		
	Individuals	39	39.4%		
Year Published				2013	3
Manual vs Electronic	Manual	49	49.5%		
	Electronic	47	47.5%		
	Both	3	3.0%		
Duration in Seconds				254	237
Views				153472.72	588905.17
Comments				25.88	93.78
Likes				325.78	1096.10
Dislikes				44.88	174.96
Referenced	No	77	77.8%		
	Yes	22	22.2%		
Presence of Ads	No	11	11.1%		
	Yes	88	88.9%		
Scientific Quality				4.87	2.28
Maximum Video Quality	0	3	3.0%		
	144	1	1.0%		
	240	13	13.1%		
	360	7	7.1%		
	480	29	29.3%		
	720	20	20.2%		
	1080	24	24.2%		
	2160	2	2.0%		

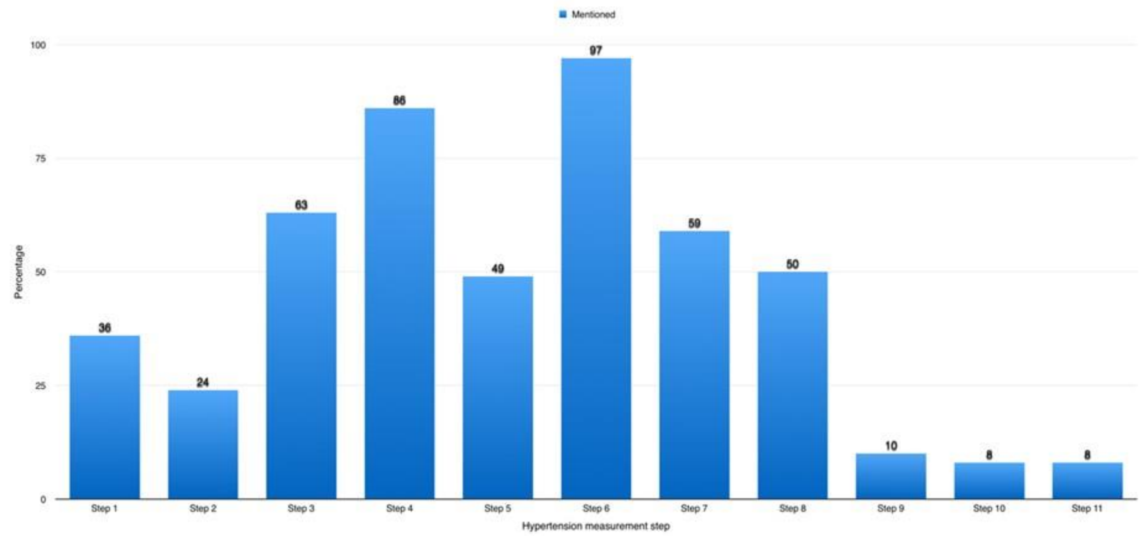

Figure 1. Percentage of videos that performed/mentioned each step of the blood pressure measurement.

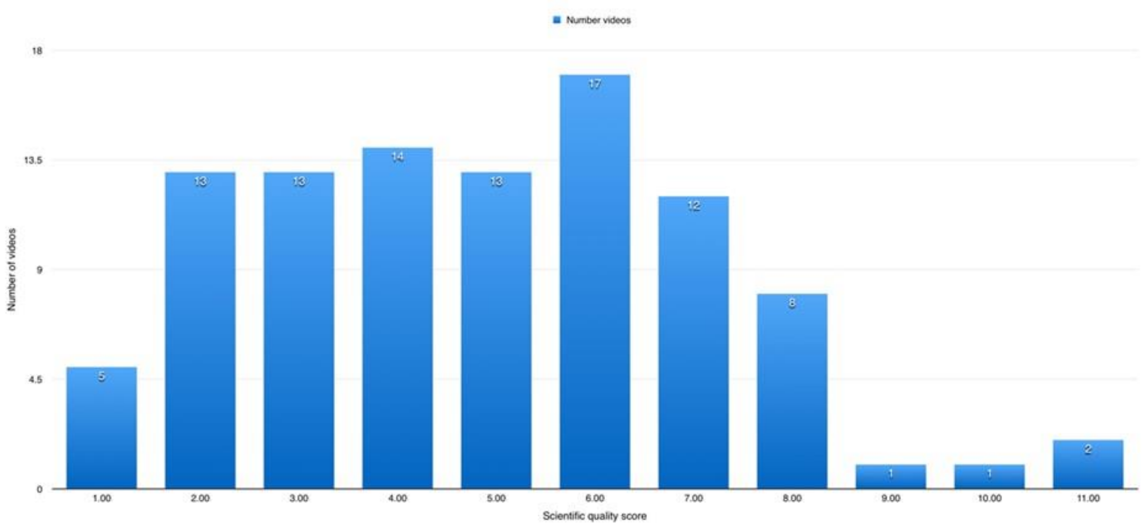

Figure 2. Frequency of videos for each scientific score.

Results

We screened 500 videos, where we included 99 videos in this study, with a total duration of 419 minutes and 53 seconds. The included videos were published between 2007 and 2018, with 2013-2015 containing 42 (42.4%) videos. 48 (48.5%) videos used a manual device for BP measurement, and 47 (47.5%) used electronic devices, and 3 (3%) used both devices.

Table 1 details the characteristics of the included sample.

Of the 11 blood pressure measurement steps, step 6 is the most commonly performed/mentioned step, whereas steps 9, 10, and 11 were the least commonly performed/mentioned. Figure 1 shows the percentage of videos that performed/mentioned each step of the blood pressure measurement.

Only 4 (4%) videos had a scientific quality score of 9 and above, where 2 videos had a scientific score of 11/11 and both were uploaded by a Health information website. Figure 2 shows the frequency of videos for each scientific score.

We found a significant difference in mean scientific quality with source upload (p=0.015), with the highest quality for health information websites (8.6 ±2.3), followed by governmental/university (5.1 ±2.28), then for-profit companies (4.6 ±1.96), and finally individuals (4.3 ±1.99).

We found a significant correlation between scientific quality and duration of video (p<0.001) with a positive correlation coefficient of 0.464. No significant correlation with views, likes, dislikes, or number of comments.

No significant difference was found between scientific quality and each of presence of reference, presence of ads, type of device used, maximal video quality.

Discussion

The appearance of online educational healthcare videos is a relatively new phenomenon, with the oldest video in our sample dating back to 2007. Therefore, studies on the use of YouTube for communicating and sharing healthcare information are still in their developing stages and effective distribution methods of YouTube videos for healthcare education are still lacking [1]. With a mean view count of (153,472.72) among our sampled videos, it is clear that many patients and physicians have hastily adopted YouTube as a resource for healthcare information although very little information on their dependability and accuracy is available [5]. This occurrence stresses the fact that more studies setting guidelines, regulations and recommendations for YouTube healthcare videos should be conducted to insure the accurate flow of medical information to YouTube users [2, 7] and to avoid the spread of misleading information [1, 14].

A study done to analyze the source and quality of medical information available on YouTube about Ebola hemorrhagic fever after the Ebola epidemic showed that the high relevance video group had a higher median number of views, more likes, channel subscriptions, and shares [7]. However, our results in addition to various other studies analyzing YouTube content [15–18], found no significant correlation with views, likes, dislikes, or number of comments to be present. This means that viewers are watching and reacting to videos with a high scientific score with similar frequency to videos with a low scientific score. This is significant when taking into account that the view count, likes and dislikes count and number of comments of videos are shown to viewers, so the high view count of weak videos might lead individuals to watch and depend on them, further favoring these low scientific score videos to spread. The spread of these videos might have deleterious effects on blood pressure management, especially when taking into account the importance of blood pressure measurement in diagnosing hypertension and the alterations in readings that occur if the proper guidelines are not followed [13].

Opposing other YouTube content analysis studies [15–18], the study in hand has shown a significant correlation between scientific quality and duration of video (correlation coefficient of 0.464). We assume that the relative simplicity and clarity of the blood pressure measurement method compared to the methods and topics in the other studies has played a role in this by allowing longer videos to cover more areas on blood pressure measurement.

A study performed on 159 medical students measuring their knowledge on the same 11 steps of blood pressure measurement used in this study had a similar mean score of 4.1 compared to the mean score of 4.87 for the online videos in this study [11]. Also, Rakotz et al. [11] and the study in hand found that the most practiced and taught skill of the 11 was ensuring the cuff was placed over a bare arm. The survey was performed in 2015 while the mean year of upload of the videos is 2013, so these online videos might have contributed to the medical student's method of blood pressure measurement. Various other studies on medical school students and nursing students [11, 19–22] have also yielded similar levels of knowledge regarding the correct procedure for measuring blood pressure. Numerous studies have shown that many mistakes are made during the improper performance of BP procedures [9, 12] and that the errors tend to compound one another and lead to readings higher than the actual blood pressure of the patient [11], which will lead to unnecessary and potentially harmful therapy for a significant number of patients who should not be diagnosed as hypertensive, in addition to the psychological effects of misdiagnosis and unnecessary cost [10, 11].

Video-based learning has been found to be an effective way to understand a method [1]; thus, YouTube videos can be utilized to increase the level of knowledge in healthcare students, workers, patients and other viewers. This can effectively be done by government agencies, professional organizations, and healthcare professionals to participate in developing and uploading YouTube videos, as studies have identified that the information provided by such sources is trustworthy [15]. Our study supports this as a significant difference in mean scientific quality with source upload has been found, such that the highest quality is for health information websites followed by government agencies. It is notable to mention that only 2 videos had a scientific score of 11/11 and both were uploaded by a Health information website.

Based on the results of the study, we recommend that individuals who want to learn the correct blood pressure measurement via online videos look out for the following criteria: 1) Source Upload 2) Video duration, with the higher scientific score favoring videos with a long duration and 3) uploaded by healthcare websites. However, they also should keep in mind that most videos do mention all the steps for blood pressure measurement, so they should depend on other more reliable sources.

While conducting the study, numerous limitations were noted. Firstly, the videos included in the study were in the English language only and were obtained solely through YouTube. With new video sharing platforms emerging, future studies should consider assessing other video databases and platforms. To add to this, we did not assess blood pressure measurements using other new devices (e.g., mobile phones) or from areas other than arms.

Ethical Compliance

The authors have stated all possible conflicts of interest within this work. The authors have stated all sources of funding for this work. If this work involved human participants, informed consent was received from each individual. If this work involved human participants, it was conducted in accordance with the 1964 Declaration of Helsinki. If this work involved experiments with humans or animals, it was conducted in accordance with the related institutions' research ethics guidelines.

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Phone: (631) 231-7269
E-mail: nova.main@novapublishers.com